



DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT
QUALITY ASSURANCE
Question Bank with Model Answers

Course Title	Quality Assurance	Course Code	IM353IA
Semester	V	Units Covered	Unit I to Unit V
Compiled From	CIE and SEE papers, 2019–2026	Questions	Part A: 110 Part B: 116

Notes:

- Questions are compiled from the CIE and SEE question papers of 2019–2026. The unit and the source paper are shown below each question.
- Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive and numerical questions (4 to 16 marks).
- Model answers are written to the depth expected for the marks allotted. Numerical answers show the complete working, and the final result is stated after **Answer:**.
- Where a question repeats an earlier one, the answer refers to the earlier question (the reference is a link).
- M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome. Control chart factors (A_2 , A_3 , B_3 , B_4 , D_3 , D_4 , d_2 , c_4) are taken from the standard table of factors for variables control charts. Probabilities of acceptance for sampling plans are calculated from the binomial (type B) or hypergeometric (type A) distribution as stated; the values of the F distribution are taken at the 5% level.

Sl. No.	Questions	M	BT	CO
PART – A				

1	Write different definitions of quality. Source: Unit I SEE Nov/Dec 2019 (16IM53), Q1.1; SEE Mar/Apr 2023 (18IM53), Q1.1 (any one definition)	02	–	–
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- Fitness for use:** quality means fitness for use (Juran). It has two aspects, quality of design (intentional grades or levels of quality) and quality of conformance (how well the product conforms to the specifications required by the design).
- Modern definition:** quality is inversely proportional to variability; if the variability in the important characteristics of a product decreases, its quality increases.
- Conformance to requirements:** products and services must meet the requirements of those who use them (Crosby).
- Customer view:** quality is the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs (ISO).

2	List any four dimensions of quality. Source: Unit I SEE Feb/Mar 2025 (IM353IA), Q1.5	02	1	1
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Answer: Any four of Garvin's eight dimensions: performance, reliability, durability, serviceability, aesthetics, features, perceived quality and conformance to standards.

Example: for a car, performance is acceleration and mileage, reliability is how seldom it needs repair, durability is its service life and serviceability is how quickly and cheaply it can be repaired.

3	The _____ are the means of translating the term 'fitness for use' into the language of technologists. Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q1.1	01	–	–
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Answer: quality characteristics

Quality characteristics (critical-to-quality or CTQ characteristics) are the measurable parameters, such as length, weight, taste or reliability, that jointly describe what the user thinks of as quality.

4	A physical or chemical property or any other requirement used to define the nature of the product is termed as _____. Source: Unit I SEE Jan/Feb 2021 (16IM53), Q1.2	01	–	–
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Answer: a quality characteristic

Quality characteristics may be physical (length, weight, voltage, viscosity), sensory (taste, appearance, colour) or time-oriented (reliability, durability, serviceability).

5	Quality cannot be inspected into a product nor can it be advertised in; it must be designed and built in. Comment on the statement. Source: Unit I SEE Jan/Feb 2021 (16IM53), Q1.1	02	–	–
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The statement is correct. Inspection only sorts good units from bad after they are made; the cost of making the defective units has already been incurred, and inspection is never 100% effective. Advertising cannot change the actual product. Quality results from prevention: quality of design (choosing the right materials, specifications and processes) and quality of conformance (capable processes kept in statistical control). Hence quality must be designed into the product and built into it through process control, not inspected or advertised in.

6	Differentiate between defect and defective. Source: Unit I CIE-I, 26/11/2024; CIE-I, 07/11/2025; SEE Jan/Feb 2021 (16IM53), Q1.7	02	2	1
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Basis	Defect (nonconformity)	Defective (nonconforming unit)
Meaning	A specific failure of a quality characteristic to meet its specification that is serious enough to affect the safe or effective use of the product.	A unit of product that contains one or more defects.
Counted as	Number of defects per unit; a unit may have several defects.	Number or fraction of defective units in a sample.
Chart used	c chart and u chart	p chart and np chart
Example	A scratch or a missing rivet on a car door	The car door that has the scratch

7	Consider the process of costing (casting). Give an example of a defect. Source: Unit I SEE Mar/Apr 2023 (18IM53), Q1.7	01	–	–
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For a casting process, typical defects are blow holes, porosity, shrinkage cavities, cold shuts, misruns, hot tears and dimensional mismatch. **Example:** a blow hole on the machined face of a cast pump housing.

8	List the statistical methods for quality control and improvement. Source: Unit I CIE-I, 07/11/2025	02	1	1
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The three major areas of statistical methods for quality control and improvement are:

- (1) **Statistical process control (SPC):** control charts and the other SPC problem-solving tools, used on-line to monitor processes and reduce variability.
- (2) **Design of experiments (DOE):** systematically varying controllable input factors (for example, factorial designs) to find the key variables and their best levels; an off-line tool.
- (3) **Acceptance sampling:** inspection and classification of a random sample from a lot and a decision on the disposition of the lot.

Regression analysis and time-series models are also used to relate the important inputs to the output quality characteristics.

9	What is the primary goal of quality assurance? Source: Unit I CIE-I, 26/11/2024; CIE-I, 07/11/2025	02	2	1
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The primary goal of quality assurance is to ensure that the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved. It does this through a documented quality system (policy, procedures, work instructions and specifications, and records), so that customers get products that consistently meet requirements and problems are prevented rather than detected.

10	What is the first step in Quality Assurance? Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.1	01	–	–
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The first step is **quality planning**: identifying the customers (external and internal) and their needs (listening to the voice of the customer), and then setting the quality policy and objectives that the quality assurance system will document and maintain.

11	Differentiate between Quality Control (QC) and Quality Assurance (QA). Source: Unit I SEE Feb/Mar 2025 (IM353IA), Q1.1	02	2	1
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Basis	Quality Control (QC)	Quality Assurance (QA)
Meaning	Set of activities used to ensure that products and services meet requirements and are improved continuously.	Set of activities that ensures quality levels are properly maintained and supplier and customer quality issues are resolved.
Orientation	Product and process oriented; detects and corrects variation.	System oriented; prevents problems through planned, documented procedures.
Tools	SPC, control charts, inspection, designed experiments.	Quality system documentation, audits, standards (ISO 9000), training.

12	What are the main objectives of a quality audit? Source: Unit I SEE Feb/Mar 2025 (IM353IA), Q1.4	02	2	1
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- Verify that the documented quality system (policy, procedures, work instructions, records) is being followed.
- Check that the system is effective in meeting quality objectives and requirements of standards such as ISO 9001.
- Identify nonconformities and opportunities for improvement and initiate corrective and preventive action.
- Provide management with an independent assessment for management review and for certification or surveillance audits.

13	How does ISO certification benefit organizations? Source: Unit I CIE-I, 26/11/2024	01	1	2
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ISO certification demonstrates a supplier's ability to control its processes. It gives a documented, consistent quality management system; satisfies customers who require certified suppliers (often a condition for doing business, nationally and internationally); clarifies responsibilities; and builds continual improvement into the organization through internal audits and management review.

14	Write any four benefits of ISO 9000. Source: Unit I SEE Nov/Dec 2019 (16IM53), Q1.3	02	–	–
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- Documented and consistent processes, so products are made the same way every time.
- Greater customer confidence and access to customers and markets that require ISO 9000 certification.
- Clear responsibilities and authority, and better training of personnel.
- Continual improvement through internal audits, corrective action and management review, which reduces waste and cost.

15	What is the broad purview of ISO 9000 standards? Source: Unit I SEE Mar/Apr 2023 (18IM53), Q1.2	01	–	–
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ISO 9000 is a generic standard for quality management systems, broadly applicable to any type of organization (manufacturing or service, large or small). Its purview is the quality management system: fundamentals and vocabulary (ISO 9000), requirements (ISO 9001) and guidelines for performance improvement (ISO 9004), covering management responsibility, resource management, product or service realization, and measurement, analysis and improvement.

16	What is the primary objective of ISO 14000 standards? Source: Unit I CIE-I, 07/11/2025; SEE Feb/Mar 2025 (IM353IA), Q1.3	02	2	1
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ISO 14000 (ISO 14001) is a family of environmental management standards. Its primary objective is to help an organization establish an effective environmental management system (EMS) that identifies and controls the environmental impact of its activities, products and services, complies with environmental laws and regulations, prevents pollution and continually improves its environmental performance.

17	The concept of integrating digital technologies with QA practices is called _____. Source: Unit I SEE Feb/Mar 2026 (IM353IA), Q1.1	01	2	2
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Answer: Quality 4.0

Quality 4.0 is the application of Industry 4.0 digital technologies to quality management.

18	What is Quality 4.0? Mention two technologies used. Source: Unit I SEE Feb/Mar 2026 (IM353IA), Q1.2	02	2	1
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Quality 4.0 is the digitalization of quality management: the use of Industry 4.0 technologies to connect people, machines and data so that quality is monitored, analysed and improved in real time. **Two technologies:** Internet of Things (IoT) sensors for real-time process data, and artificial intelligence / machine learning for predictive quality analytics (others: big data analytics, cloud computing).

19	Mention four key core technologies that drive Quality 4.0. Source: Unit I CIE-I, 26/11/2024	02	2	3
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- **Internet of Things (IoT) and connected sensors** for real-time collection of process and product data.
- **Big data analytics and cloud computing** to store and analyse large quality data sets.
- **Artificial intelligence and machine learning** for prediction of defects and automated (vision) inspection.
- **Digital twins, automation and blockchain** for simulation, traceability and paperless quality records.

20	The cost associated with measuring, evaluating or auditing the product comes under the costs of _____. Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q1.2	01	–	–
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Answer: appraisal

Appraisal costs are the costs of measuring, evaluating or auditing products, components and purchased materials to ensure conformance to standards.

21	The costs related to formal technical reviews and test equipment reviews are related to _____ type of quality cost. Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.11	01	–	–
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Answer: appraisal

Reviews and the maintenance and calibration of test equipment are appraisal costs incurred to determine whether the product conforms to specifications.

22	List the 7 QC tools. Source: Unit I CIE-I, 26/11/2024; SEE Nov/Dec 2019 (16IM53), Q1.4	02	1	3
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The seven major SPC problem-solving tools (the magnificent seven) are:

- (1) Histogram or stem-and-leaf plot
- (2) Check sheet
- (3) Pareto chart
- (4) Cause-and-effect diagram
- (5) Defect concentration diagram
- (6) Scatter diagram
- (7) Control chart

23	State the central limit theorem. Source: Unit I SEE Jan/Feb 2021 (16IM53), Q1.3; SEE Fast Track Jul 2019 (16IM53), Q1.3	01	–	–
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If $x_1, x_2, \dots, x_n$ are independent random variables with means μ and variances σ^2 , then the distribution of their sum, and hence of the sample mean, approaches the normal distribution as n becomes large, regardless of the distributions of the individual variables. The sample mean is then approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$; this is the basis of the control limits on the $\bar{x}$ chart.

24	Differentiate between chance and assignable causes of variation. Source: Unit I SEE Feb/Mar 2025 (IM353IA), Q1.2; SEE Jan/Feb 2021 (16IM53), Q1.4; SEE Fast Track Oct 2020 (16IM53), Q1.1	02	2	1
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Basis	Chance (common) causes	Assignable (special) causes
Nature	Natural, inherent variability: the cumulative effect of many small, essentially unavoidable causes ("background noise").	Large variability that is not part of the chance-cause pattern.
Sources	Part of the process itself; a stable system of chance causes.	Improperly adjusted machines, operator errors, defective raw material.
Process state	Process is in statistical control.	Process is out of control.
Action	Reduced only by changing the process (management action).	Detected by control charts and removed by corrective action.

25	Differences among materials, workers and machines is _____ type of cause of variation. Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q1.4	01	–	–
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Answer: assignable

Differences among materials, workers and machines (defective raw material, operator errors, improperly adjusted machines) are the three typical sources of assignable causes.

26	Consider an operation on a CNC machine. Give an example of a possible assignable cause. Source: Unit I SEE Mar/Apr 2023 (18IM53), Q1.3	01	–	–
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Tool wear or a broken cutting tool, a wrong tool offset entered by the operator, loose work holding, or a batch of material with different hardness. Each produces a shift or trend in the machined dimension.

27	Statistical process control helps identify _____ type of causes that affect the process. Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.2	01	–	–
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Answer: assignable (special)

28	A control chart contains a centre line, _____ and _____. Source: Unit I SEE Feb/Mar 2026 (IM353IA), Q1.3	01	1	1
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Answer: upper control limit (UCL) and lower control limit (LCL)

29	In control charts, the dividing lines between random and nonrandom deviations from the mean of the distribution are known as _____. Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.8	01	–	–
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Answer: control limits

30	The mean of the sampling distribution is equal to _____ of the process distribution. Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.6	01	–	–
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Answer: the mean

$E(\bar{x}) = \mu$; the standard deviation of the sampling distribution is $\sigma/\sqrt{n}$.

31	The percent of the sample means that will have values within ± 3 standard deviations of the distribution mean is _____. (What proportion of the normal distribution is covered under the 3σ control limits?) Source: Unit I SEE Jan/Feb 2021 (18IM53), Q1.7; SEE Mar/Apr 2023 (18IM53), Q1.4	01	–	–
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Answer: 99.73%

With three-sigma limits the probability of a point falling outside the limits when the process is in control (type I error) is 0.0027.

32	Define the term warning limit. Source: Unit I SEE Mar/Apr 2023 (18IM53), Q1.5	01	–	–
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Warning limits are inner limits, usually placed at two-sigma from the center line, inside the three-sigma action (control) limits. If one or more points fall between the warning limit and the control limit, the process is suspected and the sampling frequency or sample size is increased.

33	What is ATS and ARL? Source: Unit I CIE-I, 26/11/2024	02	1	2
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ARL (average run length): the average number of points that must be plotted before a point indicates an out-of-control condition; $ARL = 1/p$, where p is the probability that a point exceeds the control limits. For three-sigma limits, in-control $ARL_0 = 1/0.0027 = 370$.

ATS (average time to signal): the average time before a signal occurs when samples are taken h hours apart; $ATS = ARL \times h$.

34	State any two principles of rational subgrouping. Source: Unit I SEE Fast Track Oct 2020 (16IM53), Q1.2	02	–	–
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- Subgroups should be selected so that if assignable causes are present, the chance of differences **between** subgroups is maximized while the chance of differences **within** a subgroup is minimized.
- Time order of production is a logical basis: each subgroup should consist of units produced at the same time (consecutive units), giving a snapshot of the process; units from different shifts, machines or operators should not be mixed in one subgroup.

35	Sketch the extreme variation patterns of assignable cause.	02	—	—
Source: Unit I SEE Fast Track Oct 2020 (16IM53), Q1.3				

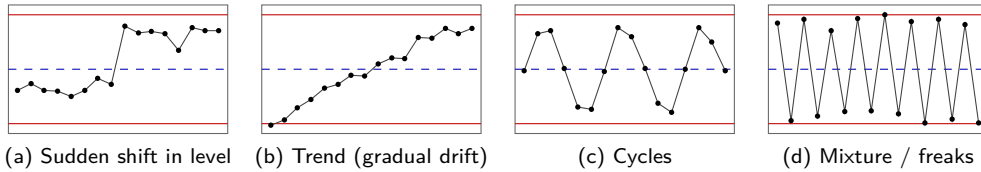


Figure 1: Typical nonrandom patterns caused by assignable causes

A sudden shift in level (new material, operator or machine setting), a trend (tool wear, operator fatigue), cycles (temperature, rotation of operators or machines) and mixtures or freaks (two populations, over-adjustment) all indicate assignable causes even if points lie within the limits.

36	_____ plans detail the action to be taken once an out-of-control situation is detected.	01	—	—
Source: Unit I SEE Jan/Feb 2021 (16IM53), Q1.6				

Answer: Out-of-control action (OCAP)

An OCAP is a flow chart or text description of the checkpoints (potential assignable causes) and terminators (actions) that follow an out-of-control signal.

37	Differentiate between attribute control chart and variable control chart.	02	L2	CO1
Source: Unit II CIE-II, 07/01/2025				

Basis	Variable control chart	Attribute control chart
Data	Measurements on a continuous scale (length, weight, diameter).	Counts or classification: conforming / nonconforming, number of defects.
Charts	$\bar{x}$ and R, $\bar{x}$ and s, individuals and moving range	p, np, c and u charts
Information	More information per unit; detects shifts with smaller samples.	Less information per unit; needs larger samples but data are cheap to collect.

38	When do you use attribute control charts?	02	—	—
Source: Unit II SEE Nov/Dec 2019 (16IM53), Q1.7				

Attribute control charts are used when the quality characteristic cannot easily be measured on a continuous scale, so each unit is judged conforming or nonconforming (p, np charts) or the number of nonconformities is counted (c, u charts); when many characteristics are to be checked together; and when go/no-go inspection data are already available.

39	The number of invoices that contain errors is an example of _____ data.	01	—	—
Source: Unit II SEE Jan/Feb 2021 (16IM53), Q1.9				

Answer: attribute (discrete count)

40	What does an R-chart measure, and why is it important in statistical process control? Source: Unit II CIE-II, 07/01/2025	02	L1	CO2
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The R chart plots the range of each subgroup and measures the **variability (dispersion)** of the process. It is important because the $\bar{x}$ chart limits are calculated from $\bar{R}$; if the variability is not in control the $\bar{x}$ limits are meaningless. The R chart signals any gain or loss in process uniformity.

41	The usual purpose of an R-chart is to signal whether there has been a gain or loss in _____. Source: Unit II SEE Jan/Feb 2021 (16IM53), Q1.5; SEE Fast Track Jul 2019 (16IM53), Q1.5	01	–	–
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Answer: uniformity (process dispersion or variability)

42	Control limits for the R chart are _____ and _____. Source: Unit II SEE Fast Track Oct 2020 (16IM53), Q1.4	02	–	–
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Answer: $UCL = D_4 \bar{R}$ and $LCL = D_3 \bar{R}$ (center line $\bar{R}$)

43	The factors A_1, A_2, D_1, D_2, D_3 and D_4 used to calculate control limits depend upon _____. Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q1.7	01	–	–
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Answer: the sample (subgroup) size n

44	Central tendency of a process is monitored in _____ control chart. Source: Unit II SEE Jan/Feb 2021 (18IM53), Q1.9	01	–	–
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Answer: $\bar{x}$ (X-bar)

45	Jars of pickles are sampled and weighed. Sample measures are plotted on control charts. The ideal weight should be precisely 11 oz. Which type of chart(s) would you recommend? Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q1.6	01	–	–
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Weight is a variable measured on a continuous scale with a target of 11 oz, so $\bar{x}$ and **R charts** are recommended: the $\bar{x}$ chart monitors the mean fill weight and the R chart monitors the variability.

46	Write the steps followed in constructing the individual chart. Source: Unit II CIE-I, 07/11/2025	02	2	2
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- (1) Collect individual measurements $x_1, x_2, \dots$ ($n = 1$) in time order.
- (2) Compute the moving ranges $MR_i = |x_i - x_{i-1}|$ and their average $\overline{MR}$; compute $\bar{x}$.
- (3) Control limits: $\bar{x} \pm 3 \overline{MR}/d_2 = \bar{x} \pm 2.66 \overline{MR}$ ($d_2 = 1.128$); MR chart: $UCL = 3.267 \overline{MR}$, $LCL = 0$.
- (4) Plot the individual values and moving ranges and interpret for points outside limits and patterns.

47	Mention any one of the uses of the chart for individual measurements. Source: Unit II SEE Mar/Apr 2023 (18IM53), Q1.9	01	–	–
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It is used when the sample size is one, for example when every unit is measured by automated inspection, when the production rate is very slow, or for chemical batches and accounting or financial data where repeated measurements differ only by measurement error.

48	When do you prefer to use a group control chart for monitoring the process quality? Source: Unit II SEE Mar/Apr 2023 (18IM53), Q1.10	01	–	–
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A group control chart is used when a process has several parallel streams (multiple spindles, heads, cavities or machines feeding one output). Only the largest and smallest stream means are plotted each time, so all streams are monitored on one chart.

49	What is the key difference between a p-chart and an np-chart? Source: Unit II CIE-II, 07/01/2025	02	L2	CO3
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The **p chart** plots the fraction nonconforming $\hat{p} = D/n$ and can be used with variable sample sizes. The **np chart** plots the number nonconforming $D = n\hat{p}$ and is used when the sample size is constant; it is easier for operators to interpret.

50	Explain the purpose of a p-chart. Source: Unit II CIE-II, 16/12/2025	02	3	1
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The p chart monitors the fraction nonconforming (proportion defective) produced by a process. Each sample's fraction defective is plotted against limits $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$; points outside the limits signal an assignable cause, so the chart is used to bring the process into control and to judge improvement in quality.

51	What can you conclude when a point falls below the lower control limit on a control chart for fraction nonconforming (p-chart)? Source: Unit II SEE Mar/Apr 2023 (18IM53), Q1.6	01	–	–
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A point below the LCL does not necessarily mean improvement. It may indicate a real improvement in quality, but often it is due to an error in inspection (poorly trained inspector, wrong gauge) or in recording the data. The cause should be investigated; if it is a genuine improvement it should be made permanent.

52	The local newspaper receives several complaints per day about typographic errors. Over a seven-day period, the publisher has received calls from readers reporting the following number of errors: 4, 3, 2, 6, 7, 3 and 9. Based on these data alone, what type of control chart(s) should the publisher use? Source: Unit II SEE Jan/Feb 2021 (16IM53), Q1.8; SEE Fast Track Jul 2019 (16IM53), Q1.8	02	–	–
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Answer: a c chart (number of errors, constant inspection unit: one day's paper)

$$\bar{c} = 34/7 = 4.857; UCL = 4.857 + 3\sqrt{4.857} = 11.47, LCL = 0.$$

53	The control chart used for the number of defects per unit is _____. Source: Unit II SEE Jan/Feb 2021 (18IM53), Q1.10	01	–	–
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Answer: u chart

54	In a lot of 50 parts, 4 parts were rejected because of 12 non-conformities found in them. The proportion defective is _____ and the average number of non-conformities per part in the lot is _____. Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q1.9	02	–	–
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Answer: proportion defective = $4/50 = 0.08$; nonconformities per part = $12/50 = 0.24$

55	Define C_p . How is it interpreted? Source: Unit II SEE Feb/Mar 2026 (IM353IA), Q1.5	02	2	2
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The process capability ratio $C_p = (USL - LSL)/6\sigma$ compares the specification width with the natural tolerance (6σ) of the process. $C_p > 1$ means the process uses less than 100% of the tolerance band, so few nonconforming units are produced (if centered); $C_p = 1$ means 0.27% nonconforming; $C_p < 1$ means the process is not capable. Recommended minimum values are 1.33 for existing and 1.50 for new processes.

56	Define C_{pk} . How is it different from C_p ? Source: Unit II SEE Feb/Mar 2026 (IM353IA), Q1.4	02	2	2
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$C_{pk} = \min(C_{pu}, C_{pl}) = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$. C_p measures **potential** capability and ignores the location of the mean; C_{pk} measures **actual** capability and takes process centering into account. $C_{pk} = C_p$ only when the process is centered at the middle of the specifications; otherwise $C_{pk} < C_p$.

57	Write down the formula for the process performance index C_{pk} taking into account process centering. Source: Unit II SEE Mar/Apr 2023 (18IM53), Q1.8; SEE Fast Track Oct 2020 (16IM53), Q1.8	02	–	–
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$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right)$$

58	In Six Sigma, the value of C_p = _____. Source: Unit II SEE Jan/Feb 2021 (18IM53), Q1.3	01	–	–
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Answer: 2.0

A six-sigma process has the specification limits at $\pm 6\sigma$ from the target, so $C_p = 12\sigma/6\sigma = 2$.

59	What is AOQ? Source: Unit III CIE-II, 16/12/2025; SEE Feb/Mar 2026 (IM353IA), Q1.7; SEE Nov/Dec 2019 (16IM53), Q1.6	02	2	2
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Average outgoing quality (AOQ) is the quality in the lot that results from the application of rectifying inspection (rejected lots are 100% inspected and the defectives are replaced). It is the average value of the lot quality over a long sequence of lots from a process with fraction defective p :

$$AOQ = \frac{P_a p (N - n)}{N} \approx P_a p \quad (N \gg n)$$

60	What is AOQL? Source: Unit III CIE-II, 16/12/2025; SEE Feb/Mar 2026 (IM353IA), Q1.6	02	3	2
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The curve of the AOQ against the incoming lot quality p is the AOQ curve. Its maximum ordinate is the **average outgoing quality limit (AOQL)**, the worst possible average quality that results from the rectifying inspection program. No matter how bad the incoming lots are, the outgoing quality will not on average be worse than the AOQL; it is an average over a large stream of lots, not a guarantee for an isolated lot.

61	Expand the acronyms AQL, AOQL and LTPD. Source: Unit III CIE-II, 07/01/2025 (AQL, LTPD); SEE Feb/Mar 2025 (IM353IA), Q1.6 (AQL, AOQL)	02	2	3
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- **AQL:** acceptable quality level, the poorest level of quality for the supplier's process that the consumer would consider acceptable as a process average.
- **AOQL:** average outgoing quality limit, the maximum of the AOQ curve of a rectifying inspection plan.
- **LTPD:** lot tolerance percent defective, the poorest level of quality that the consumer is willing to accept in an individual lot (also called the rejectable quality level, RQL, or limiting quality level, LQL).

62	State the difference between single and double sampling plans. Source: Unit III CIE-II, 16/12/2025	02	2	2
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Basis	Single sampling plan	Double sampling plan
Samples	One sample of n units; the lot is accepted if the number of defectives $d \leq c$, otherwise rejected.	A first sample n_1 may lead to acceptance, rejection or a second sample n_2 ; the combined result then decides.
Parameters	n and c	n_1, c_1, n_2, c_2
Inspection	Constant amount of inspection per lot	Smaller average inspection (many lots are decided on the first sample), but more complex to administer

63	What are multiple sampling plans? Source: Unit III CIE-II, 16/12/2025	02	2	2
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A **multiple-sampling plan** is an extension of the double-sampling concept in which more than two samples may be required to reach a decision on the lot. At each stage the lot is accepted if the cumulative number of defectives is less than or equal to the acceptance number, rejected if it equals or exceeds the rejection number, and otherwise the next sample is taken, up to a last stage at which a decision must be made. The samples are usually smaller than in single or double sampling, but the plan is more complex to administer. Its limit, taking one unit at a time, is sequential sampling.

64	Write two applications of acceptance sampling. Source: Unit III CIE-II, 07/01/2025	02	2	2
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- **Receiving (incoming) inspection** of components and raw materials from a supplier, where the lot is accepted or rejected on the result of a sample.
- **In-process and final inspection:** a manufacturer samples its own product at various stages of production, or at final inspection of finished goods; accepted lots are sent forward, rejected lots are reworked or scrapped (rectifying inspection).

65	Write any four advantages of the sampling technique. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q1.5	02	–	–
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Compared with 100% inspection, acceptance sampling (any four):

- It is usually less expensive, because there is less inspection.
- There is less handling of the product, hence reduced damage.
- Fewer personnel are involved in the inspection activities.
- It often greatly reduces the amount of inspection error.
- The rejection of entire lots, instead of the simple return of defectives, gives the supplier a stronger motivation for quality improvement.

66	Write the OC curve for a single sampling plan. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q1.6	02	–	–
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The **operating-characteristic (OC) curve** of a single-sampling plan (n, c) plots the probability of accepting the lot, P_a , against the lot fraction defective p . It shows the discriminatory power of the plan. For lots from a stream with fraction defective p , the number of defectives d in the sample is binomial and

$$P_a = P(d \leq c) = \sum_{d=0}^c \binom{n}{d} p^d (1-p)^{n-d}$$

The curve falls from $P_a = 1$ at $p = 0$ towards 0 as p increases; the steeper the curve, the better the plan discriminates between good and bad lots (an ideal OC curve would be a vertical step at the borderline quality).

67	Differentiate between consumer's risk and producer's risk. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q1.7	02	–	–
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Basis	Producer's risk (α)	Consumer's risk (β)
Meaning	The risk (probability) that a lot of good quality (at the AQL, p_1) is rejected.	The risk (probability) that a lot of poor quality (at the LTPD, p_2) is accepted.
Point on the OC curve	$P_a = 1 - \alpha$ at $p = \text{AQL}$	$P_a = \beta$ at $p = \text{LTPD}$
Suffered by	The supplier (producer)	The buyer (consumer)

68	Define the term producer's risk. Source: Unit III SEE Mar/Apr 2023 (18IM53), Q1.11	01	—	—
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The **producer's risk** (α) is the probability that a lot of acceptable quality (fraction defective equal to the AQL, p_1) is rejected by the sampling plan; the OC curve passes through the point ($p_1, 1 - \alpha$).

69	When is a type-A OC curve used? Source: Unit III SEE Nov/Dec 2019 (16IM53), Q1.5	02	—	—
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The **type-A OC curve** is used to calculate the probabilities of acceptance for an **isolated lot of finite size** N . The exact distribution of the number of defectives in the sample is hypergeometric. (The type-B curve, based on the binomial distribution, is used for lots drawn from a continuing stream or a large lot.) The type-A curve always lies below the type-B curve; the difference matters only when the lot is small relative to the sample ($n/N > 0.10$).

70	When do you use a Cusum chart? Source: Unit III SEE Nov/Dec 2019 (16IM53), Q1.8	02	—	—
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A cusum (cumulative sum) chart is used to monitor the process mean when **small shifts** (about 1.5σ or less) are important: it uses all the information in the sequence of points by accumulating the deviations from the target, $C_i = \sum(x_j - \mu_0)$, so it detects small shifts much faster than a Shewhart chart. It is especially effective for samples of size $n = 1$ and is excellent for phase II monitoring in the chemical and process industries.

71	Give an example of any one of the published sampling plans. Source: Unit III SEE Mar/Apr 2023 (18IM53), Q1.12	01	—	—
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MIL STD 105E (civilian counterparts ANSI/ASQC Z1.4 and ISO 2859), an AQL-based system of single, double and multiple sampling plans with normal, tightened and reduced inspection. Another published system is the **Dodge–Romig** tables, which are based on the LTPD or the AOQL.

72	In a double sampling plan, if the number of defects is in between the two cut-off numbers C_1 and C_2 , then the next step is to _____. Source: Unit III SEE Jan/Feb 2021 (18IM53), Q1.12	01	—	—
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Answer: take a second sample (of size n_2)

If $c_1 < d_1 \leq c_2$, no decision is possible on the first sample, so a second sample is drawn and the combined number of defectives $d_1 + d_2$ is compared with c_2 .

73	In _____ sampling plan, the samples are accumulated at each stage till the decision of accepting or rejecting a lot is taken. Source: Unit III SEE Jan/Feb 2021 (18IM53), Q1.13	01	—	—
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Answer: multiple (or sequential)

In multiple sampling (and, in the limit of one unit at a time, in sequential sampling) the number of samples depends on the results; the cumulative defectives are compared with the acceptance and rejection numbers after each stage.

74	In addition to the OC curve, _____ and _____ curves are used to select an appropriate sampling plan. Source: Unit III SEE Jan/Feb 2021 (18IM53), Q1.14	02	–	–
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Answer: average outgoing quality (AOQ) and average total inspection (ATI); the average sample number (ASN) curve is also used

The AOQ curve shows the outgoing quality of a rectifying plan, and the ATI curve the average inspection per lot; the ASN curve shows the average sample size of double and multiple plans.

75	The average total inspection under a single sampling plan in which rejected lots are 100% inspected is given by the formula $ATI = \text{_____}$. Source: Unit III SEE Jan/Feb 2021 (16IM53), Q1.10	01	–	–
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Answer: $ATI = n + (1 - P_a)(N - n)$

If a lot is accepted only the n sample units are inspected; if it is rejected (probability $1 - P_a$) the remaining $N - n$ units are also inspected (100% inspection).

76	State the three parameters of the single sampling plan. Source: Unit III SEE Jan/Feb 2021 (16IM53), Q1.11; SEE Fast Track Jul 2019 (16IM53), Q1.10	02	–	–
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A single-sampling plan is defined by the **lot size N** , the **sample size n** and the **acceptance number c** (the lot is accepted if the sample contains c or fewer defectives and rejected otherwise).

77	MIL STD 105 plans are based on _____. Source: Unit III SEE Fast Track Jul 2019 (16IM53), Q1.11	01	–	–
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Answer: the acceptable quality level (AQL)

MIL STD 105E has an AQL focus: it is indexed with a series of AQLs and, with the lot size and the inspection level, gives the sample size code letter and the normal, tightened and reduced plans.

78	List the advantages of the CRD (completely randomized design). Source: Unit IV Improvement CIE, 28/01/2025, Q4	02	2	3
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In a completely randomized design (CRD) the experimental runs are made in random order and the environment (experimental units) is as uniform as possible. Its advantages are:

- **Randomization balances out nuisance variables:** the effect of any nuisance variable that might influence the response (for example a warm-up effect of the test machine) is approximately balanced out among the treatments.
- **Simple analysis:** the single-factor model $y(ij) = \mu + \tau(i) + \varepsilon(ij)$ is analyzed by a one-way analysis of variance; no extra grouping of the units is required.
- **Flexible:** any number of treatments and replicates can be used, and the numbers of observations per treatment need not be equal.
- **Maximum degrees of freedom for error,** since no degrees of freedom are used for blocks.

79	State any two principles of experimentation. Source: Unit IV Improvement CIE, 13/01/2026, Q1	02	3	1
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Two principles of experimentation (any two):

- **Randomization:** the runs are made in random order so that the effect of nuisance variables is balanced out.
- **Replication:** the repeated runs of each treatment provide an estimate of the experimental error.
- **Blocking:** using homogeneous blocks to remove the effect of a known nuisance factor.

80	What is a factorial experiment? Source: Unit IV Improvement CIE, 13/01/2026, Q2; SEE Feb/Mar 2026 (IM353IA), Q1.9	02	2	1
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A **factorial experiment** is one in which, in each complete trial or replicate, **all possible combinations of the levels of the factors are investigated**; the factors are varied together. With factors A and B having a and b levels, each replicate contains all ab combinations. It allows the main effects of the factors and their interactions to be estimated.

81	What is replication and why is it important? Source: Unit IV SEE Feb/Mar 2026 (IM353IA), Q1.8	02	2	2
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Replication means repeating the experimental run of a treatment (factor-level combination) more than once. The n replicates of each treatment provide the degrees of freedom with which to estimate the **experimental error** (pure error), without which the significance of the effects cannot be tested by ANOVA; replication also makes the estimates of the effects more precise (the average of n observations has variance σ^2/n).

82	What is a completely randomized design (CRD) in experimentation? Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q1.7	02	2	1
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A **completely randomized design** is a design in which the observations are taken in random order and the environment (the experimental units) in which the treatments are used is as uniform as possible. The levels of the factor are called treatments, each with a number of replicates (for example the paper tensile-strength experiment: 4 hardwood concentrations $\times$ 6 specimens tested in random order). It is analyzed by the fixed effects analysis of variance.

83	What is the primary advantage of using factorial experiments? Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q1.8	02	2	1
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The primary advantage of the factorial experiment is that the factors are varied together, so it detects **interactions** between the factors, which the one-factor-at-a-time method cannot do, and it uses the data more efficiently (every observation gives information on every factor effect), so fewer runs are needed. It also gives conclusions that hold over a range of the other factors.

84	Graphically indicate the presence of an interaction effect between two factors. Source: Unit IV SEE Mar/Apr 2023 (18IM53), Q1.13	01	—	—
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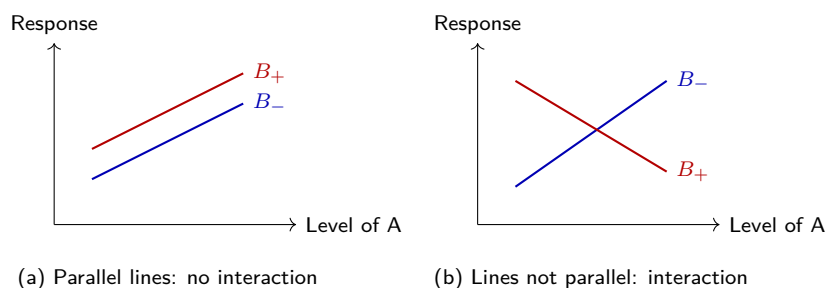


Figure 2: Interaction graph: parallel lines mean no interaction; lines that are not parallel (or that cross) show an interaction between A and B

85	What is the meaning of the term randomization? Source: Unit IV SEE Mar/Apr 2023 (18IM53), Q1.14	01	–	–
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Randomization means that the order in which the experimental runs are made, and the allocation of the treatments to the experimental units, are decided by chance (for example with random numbers). It balances out the effect of nuisance variables that may influence the response (such as a warm-up effect of the testing machine) and gives the unbiased errors that the statistical analysis assumes.

86	In experimental designs, the process in which every experimental unit has the same chance of being allocated to the treatments is known as _____. Source: Unit IV SEE Fast Track Jul 2019 (16IM53), Q1.13	01	–	–
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Answer: randomization

Random allocation of the treatments to the experimental units (and random run order) gives every unit the same chance of receiving each treatment and balances out the effect of nuisance variables.

87	How many experimental runs are to be carried out in the case of a 2^5 factorial design? Source: Unit IV SEE Mar/Apr 2023 (18IM53), Q1.15	01	–	–
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Answer: $2^5 = 32$ runs (per replicate)

A 2^k factorial design has k factors at two levels each, so one complete replicate has 2^k runs; for $k = 5$ this is $2 \times 2 \times 2 \times 2 \times 2 = 32$ runs.

88	If there are m levels for a factor A and n levels for a factor B in a factorial design of experiments, then there shall be _____ combinations. Source: Unit IV SEE Fast Track Jul 2019 (16IM53), Q1.14	01	–	–
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Answer: $m \times n$

A full factorial experiment contains all combinations of the levels of the factors, i.e. m levels of A times n levels of B .

89	What is the meaning of a full factorial design? Source: Unit IV SEE Mar/Apr 2023 (18IM53), Q1.16	01	–	–
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A **full factorial design** is a design in which every possible combination of the levels of all the factors is run (in each replicate); for k factors with two levels this is the 2^k design. A fractional factorial design, in contrast, runs only a fraction of these combinations.

90	_____ is a method to find factor effects and interaction effects. Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q1.4	01	–	–
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Answer: Factorial design (analyzed by the analysis of variance, ANOVA)

A factorial experiment varies the factors together, so the main effects and the interaction effects can be estimated, and the ANOVA tests their significance.

91	The effect of one factor depends on where some other factor is set. This is called _____.	01	–	–
Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q1.5				

Answer: interaction

When the difference in response between the levels of one factor is not the same at all levels of the other factor, there is an interaction between the factors; it is revealed by non-parallel lines on an interaction graph.

92	To find out whether noise affects the capability to solve the assignment of Quality Management, one group solves the assignment in a silent room and another group in a noisy room. The group in the silent room completes 22 problems in one hour and the group in the noisy room completes 15 problems in one hour. In this experiment, the independent variable is _____ and the dependent variable is _____.	02	–	–
Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q1.15				

Answer: noise level (silent room versus noisy room); number of problems solved in one hour

The independent variable is the factor that the experimenter changes (the noise level of the room), and the dependent variable is the response that is measured (number of problems completed in an hour).

93	What statistical procedure is used to assess the statistical significance of the main effects and the interaction(s) in a factorial design? What is the test statistic used for this?	02	–	–
Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q1.16				

The **analysis of variance (ANOVA)** is used. The total variability is partitioned into the sums of squares for the main effects, the interaction and the error, and each effect is tested with the **F statistic**, $F_0 = MS(\text{effect})/MS(\text{error})$, which is compared with the tabulated F value (for the stated significance level and degrees of freedom); the effect is significant when F_0 exceeds it.

94	The uncontrollable input factors that affect the output of a process are termed _____.	01	–	–
Source: Unit IV SEE Fast Track Jul 2019 (16IM53), Q1.12; SEE Jan/Feb 2021 (16IM53), Q1.12				

Answer: noise factors

The inputs of a process are the controllable factors $x_1, \dots, x_p$ and the uncontrollable factors $z_1, \dots, z_q$ (sometimes called noise factors), although the latter may be controlled for the purposes of a test.

95	The experiment in which two factors are varied at two levels is called a _____ experiment.	01	–	–
Source: Unit IV SEE Jan/Feb 2021 (16IM53), Q1.13				

Answer: 2^2 factorial (two-factor, two-level)

It has $2 \times 2 = 4$ treatment combinations, denoted (1), a, b and ab.

96	Define level and factor. Source: Unit IV SEE Fast Track Oct 2020 (16IM53), Q1.9	02	–	–
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Factor: a controllable input variable of the process that is varied in the experiment (for example temperature or reaction time).

Level: one of the values or settings at which a factor is run; with two levels they are called “low” (–) and “high” (+).

97	List two applications of statistical experimental design in engineering design. Source: Unit IV SEE Nov/Dec 2019 (16IM53), Q1.9	02	–	–
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- Evaluation and comparison of basic design configurations.
- Determination of key product design parameters that impact the performance (for example, which tolerances of a circuit most affect the measurement capability, or which factors influence the check effort of a door hinge).

98	Expand ANOVA and ANCOVA. Source: Unit IV Improvement CIE, 28/01/2025, Q5	02	2	2
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Answer: ANOVA: analysis of variance; ANCOVA: analysis of covariance

ANOVA partitions the total variability in the data into components (between treatments and error) so that the equality of the means of more than two populations can be tested with the F ratio. (ANCOVA combines the analysis of variance with regression on a covariate; the prescribed textbook develops only the analysis of variance.)

99	Three components, each with a reliability of 0.9, are placed in series. What is the reliability of the system? Source: Unit V Improvement CIE, 28/01/2025, Q1	02	2	1
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For independent components in series the system works only if all the components work, so the system reliability is the product of the component reliabilities:

$$R_s = R_1 R_2 R_3 = (0.9)(0.9)(0.9) = 0.729$$

Answer: $R_s = 0.729$

100	There are two components that are identical in a parallel configuration where the second unit is in standby and will be activated only upon failure of the primary unit. The switching to the backup unit is considered certain. The failure rate for each component is 0.002 failures/hour. What is the reliability for 200 hours? Source: Unit V Improvement CIE, 28/01/2025, Q2	02	1	2
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With a standby unit that is switched on at the moment the first unit fails (perfect switching), the system life is the sum of two independent exponential lives, each with failure rate λ , which is a gamma (Erlang) distribution with $r = 2$. Its cumulative distribution is the sum of Poisson terms, so the reliability is

$$R(t) = e^{-\lambda t} (1 + \lambda t)$$

$$\lambda t = 0.002(200) = 0.4$$

$$R(200) = e^{-0.4}(1 + 0.4) = 0.6703 \times 1.4 = 0.9384$$

Answer: $R(200 \text{ h}) = 0.9384$

101	What is a parallel system in reliability? Source: Unit V Improvement CIE, 13/01/2026, Q5	02	2	2
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A **parallel system** is one in which the system works if **at least one** of its components works; the system fails only when all the components fail. The components provide redundancy, so the system reliability is higher than that of any single component. For independent components the unreliability of the system is the product of the component unreliabilities:

$$R_s = 1 - (1 - R_1)(1 - R_2) \cdots (1 - R_n)$$

102	What is failure rate? Source: Unit V SEE Feb/Mar 2026 (IM353IA), Q1.10	02	2	2
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If the time to failure of a component is exponentially distributed, $f(t) = \lambda e^{-\lambda t}$, the parameter λ is called the **failure rate** of the system or component (failures per unit time, for example failures per hour). It is constant, and the mean time to failure is $1/\lambda$. For example, a failure rate of 10^{-4} per hour means a mean time to failure of 10,000 hours.

103	What does MTTF stand for and why is it important? Source: Unit V SEE Feb/Mar 2025 (IM353IA), Q1.10	02	1	1
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MTTF stands for **mean time to failure**, the mean of the distribution of the time to failure of a component or system. For the exponential model it equals $1/\lambda$, where λ is the failure rate. It is important because it summarizes the expected life of an item in one number, is used to compare designs and to estimate the reliability $R(t) = e^{-t/MTTF}$, and to plan replacement and spare parts. (For the exponential distribution the probability that a unit fails before its mean life is 0.632, whatever the value of λ .)

104	A system consists of two components arranged in parallel. The reliabilities of the components are $R_1 = 0.80$ and $R_2 = 0.85$. What is the overall reliability of the system? Source: Unit V SEE Feb/Mar 2025 (IM353IA), Q1.9	02	3	1
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$$R_s = 1 - (1 - R_1)(1 - R_2) = 1 - (0.20)(0.15) = 0.97$$

Answer: $R_s = 0.97$

105	Consider a system consisting of three components in parallel, each with a reliability of 0.65. Find the reliability of the system. Source: Unit V SEE Nov/Dec 2019 (16IM53), Q1.10	02	—	—
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$$R_s = 1 - (1 - 0.65)^3 = 1 - (0.35)^3 = 1 - 0.0429 = 0.9571$$

Answer: $R_s = 0.9571$

106	Define reliability. Source: Unit V SEE Nov/Dec 2019 (16IM53), Q1.2	02	—	—
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Reliability $R(t)$ of a component or system is the probability that it does not fail (survives) beyond time t under the stated operating conditions, $R(t) = P(T > t) = 1 - F(t)$, where T is the time to failure and $F(t)$ its cumulative distribution function. For the exponential model, $R(t) = e^{-\lambda t}$.

107	Sketch the time versus failure rate curve for the failure of components according to the exponential distribution. Source: Unit V SEE Mar/Apr 2023 (18IM53), Q1.17	01	–	–
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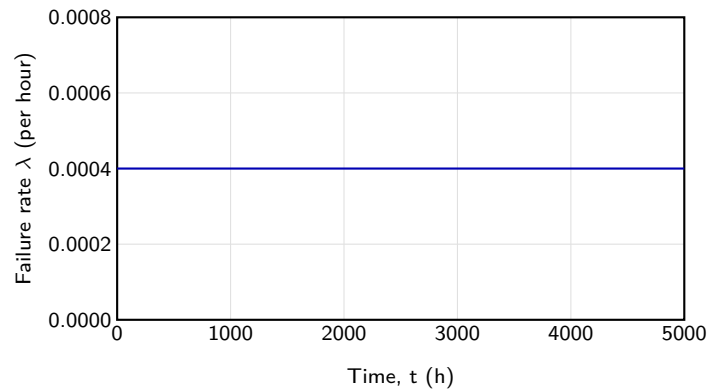


Figure 3: For the exponential distribution the failure rate is constant, independent of time (here $\lambda = 0.0004$ per hour)

The time versus failure rate curve of an exponentially distributed component is a **horizontal straight line at the value λ** : the probability of failing in the next short interval does not depend on the age of the component. The reliability, in contrast, falls exponentially, $R(t) = e^{-\lambda t}$.

108	Give a practical example of a series device configuration. Source: Unit V SEE Mar/Apr 2023 (18IM53), Q1.18	01	–	–
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In a series configuration the failure of any one component fails the whole system. **Example:** a string of old-style decorative lights wired in series (one failed bulb puts out the entire string); also a car's starting system (battery, starter motor, ignition switch and starter relay) or a conveyor line of several machines each of which must work, such as the blanking press, welder and packer in a production line.

109	If there are n components in a parallel configuration and all the components are identical, what is the formula for estimating the reliability of the system, given that the reliability of each component is R ? Source: Unit V SEE Mar/Apr 2023 (18IM53), Q1.19	01	–	–
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$$R_s = 1 - (1 - R)^n$$

The system fails only if all n identical components fail, which has probability $(1 - R)^n$.

110	If θ is the MTBF, then the failure rate λ is equal to _____ and the reliability of the unit over the time t is equal to _____. Source: Unit V SEE Fast Track Jul 2019 (16IM53), Q1.15	02	–	–
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Answer: $\lambda = 1/\theta$; $R(t) = e^{-t/\theta}$

For an exponentially distributed time to failure the mean is $\theta = 1/\lambda$, so the failure rate is $\lambda = 1/\theta$ and the reliability over a period t is $R(t) = e^{-\lambda t} = e^{-t/\theta}$.

Sl. No.	Questions	M	BT	CO
PART – B				
1	Identify the various dimensions of quality as proposed by Garvin. Explain their significance with respect to product quality, taking suitable examples. (List and explain the eight dimensions of quality.) Source: Unit I SEE Jan/Feb 2021 (16IM53), Q2a; SEE Nov/Dec 2019 (16IM53), Q2a; SEE Fast Track Oct 2020 (16IM53), Q2a (discuss any two); SEE Feb/Mar 2026 (IM353IA), Q2a (any four)	08	2	3

Garvin (1987) described eight components or dimensions of quality. Quality is a multifaceted entity, and it is important to differentiate these dimensions when a product or service is described and evaluated.

Dimension	Question it answers	Meaning and example
1. Performance	Will the product do the intended job?	How well the product performs its specific functions; e.g. execution speed of spreadsheet software, mileage and acceleration of a car.
2. Reliability	How often does the product fail?	Complex products (appliances, automobiles, airplanes) need some repair; a car that needs frequent repair is unreliable.
3. Durability	How long does the product last?	Effective service life; very important for automobiles and major appliances.
4. Serviceability	How easy is it to repair the product?	How quickly and economically repair or maintenance is done; e.g. how long a credit card company takes to correct a billing error.
5. Aesthetics	What does the product look like?	Visual and sensory appeal: style, colour, shape, packaging; soft-drink makers use packaging to differentiate their product.
6. Features	What does the product do?	Added features beyond basic performance; e.g. spreadsheet software with built-in statistical analysis.
7. Perceived quality	What is the reputation of the company or its product?	Customers rely on past reputation; e.g. preferring an airline that is always on time and never loses luggage.
8. Conformance to standards	Is the product made exactly as the designer intended?	Meeting the requirements placed on it; e.g. a car hood fitting flush with the fender with equal gaps on all sides.

Significance

- The dimensions translate the customer's idea of quality into specific characteristics that can be designed, measured and improved.
- Quality planning uses them to develop products and services that meet or exceed customer expectations (voice of the customer).
- A firm can compete on a few chosen dimensions (for example, reliability for a car maker, aesthetics for a beverage maker) instead of all of them.
- Failures that are visible to the public lower perceived quality, so conformance and reliability directly affect reputation and repeat business.

2	Choose a specific product (e.g., a smartphone, automobile, or any consumer electronics) and analyze how the Eight Dimensions of Quality influence its acceptance by customers. Break down each dimension—such as performance, features, reliability, and serviceability—and evaluate their relative importance in shaping customer perceptions. How do these dimensions interact with each other to create a competitive advantage for the product in the market? Use relevant examples to support your analysis. Source: Unit I CIE-I, 26/11/2024; SEE Mar/Apr 2023 (18IM53), Q2a (take a simple product and identify the 8 dimensions)	10	4	4
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Product chosen: a smartphone. Each of Garvin's eight dimensions influences whether customers accept the product.

Dimension	How it shapes customer acceptance (smartphone)	Relative importance
Performance	Processor speed, camera quality, battery life, network reception and display quality decide whether the phone does its job well.	Very high: main reason for purchase
Features	Fast charging, multiple cameras, face unlock, 5G, water resistance, AI assistants differentiate it from competitors.	High: drives upgrades
Reliability	Freedom from hangs, battery failures, overheating and app crashes during use.	High: failures cause returns
Durability	Number of years the phone remains usable: glass strength, battery ageing, years of software updates.	Medium to high
Serviceability	Ease, speed and cost of repair: service-centre network, spare-part availability, warranty handling.	Medium
Aesthetics	Design, colour, thinness, materials (glass, metal) and packaging; a strong reason for premium buyers.	Medium to high
Perceived quality	Brand reputation, reviews and past recalls (for example, battery-fire recalls damage the brand).	High: affects loyalty
Conformance	Every unit built exactly to design: no gaps in the body, display without dead pixels, specified battery capacity.	Basic requirement

Interaction of the dimensions and competitive advantage

- Performance and features attract the first purchase, but reliability, durability and serviceability decide satisfaction over the phone's life and hence repeat purchase.
- Conformance supports reliability: tighter manufacturing variability gives fewer early failures, which builds perceived quality.
- Aesthetics and perceived quality allow a price premium; a strong brand can win even with similar performance.
- Trade-offs exist: more features and a thinner body (aesthetics) can reduce battery life (performance) and durability, so a firm chooses the combination its target segment values most.
- A product that is strong on the dimensions its customers rank highest, and adequate on the rest, gains a competitive advantage.

3	Analyze how the Eight Dimensions of Quality can be used to evaluate the performance of a consumer product. Examine each dimension's effect on customer satisfaction and illustrate your analysis with a detailed case study from the consumer goods or electronics industry. Source: Unit I CIE-I, 07/11/2025	10	4	3
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The eight dimensions provide a checklist for evaluating a consumer product, because each dimension corresponds to a question the customer asks about the product.

Case study: a front-loading washing machine

Dimension	Evaluation of the washing machine	Effect on customer satisfaction
Performance	Cleaning effectiveness, spin speed, water and energy consumption, noise.	Core satisfaction; poor cleaning means rejection.
Features	Inverter motor, steam wash, Wi-Fi app control, quick-wash programmes.	Delight factor; justifies a higher price.
Reliability	Frequency of breakdowns of the drum bearing, pump and control board.	Each failure causes strong dissatisfaction.
Durability	Expected service life (10 years or more), motor warranty.	Long life reinforces value for money.
Serviceability	Availability of service engineers, time to repair, cost of parts.	Fast service can recover a dissatisfied customer.
Aesthetics	Colour, door design, display panel, fit in the home.	Influences choice among similar models.
Perceived quality	Brand reputation, star energy rating, reviews.	Builds trust and loyalty.
Conformance	Each unit meets design specifications: no leaks, correct drum balance.	Expected by all customers; any lapse is noticed.

Analysis: customers rank performance, reliability and serviceability highest for appliances, so a manufacturer that reduces variability in critical components (conformance), offers long warranties (durability) and has a strong service network earns high perceived quality and repeat business. Features and aesthetics differentiate models within the same price segment.

4	Define 'quality'. Giving two examples each, explain the five quality characteristics. Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q2a	06	–	–
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Quality means fitness for use; in modern terms, quality is inversely proportional to variability. Every product possesses a number of quality characteristics that jointly describe what the user thinks of as quality. Juran classifies them into five types:

Characteristic	Meaning	Two examples
1. Technological (physical)	Measurable physical or chemical properties.	Length or diameter of a shaft; hardness of steel
2. Psychological (sensory)	Characteristics judged by the senses and taste.	Taste of food; appearance and colour of a car
3. Time-oriented	Behaviour of the product over time.	Reliability of a television; durability of a tyre
4. Contractual	Provisions guaranteed to the customer.	Warranty period of a refrigerator; free-service clauses
5. Ethical	Conduct of the people who deliver the product or service.	Courtesy of sales staff; honesty of a repair shop

5	What do you mean by quality cost? Why is it so important? Differentiate between failure cost and appraisal cost. Source: Unit I SEE Jan/Feb 2021 (16IM53), Q2b	08	–	–
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Quality costs are the categories of costs associated with producing, identifying, avoiding or repairing products that do not meet requirements. They are classified as prevention, appraisal, internal failure and external failure costs.

Importance of quality costs

- The cost of quality has increased because of the increasing complexity of products associated with advances in technology.
- There is increasing awareness of life-cycle costs, including maintenance, spare parts and the cost of field failures.
- Quality engineers can communicate quality issues in the language management understands, money; quality costs become a financial control tool.
- Quality-cost analysis (often a Pareto analysis) identifies opportunities for cost reduction. Because of the leverage effect, a rupee invested in prevention and appraisal can save many rupees of failure costs.

Failure cost versus appraisal cost

Basis	Appraisal cost	Failure cost
Meaning	Costs of measuring, evaluating or auditing products, components and purchased materials to ensure conformance to standards.	Costs incurred because products, components or services fail to meet requirements.
When incurred	Before and during production, to determine the condition of the product.	Internal failure: found before delivery. External failure: found after delivery to the customer.
Examples	Incoming material inspection, product inspection and test, materials consumed in testing, calibration of test equipment.	Internal: scrap, rework, retest, downtime, yield losses, downgrading. External: complaint adjustment, returned product, warranty charges, liability costs.
Behaviour	A cost of conformance; planned.	A cost of nonconformance; would disappear if there were no defects.

6	List and explain the four (major) categories of quality cost. Source: Unit I SEE Nov/Dec 2019 (16IM53), Q2b; SEE Fast Track Oct 2020 (16IM53), Q2b	08	–	–
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Quality costs are those categories of costs associated with producing, identifying, avoiding or repairing products that do not meet requirements. Many organizations use four categories:

Category	Definition	Subcategories
Prevention costs	Costs of efforts in design and manufacturing directed toward preventing nonconformance; costs to “make it right the first time”.	Quality planning and engineering, new products review, product/process design, process control, burn-in, training, quality data acquisition and analysis
Appraisal costs	Costs of measuring, evaluating or auditing products, components and materials to ensure conformance.	Inspection and test of incoming material, product inspection and test, materials and services consumed, maintaining accuracy of test equipment
Internal failure costs	Costs when products fail to meet requirements and the failure is discovered before delivery.	Scrap, rework, retest, failure analysis, downtime, yield losses, downgrading (off-specing)
External failure costs	Costs when the product does not perform satisfactorily after delivery to the customer.	Complaint adjustment, returned product/material, warranty charges, liability costs, indirect costs (loss of reputation and market share)

Quality costs may be 4 to 5 percent of sales in some organizations and 35 to 40 percent in others. Many organizations spend too much on appraisal and too little on prevention; shifting effort to prevention reduces total quality cost.

7	Classify the Cost of Good Quality and Cost of Poor (Bad) Quality in the automotive industry. Provide examples of each cost category and explain how they can be managed effectively to improve product performance and reduce overall expenses. Source: Unit I CIE-I, 26/11/2024; SEE Feb/Mar 2025 (IM353IA), Q2b	10	2	1
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Classification

Group	Category	Examples in the automotive industry
Cost of good quality (cost of conformance)	Prevention	Quality planning and APQP, design FMEA and design reviews, supplier development, poka-yoke on assembly lines, SPC on critical dimensions, operator training, burn-in and endurance testing
	Appraisal	Incoming inspection of castings and bought-out parts, in-process gauging, end-of-line testing (wheel alignment, emission, leak and road tests), calibration of gauges, audits
Cost of poor quality (cost of nonconformance)	Internal failure	Scrap of rejected engine blocks, rework of paint defects, retesting, line stoppage (downtime), yield losses, downgrading of parts
	External failure	Warranty claims, product recalls (for example, airbag or brake recalls), complaint handling, returned vehicles, product liability suits, loss of reputation and market share

How the costs can be managed effectively

- **Invest in prevention:** robust design, design of experiments, supplier quality programmes and mistake-proofing; a rupee spent in prevention may save ten to a hundred rupees of failure cost (leverage effect).
- **Use SPC to reduce variability:** control charts on critical processes reduce scrap and rework and allow appraisal (100% inspection) to be reduced to sampling.
- **Pareto analysis of quality costs:** attack the few defect types (for example, paint defects or weld failures) that cause most of the failure cost.
- **Track quality costs** as a percentage of sales and report them to management so that improvement projects are selected on business impact.
- **Root-cause corrective action** on field failures (warranty data) prevents recalls and protects perceived quality.

As prevention increases, failure costs fall sharply and total quality cost falls, while product performance and customer satisfaction improve.

8	“Spending on prevention of poor quality leads to decrements in appraisal costs.” Do you agree with this statement? Why or why not? Discuss this with the focus on components of these quality costs. Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q2b	04	—	—
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Yes, generally. Prevention costs (quality planning, product/process design, process control, training) are aimed at making the product right the first time. When processes are capable and in statistical control, fewer defects are produced, so less inspection and testing is needed: 100% inspection can be replaced by sampling or by control charts, and incoming inspection is reduced when suppliers are certified. Thus appraisal costs (inspection and test, materials consumed, test equipment) fall, and internal and external failure costs fall even more.

However, appraisal cannot be reduced to zero. Some inspection, audits and calibration remain necessary to verify conformance, and before a very high quality level is reached the incremental cost of prevention rises faster than the savings. The statement holds over the normal working range: many organizations spend too much on appraisal and too little on prevention.

9	Evaluate the organizational, operational, and strategic benefits of implementing the ISO 9000 quality standard, and explain how it contributes to continuous improvement and competitive advantage. Source: Unit I CIE-I, 07/11/2025	10	3	3
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ISO 9000 is a generic standard for quality management systems, applicable to any organization and widely used to demonstrate a supplier's ability to control its processes. ISO 9001 specifies requirements under: quality management system, management responsibility, resource management, product or service realization, and measurement, analysis and improvement.

Organizational benefits

- Management commitment, a quality policy and measurable quality objectives give the organization a common direction.
- Responsibilities and authorities are defined and communicated; employees are trained and competent.
- A documented system (quality manual, procedures, records) preserves knowledge and reduces dependence on individuals.

Operational benefits

- Processes are planned and carried out under controlled conditions, giving consistent output and fewer errors.
- Purchased material is controlled, nonconforming product is identified and prevented from use.
- Records allow traceability of batches, which helps in complaints, corrective actions and recalls.
- Reduced scrap, rework and customer complaints lower operating costs.

Strategic benefits

- Certification is often a condition for supplying large customers and exporting; industry versions (ISO/TS 16949 for automotive, AS 9100 for aerospace) build on it.
- It builds customer confidence and improves the company's image.

Contribution to continuous improvement and competitive advantage

Clause 8 requires monitoring of customer perception, internal audits, data analysis (customer satisfaction, conformance, trends, suppliers), corrective and preventive action, and continual improvement of the effectiveness of the system. Management reviews these results regularly, so the Plan-Do-Check-Act cycle is built into the organization. Certification by itself does not guarantee good products; the competitive advantage comes when the documented system is combined with real variability reduction using statistical methods.

10	What are the key benefits of implementing the ISO 14000 standard for an organization? Source: Unit I CIE-I, 26/11/2024	10	2	2
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ISO 14000 is a family of international standards on environmental management; ISO 14001 specifies the requirements of an environmental management system (EMS). Key benefits:

- (1) **Systematic environmental management:** identifies the environmental aspects and impacts of activities, products and services and sets objectives and targets to control them.
- (2) **Legal compliance:** ensures that environmental laws and regulations are identified and met, reducing the risk of penalties and litigation.
- (3) **Pollution prevention and resource efficiency:** reduced waste, emissions, energy and water use lower operating costs.
- (4) **Market access and customer confidence:** many customers and export markets prefer or require ISO 14001-certified suppliers.
- (5) **Improved image and stakeholder relations:** demonstrates corporate social responsibility to the public, investors and regulators.
- (6) **Risk management:** better preparedness for environmental emergencies and lower insurance and liability costs.
- (7) **Continual improvement:** the Plan-Do-Check-Act structure with audits and management reviews improves environmental performance over time.

- (8) **Integration:** its structure is compatible with ISO 9001, so quality and environmental systems can be managed together.

11	What is the true meaning of a six sigma process? List out the percent inside specifications and the defectives in terms of PPM level for sigma limits of specifications from 1 to 6, assuming normal distribution and a perfectly centered process. What happens to these two performance measures when there is a shift in the mean of the process to the extent of $\pm 1.5\sigma$ from the target? Discuss. Source: Unit I SEE Mar/Apr 2023 (18IM53), Q2b	08	—	—
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Meaning of a six-sigma process: the process standard deviation is so small that the specification limits are six standard deviations from the mean (target) on each side, i.e. the specification width is 12σ and $C_p = 2$. Motorola's six-sigma concept recognises that a process mean does not stay exactly on target; even if the mean shifts by as much as 1.5σ , a six-sigma process produces only about 3.4 defective parts per million (ppm). Six Sigma is therefore both a quality level and a disciplined, project-based (DMAIC) programme for reducing variability.

Percent inside specifications and ppm defective (normal distribution)

Spec. limits	Centered: % inside	Centered: ppm defective	Shifted 1.5σ : % inside	Shifted: ppm defective
$\pm 1\sigma$	68.27	317,311	30.23	697,672
$\pm 2\sigma$	95.45	45,500	69.12	308,770
$\pm 3\sigma$	99.73002	2,700	93.32	66,811
$\pm 4\sigma$	99.99367	63	99.37903	6,210
$\pm 5\sigma$	99.9999427	0.573	99.97674	232.6
$\pm 6\sigma$	99.9999998	0.002	99.99966	3.4

Centered: % inside = $2\Phi(k) - 1$. Shifted by 1.5σ : % inside = $\Phi(k - 1.5) - \Phi(-k - 1.5)$.

Effect of a 1.5σ shift

When the mean shifts by 1.5σ , the percentage inside the specifications falls and the ppm defective rises sharply for every sigma level. For a three-sigma process the defect rate rises from 2,700 ppm to about 66,800 ppm, which is unacceptable for products with many components. For a six-sigma process it rises only from 0.002 ppm to 3.4 ppm, so the product remains essentially defect-free. This robustness to drift is why six-sigma capability is the goal for complex products.

12	How is the application of concepts of statistical quality control advantageous over 100% inspection? Explain. Source: Unit I SEE Jan/Feb 2021 (18IM53), Q2a	10	—	—
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Acceptance sampling and statistical process control give better protection at lower cost than 100% inspection:

- (1) **Lower cost:** only a sample is inspected, so inspection cost, labour and time are much lower.
- (2) **Less handling damage:** fewer units are handled, so less damage occurs during inspection.
- (3) **Applicable to destructive testing:** when the test destroys the item, 100% inspection is impossible.
- (4) **Higher accuracy:** 100% inspection suffers from inspector fatigue and boredom, so many defectives slip through; a smaller sample is inspected more carefully.
- (5) **Stronger motivation for quality:** rejection of a whole lot puts pressure on the supplier to improve the process.
- (6) **Prevention instead of detection:** SPC control charts detect assignable causes as soon as they occur, so corrective action is taken before many nonconforming units are made. 100% inspection only sorts good from bad after the cost of making defectives has been incurred; quality cannot be inspected into the product.
- (7) **Information for improvement:** control charts estimate process parameters and capability and give diagnostic information; 100% inspection does not reduce variability.

Sampling does carry the risk of accepting bad lots or rejecting good lots (consumer's and producer's risks), but these risks are known and controlled by the design of the plan.

13	A certain type of storage battery lasts on an average 3 years with a standard deviation of 0.5 year. Assuming normal distribution, find the percentage of batteries that last (a) less than 2 years, (b) less than 4 years, (c) between 2.5 and 4.5 years, (d) less than 4 years. Source: Unit I CIE-I, 26/11/2024	10	3	3
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Battery life $X \sim N(\mu = 3, \sigma = 0.5)$. Standardize with $z = (x - \mu)/\sigma$ and use the standard normal table $\Phi(z)$.

$$(a) P(X < 2) = P\left(z < \frac{2-3}{0.5}\right) = \Phi(-2.00) = 0.0228$$

$$(b) P(X < 4) = P\left(z < \frac{4-3}{0.5}\right) = \Phi(2.00) = 0.9772$$

$$(c) P(2.5 < X < 4.5) = \Phi\left(\frac{4.5-3}{0.5}\right) - \Phi\left(\frac{2.5-3}{0.5}\right) \\ = \Phi(3.00) - \Phi(-1.00) = 0.99865 - 0.15866 = 0.8400$$

$$(d) P(X < 4) = \Phi(2.00) = 0.9772$$

Answer: (a) 2.28% of the batteries last less than 2 years; (b) 97.72% last less than 4 years; (c) 84.00% last between 2.5 and 4.5 years; (d) 97.72% last less than 4 years (if part (d) is read as “more than 4 years”, the answer is 2.28%).

14	What is variability? Distinguish clearly between chance causes and assignable causes of variation. Source: Unit I SEE Nov/Dec 2019 (16IM53), Q3a	08	–	–
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Variability: no two units of a product are ever identical. There is a certain amount of variability in every product, arising from differences in materials, in the performance and operation of manufacturing equipment, and in the way operators perform their tasks. Because variability can only be described in statistical terms, statistical methods are central to quality improvement; quality improvement is the reduction of variability in processes and products.

Chance causes and assignable causes

Basis	Chance (common) causes	Assignable (special) causes
Definition	Inherent or natural variability: the cumulative effect of many small, essentially unavoidable causes (“background noise”).	Variability that is not part of the chance-cause pattern; generally large compared with the background noise.
Sources	Slight variations in material, machine vibration, ambient conditions, measurement error.	Improperly adjusted or controlled machines, operator errors, defective raw material.
Process state	Process operating with only chance causes is in statistical control (stable system of chance causes).	Process operating with assignable causes present is out of control.
Effect	Predictable within limits; part of the process.	Shifts the mean and/or increases the standard deviation; more output falls outside specifications.
Detection and removal	Cannot be removed without changing the process; requires management action.	Detected quickly by control charts; removed by investigation and corrective action at the process.

Shewhart developed this terminology; today chance causes are also called common causes and assignable causes special causes. The objective of SPC is to detect assignable causes quickly so that corrective action is taken before many nonconforming units are made.

15	Explain any four statistical concepts used in control charts. Source: Unit I SEE Feb/Mar 2026 (IM353IA), Q2b	08	2	3
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- (1) **Sampling distribution and the central limit theorem:** sample averages are approximately normally distributed with mean μ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, whatever the distribution of individual values; the chart plots a sample statistic $\bar{w}$.
- (2) **Three-sigma control limits:** $UCL = \mu_w + L\sigma_w$, $CL = \mu_w$, $LCL = \mu_w - L\sigma_w$ with $L = 3$. For a normal statistic, 99.73% of points fall inside the limits when the process is in control.
- (3) **Hypothesis testing and errors:** a point inside the limits corresponds to failing to reject the hypothesis of statistical control; a point outside corresponds to rejecting it. A type I error (false alarm) has probability 0.0027 with three-sigma limits; a type II error is failing to detect a real shift, shown by the operating-characteristic curve.
- (4) **Average run length (ARL):** $ARL = 1/p$, where p is the probability that a point exceeds the limits; in control $ARL_0 = 1/0.0027 = 370$. The average time to signal is $ATS = ARL \times h$.
- (5) **Rational subgroups:** samples are selected so that variation within subgroups reflects only chance causes, maximizing the chance of detecting differences between subgroups.
- (6) **Patterns and runs:** runs of 8 points on one side of the center line, trends, cycles and the Western Electric rules detect nonrandom behaviour even when all points are inside the limits.

16	Define and explain the type I and type II errors in the context of control charts. Are they related? How does the choice of control limits influence these two errors? Source: Unit I SEE Fast Track Jul 2019 (16IM53), Q2c	06	—	—
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Type I error (α): concluding that the process is out of control when it is really in control: a point falls outside the control limits although no assignable cause is present (a false alarm). With three-sigma limits and a normal statistic, $\alpha = 0.0027$.

Type II error (β): concluding that the process is in control when it is really out of control: a point falls inside the limits although the process has shifted. β is read from the operating-characteristic (OC) curve of the chart and depends on the size of the shift and the sample size.

Are they related? Effect of the control limits

Yes. For a given sample size the two risks move in opposite directions. Moving the control limits **farther** from the center line decreases the risk of type I error but increases the risk of type II error. Moving them **closer** increases the type I error and decreases the type II error. Three-sigma limits are a compromise that works well in practice; warning limits at two-sigma increase sensitivity (smaller β) at the cost of more false alarms. Increasing the sample size reduces β without increasing α .

17	Explain the concept of Rational Subgrouping and its principles. Apply these principles to real-world scenarios in manufacturing or service industries to show how subgrouping separates common and special causes of variation and improves process stability. Source: Unit I SEE Feb/Mar 2025 (IM353IA), Q2a; CIE-I, 26/11/2024; CIE-I, 07/11/2025; SEE Nov/Dec 2019 (16IM53), Q3b; SEE Jan/Feb 2021 (18IM53), Q4c (any four principles)	10	2	1
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Rational subgroup concept (Shewhart): subgroups or samples should be selected so that if assignable causes are present, the chance for differences **between** subgroups is maximized, while the chance for differences due to assignable causes **within** a subgroup is minimized. The within-subgroup variation then represents only common (chance) causes and is used to set the control limits, so special causes show up as points out of control.

Principles of rational subgrouping

- (1) **Use time order of production** as the logical basis for forming subgroups, because it allows assignable causes that occur over time to be detected.
- (2) **Snapshot approach:** each subgroup consists of units produced at the same time or consecutively. It minimizes within-sample variation, maximizes between-sample variation when assignable causes are present, gives the best estimate of the process standard deviation and is used when the main purpose is to detect process shifts.
- (3) **Random-sample approach:** each subgroup is a random sample of all output produced since the last sample. It is used when the chart supports decisions on accepting all units produced in the interval; care is needed because drift during the interval inflates the range and widens the limits.

- (4) **Do not mix sources:** do not combine output of different machines, spindles, heads, operators or shifts in one subgroup; chart each source separately or use a group chart.
- (5) **Do not spread a subgroup across a known change** (for example, the end of one shift and the start of the next), or differences between shifts will be hidden.
- (6) **Choose subgroup size and frequency** to detect the shift of interest: larger samples detect small shifts; small, frequent samples are favoured in high-volume production.

Application to real-world scenarios

- **Manufacturing (machining of shafts on a CNC lathe):** take 5 consecutive shafts every hour from each machine (snapshot). Within-subgroup variation reflects only machine noise; a tool change, wrong offset or new bar stock appears as a shift between subgroups on the $\bar{x}$ chart. Shafts from two lathes are never mixed in one subgroup.
- **Filling line with multiple heads:** form subgroups from one filling head at a time so that a blocked nozzle on one head is not hidden by the others.
- **Service (bank loan processing):** measure the processing time of 4 consecutive applications handled by the same officer in the same hour. Differences between hours, officers or days (special causes such as system downtime or staff shortage) then appear between subgroups.
- **Hospital emergency room:** subgroups of waiting times by shift separate common variation from special causes such as a night-shift staffing gap.

Proper rational subgrouping makes the control chart sensitive to special causes, prevents false alarms, gives correct estimates of process variability and so improves process stability.

18	Explain the need for the change in the control limits of the control charts.	06	—	—
Source: Unit I SEE Jan/Feb 2021 (18IM53), Q2b				

Control limits are not fixed for ever. They must be revised (changed) in the following situations:

- (1) **Trial limits in phase I:** limits computed from the first 20 to 25 subgroups are trial limits. Points outside the limits for which assignable causes are found are removed and the limits are recalculated.
- (2) **Process improvement:** when an assignable cause is permanently removed or the process is improved (new machine, better method or material), the variability or mean changes, so new limits are computed from new data.
- (3) **Change in the process:** a change in raw material, tooling, operator, method or product design creates a new process; the old limits no longer represent it.
- (4) **Periodic review:** limits are reviewed regularly (for example, every week, month or every 25, 50 or 100 samples) to check that they still represent the process.
- (5) **Change in sample size or standard values:** when the subgroup size or the target values (standards given) change, the limits must be recomputed.
- (6) **Charts for future use:** after the process is brought into control, revised limits based on the in-control data are adopted for phase II monitoring.

Limits should not be changed merely because points fall outside them; every revision should be based on a real, verified change in the process.

19	Identify and explain the causes of (i) trend and (ii) cyclic pattern in control charts.	06	—	—
Source: Unit I SEE Jan/Feb 2021 (18IM53), Q3a				

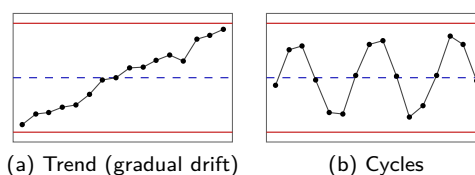


Figure 4: Trend and cyclic patterns on an $\bar{x}$ chart

(i) Trend: a continuous movement of the points in one direction (a run up or run down), also called drift. Causes: gradual wearing out or deterioration of a tool or another critical component, operator fatigue, gradual change in environmental conditions (temperature, humidity), accumulation of waste products, seasonal effects. On the R chart a trend may come

from gradual improvement in operator skill (downward) or wear (upward).

(ii) Cyclic pattern: a short trend in the data that occurs in repeated patterns. On the $\bar{x}$ chart it may be caused by systematic environmental changes (temperature), operator fatigue, regular rotation of operators or machines, fluctuation in voltage or pressure, or raw-material deliveries. On the R chart cycles may come from maintenance schedules or operator fatigue. Elimination of the cyclic source reduces variability and increases yield.

20	What are the two most common types of control charts for variables? Briefly explain their purpose. Source: Unit II SEE Feb/Mar 2025 (IM353IA), Q3a	06	2	1
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The two most common types of variables control charts are the $\bar{x}$ chart and the R chart (used together), with the $\bar{x}$ and s charts as the alternative pair.

$\bar{x}$ chart: monitors the process mean (central tendency) by plotting the sample averages. Limits: $\bar{\bar{x}} \pm A_2 \bar{R}$. It detects shifts in the mean caused by assignable causes such as wrong setting or tool wear.

R chart: monitors the process variability by plotting the sample ranges. Limits: $D_3 \bar{R}$ and $D_4 \bar{R}$. It detects a gain or loss in uniformity, for example due to a worn bearing or an untrained operator.

Both charts are needed: the mean and the variability can go out of control independently, and the $\bar{x}$ limits depend on $\bar{R}$, so the R chart is interpreted first. For large subgroups ($n > 10$) or variable sample sizes, the s chart replaces the R chart.

21	Explain the steps in constructing $\bar{X}-R$ charts and interpret $\bar{X}-R$ charts. Source: Unit II SEE Feb/Mar 2026 (IM353IA), Q4a	10	3	2
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Steps in constructing the $\bar{x}-R$ charts

- (1) Select the quality characteristic, the subgroup size (usually $n = 4$ or 5) and the sampling frequency using rational subgroups.
- (2) Collect $m = 20$ to 25 subgroups of data in time order.
- (3) For each subgroup compute the mean $\bar{x}_i$ and the range $R_i = x_{max} - x_{min}$.
- (4) Compute the grand average $\bar{\bar{x}} = \sum \bar{x}_i / m$ and the average range $\bar{R} = \sum R_i / m$.
- (5) Compute trial control limits. R chart: $UCL = D_4 \bar{R}$, $CL = \bar{R}$, $LCL = D_3 \bar{R}$. $\bar{x}$ chart: $UCL = \bar{\bar{x}} + A_2 \bar{R}$, $CL = \bar{\bar{x}}$, $LCL = \bar{\bar{x}} - A_2 \bar{R}$, with A_2, D_3, D_4 from the table of factors for the subgroup size.
- (6) Plot the R chart first, then the $\bar{x}$ chart, with the subgroup numbers on the horizontal axis.
- (7) Examine the charts for points outside the limits and nonrandom patterns. For points with identified assignable causes, discard them and revise the limits (phase I).
- (8) Use the revised limits for future monitoring (phase II) and estimate the process standard deviation $\hat{\sigma} = \bar{R} / d_2$ for capability analysis.

Interpretation of the $\bar{x}-R$ charts

- Interpret the R chart first: if variability is out of control, the $\bar{x}$ limits are not reliable.
- A process is in control when all points are inside the limits and the pattern is random.
- Out-of-control signals: a point outside a limit; runs of 8 points on one side of the center line; 2 of 3 points beyond two-sigma; 4 of 5 beyond one-sigma; 6 points steadily increasing or decreasing (Western Electric rules).
- Patterns and typical causes: cycles (environmental changes, operator rotation), trends (tool wear, fatigue), sudden shifts in level (new material, operator or setting), mixtures (two populations, over-control), stratification (points hugging the center line: wrong subgrouping or wrong limits).
- A change in the $\bar{x}$ chart only indicates a change in the process mean; a change in the R chart indicates a change in uniformity (dispersion).

22	A manufacturing company measures the diameter (in mm) of a component during production. Data are collected in 6 samples of 4 measurements: <table border="1"> <thead> <tr> <th>Sample</th><th>Meas. 1</th><th>Meas. 2</th><th>Meas. 3</th><th>Meas. 4</th></tr> </thead> <tbody> <tr><td>1</td><td>24.8</td><td>25.1</td><td>24.9</td><td>25.0</td></tr> <tr><td>2</td><td>25.3</td><td>25.2</td><td>25.4</td><td>25.1</td></tr> <tr><td>3</td><td>24.7</td><td>24.9</td><td>24.8</td><td>24.6</td></tr> <tr><td>4</td><td>25.5</td><td>25.4</td><td>25.6</td><td>25.3</td></tr> <tr><td>5</td><td>24.9</td><td>25.0</td><td>25.1</td><td>24.8</td></tr> <tr><td>6</td><td>25.2</td><td>25.3</td><td>25.1</td><td>25.0</td></tr> </tbody> </table> (a) Calculate the sample means. (b) Determine the $\bar{X}$-chart and R-chart control limits. (c) Interpret the control charts: are the process means and ranges in control? Identify any points or patterns that indicate special-cause variation. Source: Unit II CIE-I, 07/11/2025	Sample	Meas. 1	Meas. 2	Meas. 3	Meas. 4	1	24.8	25.1	24.9	25.0	2	25.3	25.2	25.4	25.1	3	24.7	24.9	24.8	24.6	4	25.5	25.4	25.6	25.3	5	24.9	25.0	25.1	24.8	6	25.2	25.3	25.1	25.0	10	4	4
Sample	Meas. 1	Meas. 2	Meas. 3	Meas. 4																																			
1	24.8	25.1	24.9	25.0																																			
2	25.3	25.2	25.4	25.1																																			
3	24.7	24.9	24.8	24.6																																			
4	25.5	25.4	25.6	25.3																																			
5	24.9	25.0	25.1	24.8																																			
6	25.2	25.3	25.1	25.0																																			

(a) Sample means and ranges

Sample	x_1	x_2	x_3	x_4	$\bar{x}$	R
1	24.8	25.1	24.9	25.0	24.950	0.3
2	25.3	25.2	25.4	25.1	25.250	0.3
3	24.7	24.9	24.8	24.6	24.750	0.3
4	25.5	25.4	25.6	25.3	25.450	0.3
5	24.9	25.0	25.1	24.8	24.950	0.3
6	25.2	25.3	25.1	25.0	25.150	0.3

(b) Control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{150.500}{6} = 25.0833, \quad \bar{R} = \frac{\sum R_i}{m} = \frac{1.800}{6} = 0.3000$$

For $n = 4$: $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$

$\bar{x}$ chart: $UCL = \bar{\bar{x}} + A_2 \bar{R} = 25.0833 + 0.729(0.3000) = 25.3020$; $CL = 25.0833$

$LCL = \bar{\bar{x}} - A_2 \bar{R} = 25.0833 - 0.729(0.3000) = 24.8646$

R chart: $UCL = D_4 \bar{R} = 2.282(0.3000) = 0.6846$; $CL = 0.3000$; $LCL = D_3 \bar{R} = 0.0000$

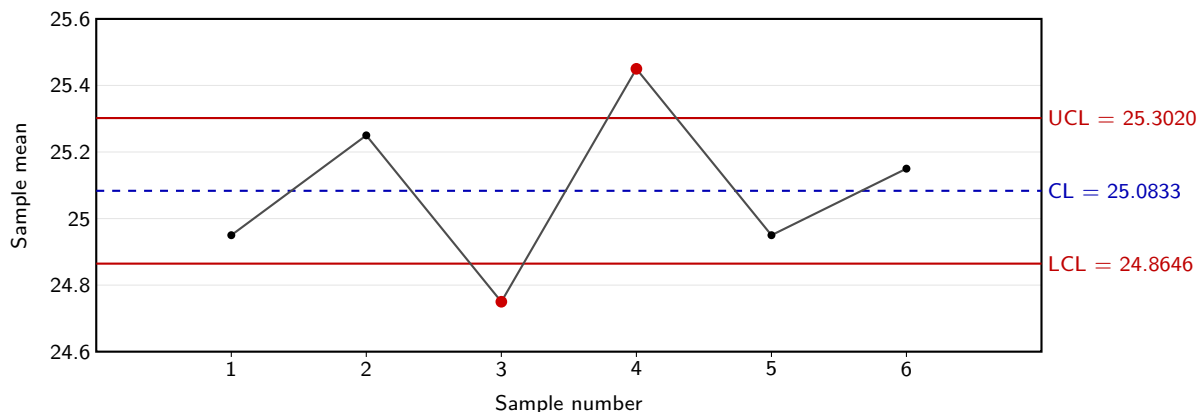
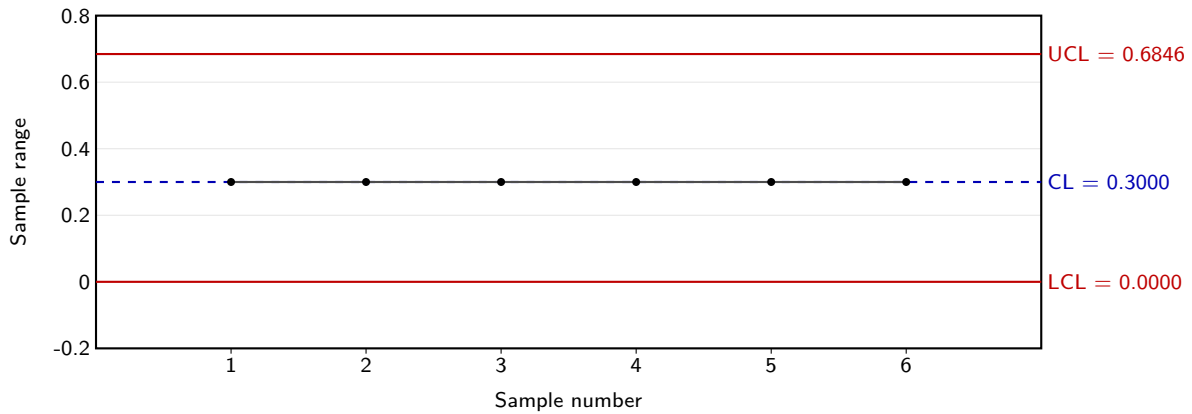


Figure 5: $\bar{x}$ chart

Figure 6: R chart**(c) Interpretation**

All the sample ranges are equal (0.3 mm) and lie well inside the R -chart limits, so the process variability is in control. On the $\bar{x}$ chart, sample 3 and 4 fall outside the control limits (sample 3 = 24.750 is below the LCL of 24.8646, and sample 4 = 25.450 is above the UCL of 25.3020). The sample means also alternate between high and low values. The process mean is therefore **out of control**: special causes such as tool wear and re-setting, a change of material lot or operator adjustment are shifting the mean between samples. These causes must be found and removed and the limits recomputed (phase I) before the chart is used for monitoring.

Answer: $\bar{x}$ chart: UCL = 25.3020, CL = 25.0833, LCL = 24.8646 mm; R chart: UCL = 0.6846, CL = 0.3000, LCL = 0. Ranges are in control; the mean is out of control (samples 3, 4).

23

A manufacturing company measures the diameter of a component in millimetres during a production process. The data are collected in 5 samples, each with 4 measurements:

Sample	Meas. 1	Meas. 2	Meas. 3	Meas. 4
1	10.2	10.3	10.1	10.4
2	10.6	10.8	10.4	10.5
3	10.4	10.2	10.2	10.3
4	10.1	10.3	10.3	10.2
5	10.5	10.4	10.6	10.5

(i) Calculate the mean ($\bar{X}$) and range (R) for each sample. (ii) Determine control limits using the standard factors $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$. (iii) Interpret the control chart.

Source: Unit II | SEE Feb/Mar 2025 (IM353IA), Q4b

10

3

3

(i) Mean and range of each sample

Sample	x_1	x_2	x_3	x_4	$\bar{x}$	R
1	10.2	10.3	10.1	10.4	10.250	0.3
2	10.6	10.8	10.4	10.5	10.575	0.4
3	10.4	10.2	10.2	10.3	10.275	0.2
4	10.1	10.3	10.3	10.2	10.225	0.2
5	10.5	10.4	10.6	10.5	10.500	0.2

(ii) Control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{51.825}{5} = 10.3650, \quad \bar{R} = \frac{\sum R_i}{m} = \frac{1.300}{5} = 0.2600$$

For $n = 4$: $A_2 = 0.729$, $D_3 = 0$, $D_4 = 2.282$

$$\bar{x} \text{ chart: } UCL = \bar{\bar{x}} + A_2 \bar{R} = 10.3650 + 0.729(0.2600) = 10.5545; \quad CL = 10.3650$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R} = 10.3650 - 0.729(0.2600) = 10.1755$$

$$R \text{ chart: } UCL = D_4 \bar{R} = 2.282(0.2600) = 0.5933; \quad CL = 0.2600; \quad LCL = D_3 \bar{R} = 0.0000$$

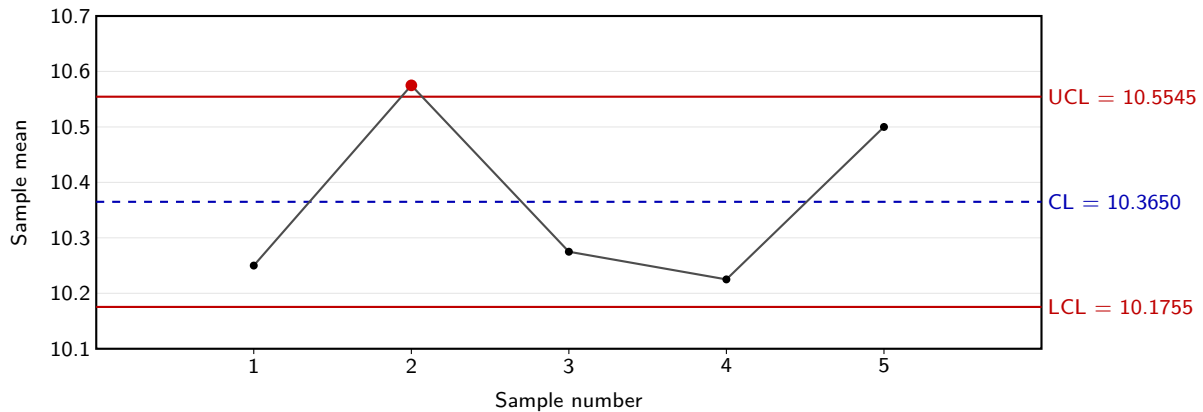


Figure 7: $\bar{x}$ chart

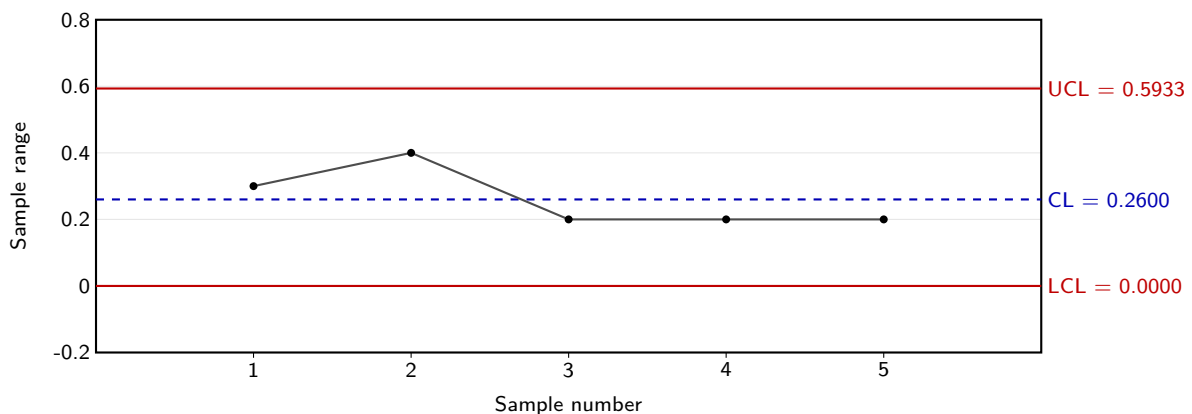


Figure 8: R chart

(iii) Interpretation

All ranges lie within the R-chart limits (0 to 0.593), so the process variability is in control. On the $\bar{x}$ chart, sample 2 ($\bar{x} = 10.575$) lies above the UCL of 10.5545, so the process mean is **out of control**; sample 5 (10.500) is also close to the UCL. An assignable cause (for example, a machine setting or material change) affected sample 2 and should be investigated. After removing it, the limits should be recomputed from the remaining samples, and more samples (20 to 25) collected before the limits are used for routine control.

Answer: $\bar{x}$ chart: UCL = 10.5545, CL = 10.3650, LCL = 10.1755; R chart: UCL = 0.5933, CL = 0.2600, LCL = 0. Process mean is out of control (sample 2); variability is in control.

24

Mean values and ranges of data from 20 samples (sample size = 4) are shown below:

S.N.	Mean	Range	S.N.	Mean	Range
1	10	4	11	12	5
2	15	4	12	13	4
3	12	5	13	12	4
4	11	4	14	12	3
5	9	5	15	11	3
6	11	6	16	15	4
7	11	4	17	12	4
8	9	4	18	15	3
9	10	4	19	11	3
10	11	6	20	10	4

Calculate the control limits for $\bar{x}$ and R charts. Draw the control charts and give suitable conclusions.

Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q3b

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$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{232.00}{20} = 11.600, \quad \bar{R} = \frac{\sum R_i}{m} = \frac{83.00}{20} = 4.150$$

$$\text{For } n = 4 : A_2 = 0.729, D_3 = 0, D_4 = 2.282$$

$$\bar{x} \text{ chart: UCL} = \bar{\bar{x}} + A_2 \bar{R} = 11.600 + 0.729(4.150) = 14.625; \quad \text{CL} = 11.600$$

$$\text{LCL} = \bar{\bar{x}} - A_2 \bar{R} = 11.600 - 0.729(4.150) = 8.575$$

$$R \text{ chart: UCL} = D_4 \bar{R} = 2.282(4.150) = 9.470; \quad \text{CL} = 4.150; \quad \text{LCL} = D_3 \bar{R} = 0.000$$

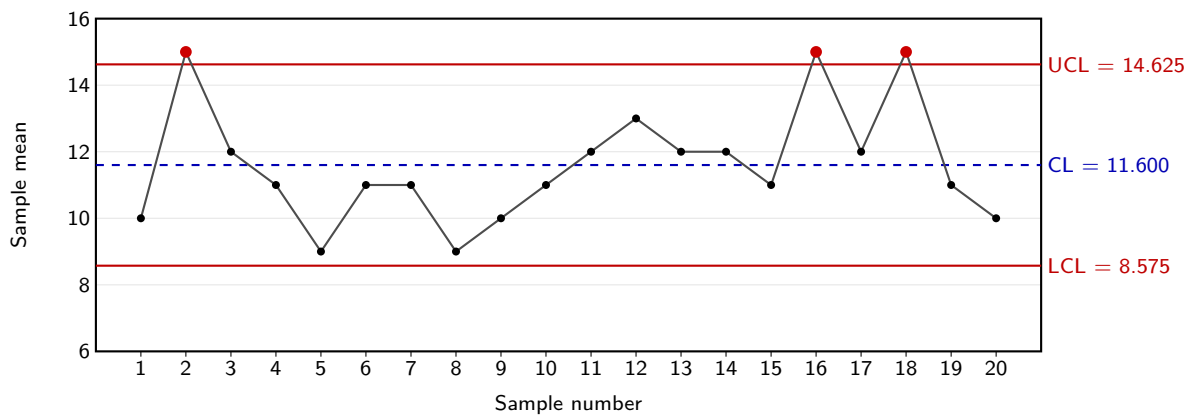


Figure 9: $\bar{x}$ chart

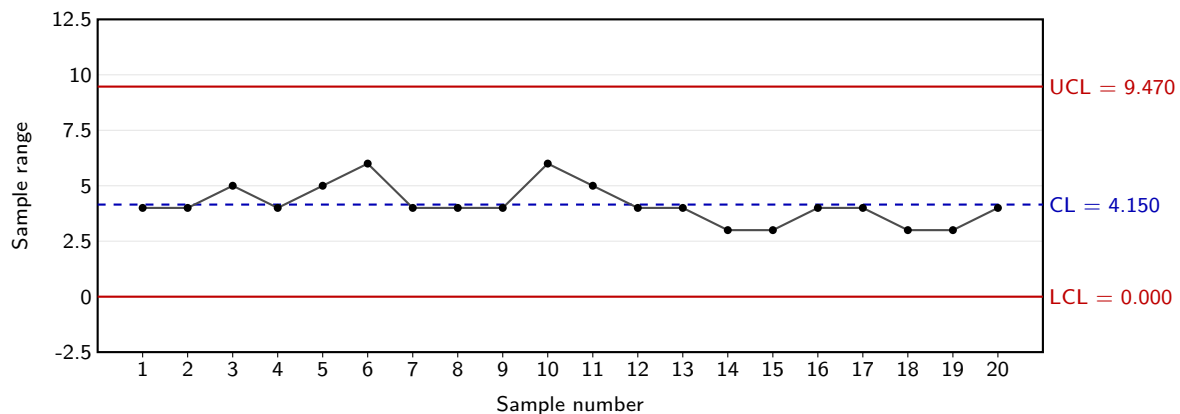


Figure 10: R chart

Conclusions

The R chart is in control: all ranges lie between 0 and 9.470. On the $\bar{x}$ chart, samples 2, 16, 18 ($\bar{x} = 15$) are above the UCL of 14.625, and samples 5 and 8 ($\bar{x} = 9$) are close to the LCL. The process variability is stable but the process mean is **not in statistical control**; assignable causes are shifting the mean upward from time to time.

Revised limits: if assignable causes are found for samples 2, 16, 18 and they are removed, the limits recomputed from the remaining 17 samples are $\bar{\bar{x}} = 11.000$, $\bar{R} = 4.235$, $\bar{x}$ chart UCL = 14.088 and LCL = 7.912, R chart UCL = 9.665. All remaining points lie within these revised limits.

Answer: Trial limits: $\bar{x}$ chart 14.625 / 11.600 / 8.575; R chart 9.470 / 4.150 / 0. Variability in control; mean out of control at samples 2, 16, 18.

25

In a foundry the specification on mould hardness is 65 ± 15 . Moulds were chosen at random and mould hardness was estimated at 4 different places per mould. The data obtained are:

Mould No.	$\bar{X}$	R	Mould No.	$\bar{X}$	R
1	68	5	13	70	10
2	59	16	14	67	10
3	60	15	15	63	15
4	57	12	16	69	10
5	74	24	17	62	5
6	68	5	18	56	10
7	66	10	19	66	10
8	56	10	20	63	8
9	78	15	21	67	15
10	55	10	22	75	10
11	67	5	23	53	5
12	57	5	24	56	12

(i) Construct the control charts for the data. (ii) Interpret the state of the manufacturing process. (iii) Can the process meet the specification? (iv) What is the current state of affairs regarding non-conformity? (v) What are the different possible ways of reducing the rejection?

Source: Unit II | SEE Feb/Mar 2026 (IM353IA), Q3

16

4

4

(i) Control charts (n = 4, 24 moulds in the table)

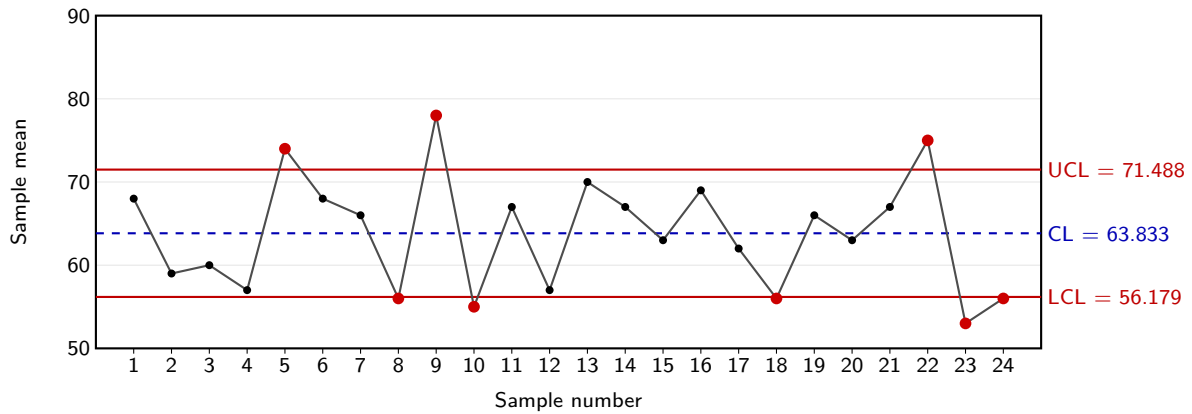
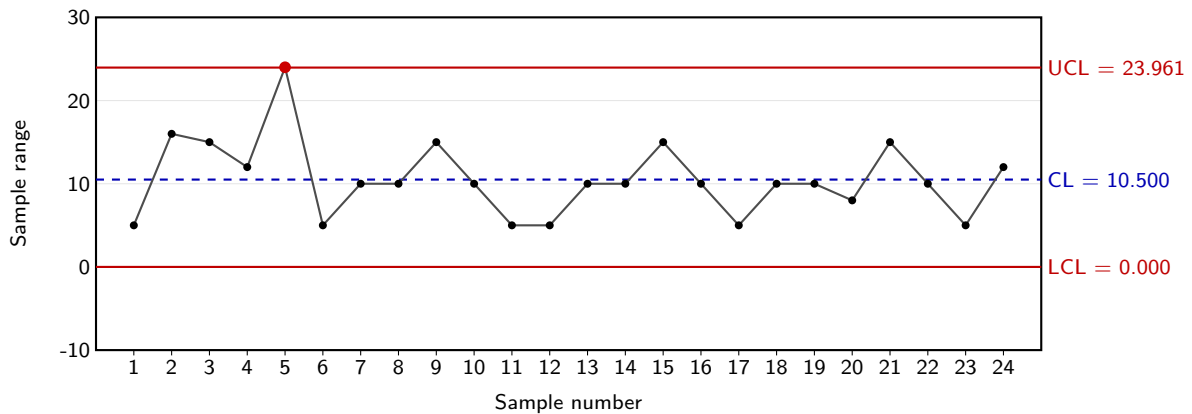
$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{1532.00}{24} = 63.833, \quad \bar{R} = \frac{\sum R_i}{m} = \frac{252.00}{24} = 10.500$$

$$\text{For } n = 4 : A_2 = 0.729, D_3 = 0, D_4 = 2.282$$

$$\bar{x} \text{ chart: UCL} = \bar{\bar{x}} + A_2 \bar{R} = 63.833 + 0.729(10.500) = 71.488; \quad \text{CL} = 63.833$$

$$\text{LCL} = \bar{\bar{x}} - A_2 \bar{R} = 63.833 - 0.729(10.500) = 56.179$$

$$R \text{ chart: UCL} = D_4 \bar{R} = 2.282(10.500) = 23.961; \quad \text{CL} = 10.500; \quad \text{LCL} = D_3 \bar{R} = 0.000$$

Figure 11: $\bar{x}$ chartFigure 12: R chart**(ii) State of the process**

R chart: mould 5 ($R = 24$) lies just above the UCL of 23.96; all other ranges are in control. The within-mould variability is therefore almost stable, with one assignable cause at mould 5. **$\bar{x}$ chart:** 8 of the 24 points lie outside the limits (moulds 5, 8, 9, 10, 18, 22, 23, 24); the hardness level swings widely from mould to mould. The process is **out of statistical control**, mainly in its mean; assignable causes (sand properties, pouring temperature, cooling time, metal composition) are present.

(iii) Can the process meet the specification of 65 ± 15 ?

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 30.60, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 48.53 \text{ to } 79.13$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{80.0 - 50.0}{30.60} = 0.980, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.904$$

$$P(x < LSL) = \Phi\left(\frac{50.0 - 63.83}{5.10}\right) = \Phi(-2.71) = 0.00334$$

$$P(x > USL) = 1 - \Phi\left(\frac{80.0 - 63.83}{5.10}\right) = 1 - \Phi(3.17) = 0.00076$$

$$\text{Total fraction nonconforming} = 0.00410 \text{ (0.41\%)}$$

No. Even on the basis of within-mould variation alone, the natural spread ($6\sigma = 30.60$) is slightly larger than the tolerance (30), so $C_p = 0.98 < 1$. Because the mean is also out of control, the actual spread of hardness is much wider than 6σ (the mould averages alone range from 53 to 78). The process cannot meet the specification in its present state.

(iv) Present state regarding non-conformity

Even if the mean were held at 63.83, about 0.41% of the hardness readings would fall outside 50 to 80 because of within-mould variation alone. Moulds whose averages are near or beyond the specification limits (moulds 5, 9, 22 and 23) are very likely to contain nonconforming hardness values, so the actual rejection is considerably higher than this estimate.

(v) Ways of reducing the rejection

- Identify and remove the assignable causes of mould-to-mould variation (control sand moisture, compactability, metal composition, pouring temperature and cooling time); use the $\bar{x}$ chart on-line.

- Keep the process mean centered at the nominal 65 so that both tails are equally far from the limits.
- Reduce within-mould variation by uniform ramming and standardized operating procedures; train operators.
- Use designed experiments to find the process settings that minimize hardness variability.
- If the process cannot be improved, review the specification with design or use a more capable process.

26 The following are $\bar{X}$ and R values of 20 subgroups of 5 readings:

Subgroup	$\bar{X}$	R	Subgroup	$\bar{X}$	R
1	34.0	4	11	38.4	4
2	31.6	2	12	34.0	14
3	30.8	3	13	35.0	4
4	33.8	5	14	33.8	7
5	31.6	3	15	31.6	5
6	33.0	5	16	33.0	7
7	28.2	13	17	32.6	3
8	33.8	19	18	31.8	9
9	37.8	6	19	35.6	6
10	35.8	4	20	33.0	4

(i) Determine the control limits of the $\bar{X}$ and R charts. (ii) What is the process capability? (iii) Is the process capable of meeting the specification limits? (iv) Determine the percentage of rejection, if any (specification limits are 33 ± 5). (v) What values of $\bar{X}$ and R do you recommend for future use?

Source: Unit II | SEE Jan/Feb 2021 (16IM53), Q4a

16

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(i) Trial control limits

$$\bar{\bar{x}} = \frac{\sum \bar{x}_i}{m} = \frac{669.20}{20} = 33.460, \quad \bar{R} = \frac{\sum R_i}{m} = \frac{127.00}{20} = 6.350$$

For $n = 5$: $A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$

$\bar{x}$ chart: $UCL = \bar{\bar{x}} + A_2 \bar{R} = 33.460 + 0.577(6.350) = 37.124$; $CL = 33.460$

$LCL = \bar{\bar{x}} - A_2 \bar{R} = 33.460 - 0.577(6.350) = 29.796$

R chart: $UCL = D_4 \bar{R} = 2.114(6.350) = 13.424$; $CL = 6.350$; $LCL = D_3 \bar{R} = 0.000$

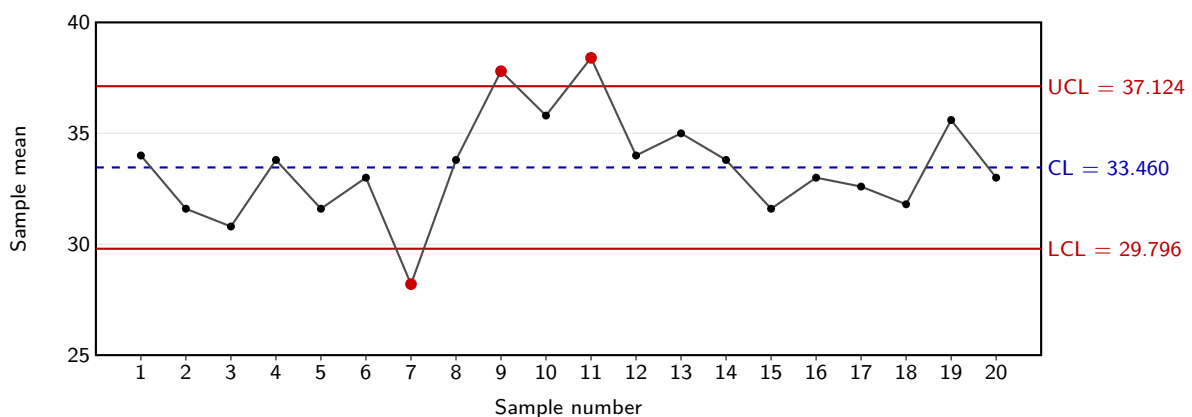


Figure 13: $\bar{x}$ chart

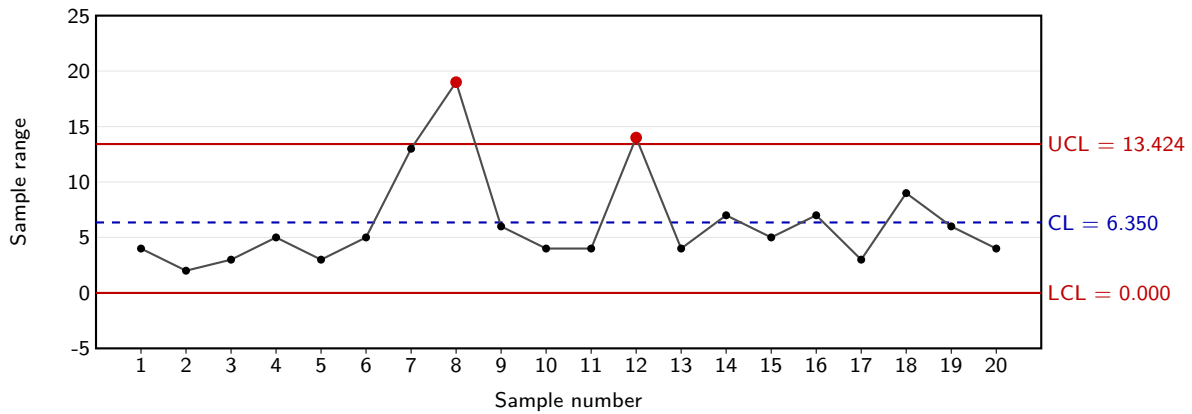


Figure 14: R chart

Out-of-control points: $\bar{x}$ chart subgroups 7, 9, 11; R chart subgroups 8, 12. Assuming assignable causes are found for them, these subgroups are discarded and the limits revised until all remaining points are in control (subgroups removed: 7, 8, 9, 11, 12).

Revised (15 subgroups): $\bar{\bar{x}} = 33.133$, $\bar{R} = 4.733$

$\bar{x}$ chart: UCL = 35.864, LCL = 30.402; R chart: UCL = 10.006, LCL = 0

(ii)–(iv) Process capability and rejection (specification 33 ± 5)

$\hat{\sigma} = \bar{R}/d_2$, $6\hat{\sigma} = 12.21$, natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma} = 27.03$ to 39.24

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{38.0 - 28.0}{12.21} = 0.819, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.797$$

$$P(x < LSL) = \Phi\left(\frac{28.0 - 33.13}{2.03}\right) = \Phi(-2.52) = 0.00583$$

$$P(x > USL) = 1 - \Phi\left(\frac{38.0 - 33.13}{2.03}\right) = 1 - \Phi(2.39) = 0.00839$$

Total fraction nonconforming = 0.01421 (1.42%)

Process capability = $6\sigma = 12.210$, which is greater than the tolerance of 10, so $C_p = 0.82$: the process is **not capable**. The mean (33.133) is slightly above the nominal 33, giving $C_{pk} = 0.80$. Even after the assignable causes are removed, about 1.42% of the output is expected to fall outside 28 to 38, so the process cannot reliably meet the specification; the process variability must be reduced (or the tolerance reviewed) and the mean centred at 33.

(v) Values recommended for future use

Answer: Use $\bar{\bar{x}} = 33.133$ and $\bar{R} = 4.733$ (revised): $\bar{x}$ chart limits 30.402 to 35.864, R chart UCL 10.006. Process capability $6\sigma = 12.210$ ($C_p = 0.82$), with about 1.42% rejection expected.

27	Samples of size $n = 5$ are taken from a manufacturing process every hour. A quality characteristic is measured and $\bar{x}$ and R are computed for each sample. After 25 samples have been analyzed, $\sum \bar{x}_i = 662.50$ and $\sum R_i = 9.00$. The quality characteristic is normally distributed. (i) Find the control limits for the $\bar{x}$ and R charts. (ii) Assume that both charts exhibit control. If the specifications are 26.40 ± 0.50, estimate the fraction nonconforming. (iii) If the mean of the process were 26.40, what fraction nonconforming would result? Source: Unit II SEE Fast Track Oct 2020 (16IM53), Q3a	08	–	–
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(i) Control limits

$$\bar{\bar{x}} = \frac{662.50}{25} = 26.5000, \quad \bar{R} = \frac{9.00}{25} = 0.3600$$

$$(n = 5 : A_2 = 0.577, D_3 = 0, D_4 = 2.114, d_2 = 2.326)$$

$$\bar{x} \text{ chart: UCL} = 26.5000 + 0.577(0.3600) = 26.7077, \quad \text{CL} = 26.5000$$

$$\text{LCL} = 26.5000 - 0.577(0.3600) = 26.2923$$

$$R \text{ chart: UCL} = 2.114(0.3600) = 0.7610, \quad \text{CL} = 0.3600, \quad \text{LCL} = 0(0.3600) = 0.0000$$

$$\hat{\sigma} = \bar{R}/d_2 = 0.3600/2.326 = 0.15477$$

(ii) Fraction nonconforming (specifications 25.90 to 26.90)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 0.929, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 26.036 \text{ to } 26.964$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{26.90 - 25.90}{0.929} = 1.077, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.861$$

$$P(x < LSL) = \Phi\left(\frac{25.90 - 26.500}{0.155}\right) = \Phi(-3.88) = 0.00005$$

$$P(x > USL) = 1 - \Phi\left(\frac{26.90 - 26.500}{0.155}\right) = 1 - \Phi(2.58) = 0.00488$$

$$\text{Total fraction nonconforming} = 0.00493 \text{ (0.49\%)}$$

(iii) If the process were centered at 26.40

$$P = 2\Phi\left(\frac{25.90 - 26.40}{0.1548}\right) = 2\Phi(-3.23) = 0.00124$$

Answer: (i) $\bar{x}$ chart 26.7077 / 26.500 / 26.2923; R chart 0.7610 / 0.360 / 0. (ii) Fraction nonconforming = 0.00493 (0.49%). (iii) With the mean at 26.40 it falls to 0.00124 (0.124%).

28	Samples of $n = 5$ units are taken from a process every hour. The $\bar{x}$ and R values for a particular quality characteristic are determined. After 25 samples have been collected, $\bar{\bar{x}} = 20$ and $\bar{\bar{R}} = 4.56$. (i) What are the three-sigma control limits? (ii) Both charts exhibit control; estimate the process standard deviation. (iii) Assuming that the process output is normally distributed, if the specifications are 19 ± 5, what are your conclusions regarding the process capability? (iv) If the process mean shifts to 24, what is the probability of not detecting this shift on the first subsequent sample? Source: Unit II SEE Fast Track Oct 2020 (16IM53), Q3b	08	—	—
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(i) Three-sigma control limits and (ii) process standard deviation

$$\bar{\bar{x}} = 20, \bar{\bar{R}} = 4.56, n = 5 : A_2 = 0.577, D_3 = 0, D_4 = 2.114, d_2 = 2.326$$

$$\bar{x} \text{ chart: } UCL = 20 + 0.577(4.56) = 22.6311, \quad CL = 20, \quad LCL = 20 - 0.577(4.56) = 17.3689$$

$$R \text{ chart: } UCL = 2.114(4.56) = 9.6398, \quad CL = 4.56, \quad LCL = 0$$

$$\hat{\sigma} = \bar{R}/d_2 = 4.56/2.326 = 1.9604$$

(iii) Process capability (specifications $19 \pm 5 = 14$ to 24)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 11.76, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 14.12 \text{ to } 25.88$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{24.0 - 14.0}{11.76} = 0.850, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.680$$

$$P(x < LSL) = \Phi\left(\frac{14.0 - 20.00}{1.96}\right) = \Phi(-3.06) = 0.00110$$

$$P(x > USL) = 1 - \Phi\left(\frac{24.0 - 20.00}{1.96}\right) = 1 - \Phi(2.04) = 0.02066$$

$$\text{Total fraction nonconforming} = 0.02176 \text{ (2.18\%)}$$

$C_p = 0.850 < 1$: the natural spread (11.763) is larger than the tolerance (10), and the mean (20) is 1 unit above the nominal 19. The process is **not capable**; about 2.18% of the output is nonconforming. Variability must be reduced and the mean re-centred at 19.

(iv) Probability of not detecting a shift to 24 on the first sample

$$\sigma_{\bar{x}} = \hat{\sigma}/\sqrt{n} = 1.9604/\sqrt{5} = 0.8767$$

$$\begin{aligned} \beta &= P(LCL \leq \bar{x} \leq UCL \mid \mu = 24) = \Phi\left(\frac{22.6311 - 24}{0.8767}\right) - \Phi\left(\frac{17.3689 - 24}{0.8767}\right) \\ &= \Phi(-1.56) - \Phi(-7.56) = 0.0592 \end{aligned}$$

Answer: Limits: $\bar{x}$ 22.631 / 20 / 17.369, R 9.640 / 4.56 / 0; $\hat{\sigma} = 1.9604$; $C_p = 0.850$, not capable (2.18% nonconforming); probability of not detecting the shift on the first sample = 0.0592.

29	Samples of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured and $\bar{X}$ and R are calculated for each sample. After 50 samples, $\sum \bar{X}_i = 2000$ and $\sum R_i = 200$. Assume that the quality characteristic is normally distributed. (i) Compute the control limits for the $\bar{X}$ and R charts. (ii) All points fall between the control limits. What are the natural tolerance limits of the process? (iii) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability of the process to produce items within these specifications? (iv) If an item exceeding the upper specification limit can be reworked and one below the lower specification limit must be scrapped, what percent of scrap and rework is the process producing? Source: Unit II SEE Mar/Apr 2023 (18IM53), Q3a	12	–	–
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(i) Control limits

$$\bar{\bar{x}} = \frac{2000.00}{50} = 40.0000, \quad \bar{R} = \frac{200.00}{50} = 4.0000$$

$$(n = 6 : A_2 = 0.483, D_3 = 0, D_4 = 2.004, d_2 = 2.534)$$

$$\bar{x} \text{ chart: } UCL = 40.0000 + 0.483(4.0000) = 41.9320, \quad CL = 40.0000$$

$$LCL = 40.0000 - 0.483(4.0000) = 38.0680$$

$$R \text{ chart: } UCL = 2.004(4.0000) = 8.0160, \quad CL = 4.0000, \quad LCL = 0(4.0000) = 0.0000$$

$$\hat{\sigma} = \bar{R}/d_2 = 4.0000/2.534 = 1.57853$$

(ii) Natural tolerance limits

$$\hat{\mu} \pm 3\hat{\sigma} = 40 \pm 3(1.5785) = 40 \pm 4.736 \Rightarrow 35.264 \text{ to } 44.736$$

(iii) Ability to meet specifications 41 ± 5 (36 to 46)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 9.47, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 35.26 \text{ to } 44.74$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{46.0 - 36.0}{9.47} = 1.056, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.845$$

$$P(x < LSL) = \Phi\left(\frac{36.0 - 40.00}{1.58}\right) = \Phi(-2.53) = 0.00564$$

$$P(x > USL) = 1 - \Phi\left(\frac{46.0 - 40.00}{1.58}\right) = 1 - \Phi(3.80) = 0.00007$$

$$\text{Total fraction nonconforming} = 0.00571 \text{ (0.57\%)}$$

The natural spread (9.471) is slightly smaller than the tolerance (10), so $C_p = 1.056$, just above 1 (marginally capable). However, the process mean (40) is 1 unit below the nominal 41, so $C_{pk} = 0.845 < 1$: the process is not centered and will produce some items below the LSL. Re-centering the mean at 41 would bring the process to its potential capability.

(iv) Scrap and rework

Answer: Scrap (below LSL 36) = 0.564%; rework (above USL 46) = 0.0072%; total = 0.571%.

30	Subgroups of 5 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$ and R values computed for each subgroup. After 25 subgroups, $\sum \bar{X} = 357.5$ and $\sum R = 8.8$. Assume that all the points are within the control limits on both the charts. The specifications are 14.4 ± 0.4. (i) Compute the control limits for the $\bar{X}$ and R charts. (ii) What is the process capability? (iii) Determine the percentage of rejections, if any. (iv) What can you conclude regarding its ability to meet the specifications? Source: Unit II SEE Nov/Dec 2019 (16IM53), Q4a	12	–	–
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(i) Control limits

$$\bar{\bar{x}} = \frac{357.50}{25} = 14.3000, \quad \bar{R} = \frac{8.80}{25} = 0.3520$$

$$(n = 5 : A_2 = 0.577, D_3 = 0, D_4 = 2.114, d_2 = 2.326)$$

$$\bar{x} \text{ chart: UCL} = 14.3000 + 0.577(0.3520) = 14.5031, \quad \text{CL} = 14.3000$$

$$\text{LCL} = 14.3000 - 0.577(0.3520) = 14.0969$$

$$R \text{ chart: UCL} = 2.114(0.3520) = 0.7441, \quad \text{CL} = 0.3520, \quad \text{LCL} = 0(0.3520) = 0.0000$$

$$\hat{\sigma} = \bar{R}/d_2 = 0.3520/2.326 = 0.15133$$

(ii) Process capability

Process capability = $6\hat{\sigma} = 6(0.1513) = 0.9080$; the natural tolerance limits are 14.30 ± 0.4540 , i.e. 13.846 to 14.754.

(iii) Percentage of rejections (specifications 14.0 to 14.8)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 0.91, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 13.85 \text{ to } 14.75$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{14.8 - 14.0}{0.91} = 0.881, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.661$$

$$P(x < LSL) = \Phi\left(\frac{14.0 - 14.30}{0.15}\right) = \Phi(-1.98) = 0.02372$$

$$P(x > USL) = 1 - \Phi\left(\frac{14.8 - 14.30}{0.15}\right) = 1 - \Phi(3.30) = 0.00048$$

$$\text{Total fraction nonconforming} = 0.02419 \text{ (2.42\%)}$$

(iv) Conclusion

The process spread (0.9080) is greater than the tolerance (0.8), so $C_p = 0.881 < 1$, and the mean (14.3) is 0.1 below the nominal 14.4, giving $C_{pk} = 0.661$. The process is **not capable** of meeting the specifications; about 2.42% will be rejected, mostly below the LSL. The mean should be re-centred at 14.4 (which reduces rejections to about 0.82%) and the process variability must be reduced.

Answer: $\bar{x}$ chart 14.5031 / 14.300 / 14.0969; R chart 0.7441 / 0.352 / 0; capability $6\sigma = 0.9080$; rejection $\approx 2.42\%$; process not capable.

31	Subgroups of 4 items each are taken from a manufacturing process at regular intervals. A certain quality characteristic is measured and $\bar{X}$, R values are computed for each subgroup. After 25 subgroups, $\sum \bar{X} = 15350$ and $\sum R = 411.4$. (i) Compute the control limits for the $\bar{X}$ - R charts. (ii) Assume all the points fall within the control limits on both charts. The specification limits are 610 ± 15 . If the quality characteristic is normally distributed, what percentage of product would fail to meet specifications? (iii) Any product that falls below the LSL will be scrapped and above the USL must be reworked. It is suggested that the process be centered at a level so that not more than 0.1% of the product will be scrapped. What should be the aimed value of $\bar{X}'$ to make the scrap exactly 0.1%?	12	–	–
Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q3a				

(i) Control limits

$$\bar{\bar{x}} = \frac{15350.00}{25} = 614.000, \quad \bar{R} = \frac{411.40}{25} = 16.456$$

$$(n = 4 : A_2 = 0.729, D_3 = 0, D_4 = 2.282, d_2 = 2.059)$$

$$\bar{x} \text{ chart: UCL} = 614.000 + 0.729(16.456) = 625.996, \quad \text{CL} = 614.000$$

$$\text{LCL} = 614.000 - 0.729(16.456) = 602.004$$

$$R \text{ chart: UCL} = 2.282(16.456) = 37.553, \quad \text{CL} = 16.456, \quad \text{LCL} = 0(16.456) = 0.000$$

$$\hat{\sigma} = \bar{R}/d_2 = 16.456/2.059 = 7.9922$$

(ii) Percentage failing the specifications 610 ± 15 (595 to 625)

$$\hat{\sigma} = \bar{R}/d_2, \quad 6\hat{\sigma} = 48.0, \quad \text{natural tolerance limits } \hat{\mu} \pm 3\hat{\sigma} = 590.0 \text{ to } 638.0$$

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{625 - 595}{48.0} = 0.626, \quad C_{pk} = \frac{\min(USL - \hat{\mu}, \hat{\mu} - LSL)}{3\hat{\sigma}} = 0.459$$

$$P(x < LSL) = \Phi\left(\frac{595 - 614.0}{8.0}\right) = \Phi(-2.38) = 0.00872$$

$$P(x > USL) = 1 - \Phi\left(\frac{625 - 614.0}{8.0}\right) = 1 - \Phi(1.38) = 0.08436$$

$$\text{Total fraction nonconforming} = 0.09308 \text{ (9.31\%)}$$

(iii) Aimed value for exactly 0.1% scrap

$$P(x < 595) = 0.001 \Rightarrow \frac{595 - \bar{X}'}{\hat{\sigma}} = -3.09$$

$$\bar{X}' = 595 + 3.09(7.992) = 619.70$$

$$\text{Rework then} = 1 - \Phi\left(\frac{625 - 619.70}{7.992}\right) = 1 - \Phi(0.664) = 0.2535$$

Answer: Limits: $\bar{X}$ chart 626.00 / 614.0 / 602.00, R chart 37.55 / 16.456 / 0. Product failing specifications = 9.31% (scrap 0.87%, rework 8.44%). Aim $\bar{X}' = 619.70$ for 0.1% scrap (rework then rises to about 25.3%).

32	A process is producing shafts with a nominal diameter of 50 mm. The specification limits are 50 ± 0.5 mm. From control chart data, the process mean is 50.1 mm and the process standard deviation (σ) is 0.1 mm. (a) Calculate the upper control limit (UCL) and lower control limit (LCL) for the process using $\pm 3\sigma$ limits. (b) Compare the control limits with the specification limits and comment on the process capability. Source: Unit II CIE-I, 07/11/2025	10	4	4
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(a) Three-sigma limits of the process

$$UCL = \mu + 3\sigma = 50.1 + 3(0.1) = 50.4 \text{ mm}$$

$$CL = 50.1 \text{ mm}$$

$$LCL = \mu - 3\sigma = 50.1 - 3(0.1) = 49.8 \text{ mm}$$

(b) Comparison with specification limits (49.5 to 50.5 mm)

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{50.5 - 49.5}{6(0.1)} = 1.667$$

$$C_{pk} = \min \left(\frac{50.5 - 50.1}{0.3}, \frac{50.1 - 49.5}{0.3} \right) = \min(1.333, 2.000) = 1.333$$

$$P(x > 50.5) = 1 - \Phi(4.0) = 0.00003, \quad P(x < 49.5) = \Phi(-6.0) \approx 0$$

The natural tolerance limits (49.8 to 50.4 mm) lie inside the specification limits (49.5 to 50.5 mm), so the process is **capable**: $C_p = 1.67$ and $C_{pk} = 1.33$, both at or above the recommended 1.33, and practically no shafts (about 30 ppm) will be outside the specification. The mean is 0.1 mm above the nominal, so the upper side has less margin; re-centering at 50.0 mm would raise C_{pk} to 1.67. Note that control limits describe the process, while specification limits describe the requirement; they should not be confused.

Answer: UCL = 50.4 mm, LCL = 49.8 mm; process is capable ($C_p = 1.67$, $C_{pk} = 1.33$) but slightly off-centre.

33	A TiW layer is deposited on a substrate using a sputtering tool. The following table contains layer thickness measurements (in Angstroms) on 10 subgroups of four substrates. Set up $\bar{X}$ and s charts on this process and compute trial control limits. <table border="1"> <thead> <tr> <th>Subgroup</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th></tr> </thead> <tbody> <tr><td>1</td><td>459</td><td>449</td><td>435</td><td>450</td></tr> <tr><td>2</td><td>443</td><td>440</td><td>442</td><td>442</td></tr> <tr><td>3</td><td>457</td><td>444</td><td>449</td><td>444</td></tr> <tr><td>4</td><td>469</td><td>463</td><td>453</td><td>438</td></tr> <tr><td>5</td><td>443</td><td>457</td><td>445</td><td>454</td></tr> <tr><td>6</td><td>444</td><td>456</td><td>456</td><td>457</td></tr> <tr><td>7</td><td>445</td><td>449</td><td>450</td><td>445</td></tr> <tr><td>8</td><td>446</td><td>455</td><td>449</td><td>452</td></tr> <tr><td>9</td><td>444</td><td>452</td><td>457</td><td>440</td></tr> <tr><td>10</td><td>432</td><td>463</td><td>463</td><td>443</td></tr> </tbody> </table> Source: Unit II CIE-II, 07/01/2025	Subgroup	x_1	x_2	x_3	x_4	1	459	449	435	450	2	443	440	442	442	3	457	444	449	444	4	469	463	453	438	5	443	457	445	454	6	444	456	456	457	7	445	449	450	445	8	446	455	449	452	9	444	452	457	440	10	432	463	463	443	10	3	CO3
Subgroup	x_1	x_2	x_3	x_4																																																							
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2	443	440	442	442																																																							
3	457	444	449	444																																																							
4	469	463	453	438																																																							
5	443	457	445	454																																																							
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8	446	455	449	452																																																							
9	444	452	457	440																																																							
10	432	463	463	443																																																							

For each subgroup $\bar{x}_i = \sum x/n$ and $s_i = \sqrt{\sum (x - \bar{x}_i)^2 / (n - 1)}$.

Subgroup	x_1	x_2	x_3	x_4	$\bar{x}$	s
1	459	449	435	450	448.25	9.912
2	443	440	442	442	441.75	1.258
3	457	444	449	444	448.50	6.137
4	469	463	453	438	455.75	13.549
5	443	457	445	454	449.75	6.801
6	444	456	456	457	453.25	6.185
7	445	449	450	445	447.25	2.630
8	446	455	449	452	450.50	3.873
9	444	452	457	440	448.25	7.676
10	432	463	463	443	450.25	15.392

$$\bar{\bar{x}} = \frac{4493.50}{10} = 449.350, \quad \bar{s} = \frac{73.413}{10} = 7.341 \quad (n = 4 : A_3 = 1.628, B_3 = 0, B_4 = 2.266)$$

$$\bar{x} \text{ chart: } UCL = \bar{\bar{x}} + A_3 \bar{s} = 449.350 + 1.628(7.341) = 461.302; \quad CL = 449.350; \quad LCL = 437.398$$

$$s \text{ chart: } UCL = B_4 \bar{s} = 2.266(7.341) = 16.635; \quad CL = 7.341; \quad LCL = B_3 \bar{s} = 0$$

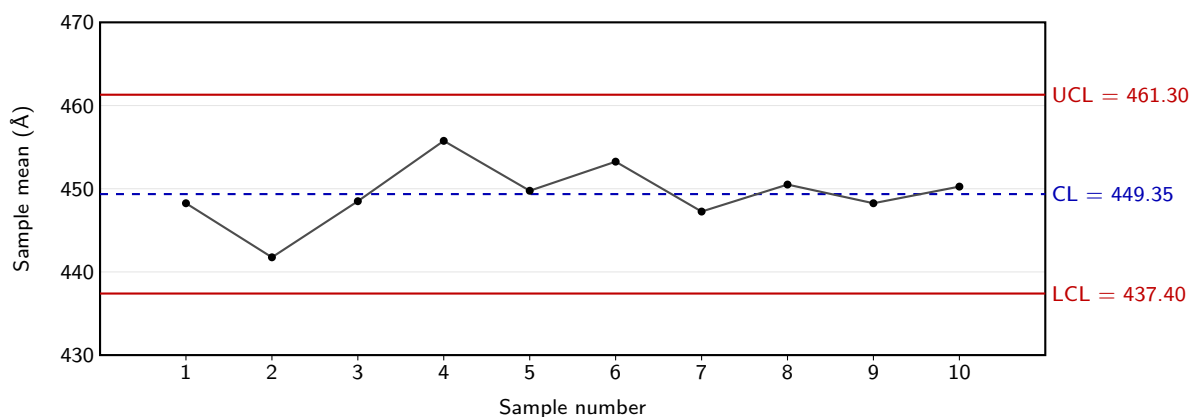


Figure 15: $\bar{x}$ chart for TiW layer thickness

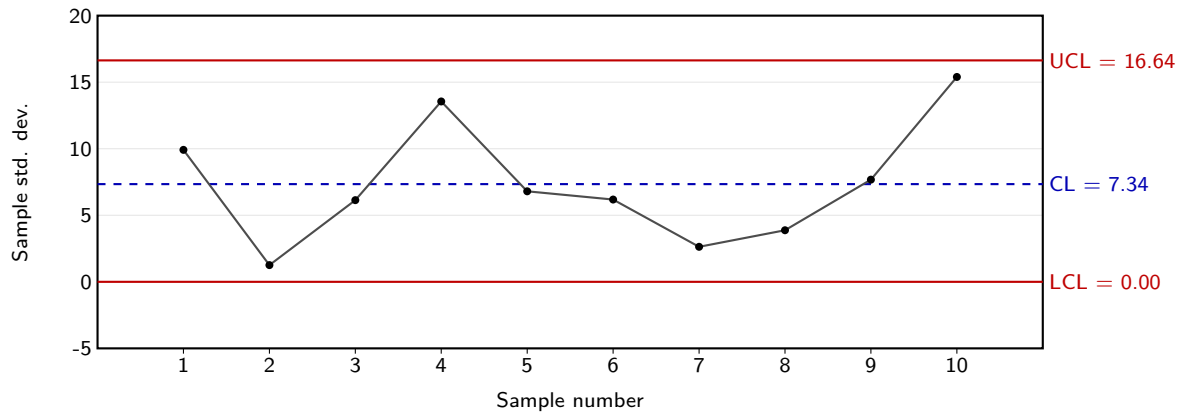


Figure 16: s chart for TiW layer thickness

All subgroup averages and standard deviations lie within the trial control limits and show no unusual pattern, so the sputtering process is in statistical control. The trial limits can be adopted for monitoring future production.

Answer: $\bar{x}$ chart: UCL = 461.30, CL = 449.35, LCL = 437.40 Å; s chart: UCL = 16.64, CL = 7.34, LCL = 0.

34	When do we use the moving range chart? Explain.	04	–	–
Source: Unit II SEE Nov/Dec 2019 (16IM53), Q4b				

The moving range chart is used together with the control chart for individual measurements when the **sample size is $n = 1$** , so that the variability cannot be estimated from within a subgroup. Typical situations:

- Automated inspection and measurement technology is used and every unit is analysed, so there is no basis for rational subgrouping.
- The production rate is very slow and it is inconvenient to wait for several units to form a subgroup (for example, one batch per day).
- Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
- Measurements on some parameter differ very little and give little useful information (for example, viscosity of paint at different points in a well-mixed batch).
- Accounting, financial and service data where only one value is available per period.

The moving range of two successive observations, $MR_i = |x_i - x_{i-1}|$, is used to estimate the variability: $\hat{\sigma} = \overline{MR}/d_2 = \overline{MR}/1.128$. Limits for the MR chart are $UCL = D_4 \overline{MR} = 3.267 \overline{MR}$ and $LCL = 0$; for the individuals chart $\bar{x} \pm 3\overline{MR}/1.128$.

35	Write a note on the standardized control chart. What is process capability? Explain any one of the techniques of process capability analysis.	10	2	CO1
Source: Unit II CIE-II, 07/01/2025				

Standardized control chart

A standardized control chart plots the statistic in standard deviation units, so the control limits are always at ± 3 and the center line at zero. For a p chart with variable sample size the standardized value is $Z_i = (\hat{p}_i - \bar{p})/\sqrt{\bar{p}(1-\bar{p})/n_i}$; for an $\bar{x}$ chart $Z_i = (\bar{x}_i - \mu)/(\sigma/\sqrt{n})$. It is useful when the sample size varies (limits do not change from sample to sample), in short production runs where different part numbers are plotted on one chart (deviation from nominal), and it makes the runs and pattern rules easy to apply.

Process capability

Process capability refers to the uniformity of the process: the variability in the critical-to-quality characteristics. It is measured by the natural tolerance limits $\mu \pm 3\sigma$ (the 6σ spread of the process) and compared with the specifications through capability ratios $C_p = (USL - LSL)/6\sigma$ and C_{pk} . A process capability analysis is an engineering study to estimate process capability.

Technique: capability analysis using a control chart

- (1) Collect 20 to 25 subgroups and construct $\bar{x}$ and R charts; bring the process into statistical control by removing assignable causes.
- (2) Estimate the process mean $\hat{\mu} = \bar{\bar{x}}$ and standard deviation $\hat{\sigma} = \bar{R}/d_2$.
- (3) Compute the natural tolerance limits $\hat{\mu} \pm 3\hat{\sigma}$ and the capability ratios C_p and C_{pk} .
- (4) Estimate the fraction nonconforming from the normal distribution: $P(x < LSL) + P(x > USL)$.

Other techniques are the histogram or probability plot (shape, centre and spread of the distribution) and designed experiments (to separate the sources of variability).

36	What is process capability? Explain the procedure of computing process capability using the average range method. Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q4b; SEE Mar/Apr 2023 (18IM53), Q3b (outline any one method of estimating process capability); SEE Nov/Dec 2019 (16IM53), Q3c (note on process capability study)	06	–	–
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Process capability is the uniformity of the process, that is, the inherent (natural) variability of the process when it operates in statistical control. It is usually expressed as the natural tolerance 6σ (natural tolerance limits $\mu \pm 3\sigma$) and compared with the specification width through $C_p = (USL - LSL)/6\sigma$ and C_{pk} .

Procedure using the average range method (control chart method)

- (1) Select the critical quality characteristic and a rational subgroup size n (usually 4 or 5).
- (2) Collect at least 20 to 25 subgroups from the process operating under normal conditions.
- (3) Compute $\bar{x}_i$ and R_i for every subgroup, then $\bar{\bar{x}}$ and $\bar{R}$.
- (4) Construct $\bar{x}$ and R charts with limits $\bar{\bar{x}} \pm A_2\bar{R}$, $D_3\bar{R}$, $D_4\bar{R}$; remove subgroups with assignable causes and revise until the process is in control.
- (5) Estimate the process standard deviation $\hat{\sigma} = \bar{R}/d_2$ (d_2 from the table for the subgroup size).
- (6) Process capability = $6\hat{\sigma}$; natural tolerance limits = $\bar{\bar{x}} \pm 3\hat{\sigma}$.
- (7) Compare with the specifications: $C_p = (USL - LSL)/6\hat{\sigma}$, $C_{pk} = \min(USL - \bar{\bar{x}}, \bar{\bar{x}} - LSL)/3\hat{\sigma}$, and estimate the percentage outside the specifications using the normal table.
- (8) Decide: if $C_p \geq 1.33$ the process is capable; if $C_p < 1$ reduce variability, change the process or widen the tolerance; if $C_{pk} < C_p$ re-centre the process.

Other methods are the histogram (or probability plot) method and designed experiments; the control chart method is preferred because it first establishes that the process is in control.

37	What are the process capability indices C_p and C_{pk} ? Explain their significance. Source: Unit II SEE Feb/Mar 2026 (IM353IA), Q4b	06	2	3
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Process capability ratio C_p : $C_p = (USL - LSL)/6\sigma$. It compares the specification width with the natural spread of the process and measures potential capability; it assumes the process mean is centered between the specifications. $1/C_p$ is the fraction of the specification band used by the process.

C_{pk} : $C_{pk} = \min(C_{pu}, C_{pl}) = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$. It takes process centering into account and measures actual capability (the one-sided ratio for the specification limit nearest to the mean).

Significance

Value	Meaning
$C_p < 1$	Process spread exceeds the tolerance; many nonconforming units; process not capable
$C_p = 1$	Spread equals the tolerance; 0.27% nonconforming (2700 ppm) if centered
$C_p \geq 1.33$	Capable; recommended minimum for existing processes (1.50 for new processes, higher for safety-critical parameters)
$C_{pk} = C_p$	Process is centered at the midpoint of the specifications
$C_{pk} < C_p$	Process is off-center; capability can be improved by re-centering
$C_{pk} < 0$	Process mean lies outside the specification limits

Together they tell management whether to reduce variability (low C_p) or to re-centre the process (low C_{pk} with adequate C_p); they are used in supplier selection, make-or-buy decisions and specification setting.

38	A stable, normally distributed process has the specification 3.5 ± 0.03 . Given $\mu = 3.51$ and standard deviation = 0.016, determine C_p and C_{pk} . What conclusions can you draw about the capability of the process? Source: Unit II SEE Jan/Feb 2021 (18IM53), Q4b	04	–	–
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$$LSL = 3.47, \quad USL = 3.53, \quad \mu = 3.51, \quad \sigma = 0.016$$

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{3.53 - 3.47}{6(0.016)} = \frac{0.06}{0.096} = 0.625$$

$$C_{pk} = \min \left(\frac{3.53 - 3.51}{3(0.016)}, \frac{3.51 - 3.47}{3(0.016)} \right) = \min(0.417, 0.833) = 0.417$$

Both indices are well below 1. The natural spread of the process ($6\sigma = 0.096$) is 1.6 times the tolerance (0.06), so the process is **not capable** even if centred; in addition the mean is 0.01 above the nominal 3.50, which lowers C_{pk} further. About 11.2% of the output will be outside the specifications (10.6% above the USL). The process variability must be reduced (or the process changed) and the mean re-centred at 3.50.

Answer: $C_p = 0.625$, $C_{pk} = 0.417$: the process is not capable.

39	What are the two most common types of control charts for attributes? Briefly explain their purpose. Source: Unit II SEE Feb/Mar 2025 (IM353IA), Q4a	06	2	1
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The two most common attribute control charts are the p chart (fraction nonconforming) and the c chart (number of nonconformities).

p chart (control chart for fraction nonconforming): plots the fraction of nonconforming units $\hat{p} = D/n$ in each sample, with limits $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$. It is used when each unit is classified as conforming or nonconforming (go/no-go), and can handle variable sample sizes. Purpose: monitor and reduce the proportion defective, e.g. the fraction of defective bushing screws in incoming lots.

c chart (control chart for nonconformities): plots the number of nonconformities (defects) in an inspection unit of constant size, with limits $\bar{c} \pm 3\sqrt{\bar{c}}$ (Poisson). Purpose: monitor the number of defects where a unit can have many defects, e.g. missing rivets on an aircraft or flaws in a roll of cloth.

Their variants are the np chart (number nonconforming, constant n) and the u chart (nonconformities per unit, variable inspection unit size).

40	What are the differences between (i) p chart and np chart, (ii) c chart and u chart? (Also distinguish p and u charts, and p and c charts.) Source: Unit II SEE Fast Track Oct 2020 (16IM53), Q4b; SEE Jan/Feb 2021 (16IM53), Q3b; CIE-II, 16/12/2025 (c and u); SEE Nov/Dec 2019 (16IM53), Q5b (p and c)	08	3	3
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(i) p chart versus np chart

Basis	p chart	np chart
Plots	Fraction nonconforming $\hat{p} = D/n$	Number nonconforming $D = n\hat{p}$
Sample size	Constant or variable	Constant only
Limits	$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$	$n\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$
Use	Comparing samples of different sizes; management reporting	Easier for operators: actual counts are plotted

(ii) c chart versus u chart

Basis	c chart	u chart
Plots	Total number of nonconformities c in an inspection unit	Average nonconformities per unit $u = c/n$
Inspection unit	Constant size (one unit or a fixed group of units)	Constant or variable number of units
Limits	$\bar{c} \pm 3\sqrt{\bar{c}}$	$\bar{u} \pm 3\sqrt{\bar{u}/n}$

Example	Missing rivets per aircraft; defects per roll of cloth	Defects per computer when the number of computers inspected varies
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p chart versus u chart, and p chart versus c chart

Basis	p chart	c chart / u chart
What is counted	Defective units (each unit either conforms or not)	Defects (nonconformities); one unit may have several
Distribution	Binomial	Poisson
Value per unit	At most 1 (0 or 1 per unit); $\hat{p} \leq 1$	Can exceed 1 per unit
Example	Proportion of rejected transformers	Number of surface flaws per transformer

41	Lots of bushing screws from a single supplier were inspected 100 percent with the following results. <table><tr><th>Lot No.</th><th>Amount in lot</th><th>Number rejected</th></tr><tr><td>1</td><td>6,000</td><td>183</td></tr><tr><td>2</td><td>6,000</td><td>131</td></tr><tr><td>3</td><td>10,000</td><td>184</td></tr><tr><td>4</td><td>1,783</td><td>23</td></tr><tr><td>5</td><td>3,089</td><td>171</td></tr><tr><td>6</td><td>5,774</td><td>65</td></tr><tr><td>7</td><td>3,000</td><td>46</td></tr><tr><td>8</td><td>2,724</td><td>85</td></tr></table> Draw an appropriate control chart for the data. What can you conclude on the state of control? Source: Unit II CIE-II, 07/01/2025; CIE-II, 16/12/2025; SEE Feb/Mar 2025 (IM353IA), Q3b	Lot No.	Amount in lot	Number rejected	1	6,000	183	2	6,000	131	3	10,000	184	4	1,783	23	5	3,089	171	6	5,774	65	7	3,000	46	8	2,724	85	10	3	3
Lot No.	Amount in lot	Number rejected																													
1	6,000	183																													
2	6,000	131																													
3	10,000	184																													
4	1,783	23																													
5	3,089	171																													
6	5,774	65																													
7	3,000	46																													
8	2,724	85																													

The lot sizes are different, so a **p chart with variable control limits** is appropriate: $\bar{p} = \sum D_i / \sum n_i$ and limits $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n_i}$ for each lot.

$$\bar{p} = \frac{\sum D_i}{\sum n_i} = \frac{888}{38,370} = 0.02314$$

$$\text{e.g. lot 1: } \sigma_{\hat{p}} = \sqrt{\frac{0.02314(1-0.02314)}{6000}} = 0.00194, \quad \text{UCL} = 0.0290, \quad \text{LCL} = 0.0173$$

Lot	n	Rejected	$\hat{p}$	$\sigma_{\hat{p}}$	LCL	UCL	Status
1	6,000	183	0.0305	0.00194	0.0173	0.0290	Above UCL
2	6,000	131	0.0218	0.00194	0.0173	0.0290	In control
3	10,000	184	0.0184	0.00150	0.0186	0.0277	Below LCL
4	1,783	23	0.0129	0.00356	0.0125	0.0338	In control
5	3,089	171	0.0554	0.00271	0.0150	0.0313	Above UCL
6	5,774	65	0.0113	0.00198	0.0172	0.0291	Below LCL
7	3,000	46	0.0153	0.00275	0.0149	0.0314	In control
8	2,724	85	0.0312	0.00288	0.0145	0.0318	In control

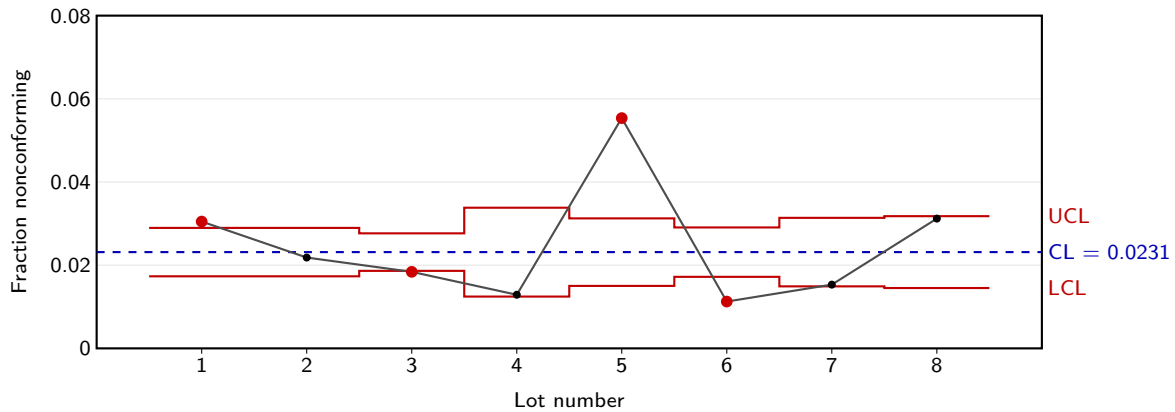


Figure 17: p chart with variable limits for bushing screws (CL = 0.0231)

Lots 1, 3, 5, 6 fall outside their control limits: lots 1 and 5 have far more rejects than expected, while lots 3 and 6 are below the LCL (better than the process average, possibly a genuine improvement or an inspection error). The supplier's process is **not in statistical control**; the quality of incoming lots varies far more than chance alone would explain. The supplier should investigate the assignable causes (especially for lot 5, with 5.5% rejects) before reliable limits can be set.

Answer: $\bar{p} = 0.0231$; lots 1, 3, 5, 6 are out of control, so the process is not in a state of statistical control.

42

Ten samples of 15 components are tested with the results below. Construct a p-chart and draw conclusions.

Sample	1	2	3	4	5	6	7	8	9	10
No. of defectives	2	1	3	2	3	1	0	1	0	1

Source: Unit II | SEE Jan/Feb 2021 (18IM53), Q4a

08

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$$\bar{p} = \frac{\sum D}{mn} = \frac{14}{10 \times 15} = 0.0933$$

$$\sigma_{\bar{p}} = \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} = \sqrt{\frac{0.0933(0.9067)}{15}} = 0.0751$$

$$UCL = 0.0933 + 3(0.0751) = 0.3187, \quad CL = 0.0933$$

$$LCL = 0.0933 - 3(0.0751) = -0.1320 < 0 \Rightarrow 0$$

Sample	1	2	3	4	5	6	7	8	9	10
$\hat{p} = D/15$	0.133	0.067	0.200	0.133	0.200	0.067	0.000	0.067	0.000	0.067

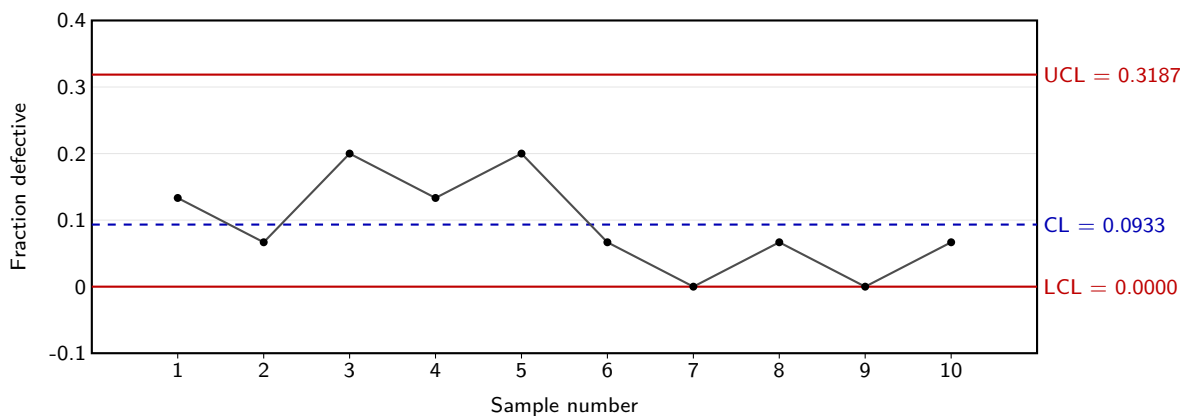


Figure 18: p chart for the components

The largest fraction defective is $3/15 = 0.200$, below the UCL of 0.3187; no point is below the LCL (zero) and there is no unusual pattern. The process is **in statistical control** at an average fraction defective of 0.0933 (9.3%). Note that with $n = 15$ the chart is not sensitive to small shifts; larger samples would improve it.

Answer: CL = 0.0933, UCL = 0.3187, LCL = 0; the process is in control.

43	In a manufacturing unit of certain transformers the units are required to meet a number of specifications and are subjected to inspection daily. At the end of twenty days 230 units have been rejected out of 4150 produced and inspected. (i) Determine the 3-sigma control limits based on an average daily production of 200 units. (ii) Only one point has crossed the control limits: on that day 30 non-conforming items were found in 200 units inspected. (iii) What is the aimed value of p? (iv) What control limits would you recommend for future use? Source: Unit II SEE Jan/Feb 2021 (16IM53), Q3a	10	–	–
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(i) Three-sigma limits with n = 200

$$\bar{p} = \frac{230}{4150} = 0.0554, \quad \sigma_{\bar{p}} = \sqrt{\frac{0.0554(1 - 0.0554)}{200}} = 0.0162$$

$$UCL = 0.0554 + 3(0.0162) = 0.1040, \quad LCL = 0.0554 - 3(0.0162) = 0.0069$$

(ii) The out-of-control day

On that day $\hat{p} = 30/200 = 0.150$, which exceeds the UCL of 0.1040, so the process was out of control on that day. Assuming the assignable cause is found and corrected, that day's data are removed.

(iii) Aimed (revised) value of p

$$p_{\text{aim}} = \frac{230 - 30}{4150 - 200} = \frac{200}{3950} = 0.0506$$

(iv) Control limits for future use

$$\sigma_{\bar{p}} = \sqrt{\frac{0.0506(1 - 0.0506)}{200}} = 0.0155$$

$$UCL = 0.0506 + 3(0.0155) = 0.0971, \quad CL = 0.0506, \quad LCL = 0.0506 - 3(0.0155) = 0.0041$$

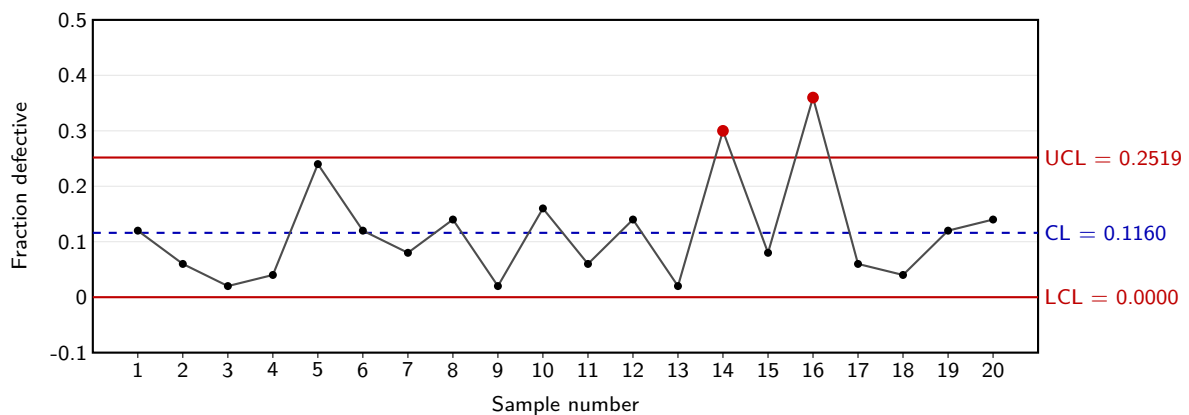
Answer: Trial limits 0.1040 / 0.0554 / 0.0069; revised (recommended) p = 0.0506 with UCL = 0.0971 and LCL = 0.0041 for a daily sample of 200.

44	The inspection results in a machine shop based on a sample size of 50 are given below. <table><tr><th>Serial No.</th><th>Defectives</th><th>Serial No.</th><th>Defectives</th></tr><tr><td>1</td><td>6</td><td>11</td><td>3</td></tr><tr><td>2</td><td>3</td><td>12</td><td>7</td></tr><tr><td>3</td><td>1</td><td>13</td><td>1</td></tr><tr><td>4</td><td>2</td><td>14</td><td>15</td></tr><tr><td>5</td><td>12</td><td>15</td><td>4</td></tr><tr><td>6</td><td>6</td><td>16</td><td>18</td></tr><tr><td>7</td><td>4</td><td>17</td><td>3</td></tr><tr><td>8</td><td>7</td><td>18</td><td>2</td></tr><tr><td>9</td><td>1</td><td>19</td><td>6</td></tr><tr><td>10</td><td>8</td><td>20</td><td>7</td></tr></table> (i) Calculate the control limits for the p-chart using 3σ limits. (ii) Plot the data and offer your comments on the behaviour of the process. (iii) What standard fraction defective do you recommend for the future period? Source: Unit II SEE Nov/Dec 2019 (16IM53), Q5a	Serial No.	Defectives	Serial No.	Defectives	1	6	11	3	2	3	12	7	3	1	13	1	4	2	14	15	5	12	15	4	6	6	16	18	7	4	17	3	8	7	18	2	9	1	19	6	10	8	20	7	12	—	—
Serial No.	Defectives	Serial No.	Defectives																																													
1	6	11	3																																													
2	3	12	7																																													
3	1	13	1																																													
4	2	14	15																																													
5	12	15	4																																													
6	6	16	18																																													
7	4	17	3																																													
8	7	18	2																																													
9	1	19	6																																													
10	8	20	7																																													

(i) Control limits

$$\bar{p} = \frac{116}{20 \times 50} = 0.1160, \quad \sigma_{\hat{p}} = \sqrt{\frac{0.1160(0.8840)}{50}} = 0.0453$$

$$UCL = 0.1160 + 3(0.0453) = 0.2519, \quad LCL = 0.1160 - 3(0.0453) < 0 \Rightarrow 0$$

(ii) Plot and comments**Figure 19:** p chart for the machine shop ($n = 50$)

Samples 14, 16 (fraction defective 0.30 and 0.36) are above the UCL, so the process is **not in statistical control**; sample 5 (0.24) is also close to the limit. These samples indicate assignable causes (for example, tool breakage, a defective material lot or a new operator) that must be investigated and removed.

(iii) Standard fraction defective for the future

$$m = 20 : \bar{p} = \frac{116}{1000} = 0.1160, \quad \sigma_{\hat{p}} = 0.0453, \quad UCL = 0.2519, \quad LCL = 0.0000 \quad (\text{out: 14, 16})$$

$$m = 18 : \bar{p} = \frac{83}{900} = 0.0922, \quad \sigma_{\hat{p}} = 0.0409, \quad UCL = 0.2150, \quad LCL = 0.0000 \quad (\text{out: 5})$$

$$m = 17 : \bar{p} = \frac{71}{850} = 0.0835, \quad \sigma_{\hat{p}} = 0.0391, \quad UCL = 0.2009, \quad LCL = 0.0000 \quad (\text{out: none})$$

Answer: Trial limits: CL = 0.1160, UCL = 0.2519, LCL = 0. After removing samples 5, 14, 16, the recommended standard fraction defective is $p = 0.0835$, with UCL = 0.2009 and LCL = 0.

45	A process is being controlled with a fraction nonconforming control chart. The process average has been shown to be 0.07. Three-sigma control limits are used and the procedure calls for taking daily samples of 400 items. (i) Calculate the upper and lower control limits. (ii) If the process average should suddenly shift to 0.10, what is the probability that the shift would be detected on the first subsequent sample? (iii) What is the probability that the shift in part (ii) would be detected on the first or second sample taken after the shift? Source: Unit II SEE Fast Track Oct 2020 (16IM53), Q4a; SEE Mar/Apr 2023 (18IM53), Q4a (parts i and ii)	08	–	–
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(i) Control limits

$$\sigma_{\hat{p}} = \sqrt{\frac{0.07(0.93)}{400}} = 0.01276$$

$$UCL = 0.07 + 3(0.01276) = 0.1083, \quad CL = 0.07, \quad LCL = 0.07 - 3(0.01276) = 0.0317$$

(ii) Probability of detecting a shift to $p = 0.10$ on the first sample

$$\text{With } p = 0.10 : \sigma_{\hat{p}} = \sqrt{\frac{0.10(0.90)}{400}} = 0.015$$

$$P(\text{detect}) = P(\hat{p} > UCL) + P(\hat{p} < LCL) = 1 - \Phi\left(\frac{0.1083 - 0.10}{0.015}\right) + \Phi\left(\frac{0.0317 - 0.10}{0.015}\right)$$

$$= 1 - \Phi(0.551) + \Phi(-4.55) = 0.2907$$

(iii) Detection on the first or second sample

$$P = 1 - (1 - 0.2907)^2 = 1 - (0.7093)^2 = 0.4968$$

Answer: UCL = 0.1083, LCL = 0.0317; P(detect on first sample) = 0.2907; P(detect by the second sample) = 0.4968.

46	An item is made in large lots and a sample of 200 items is drawn and given 100% inspection. The record sheet for the first 25 samples showed that a total of 75 items were defective. (i) Determine the trial control limits for the np-chart. (ii) Assuming that all the points fall within the control limits, what is the process average fraction defective? (iii) If p remains constant, what is the probability that the 26th sample will contain exactly 7 defectives? (iv) What is the probability that the 26th sample will contain 7 or more defectives? Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q4a	10	–	–
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(i) Trial control limits for the np chart

$$\bar{p} = \frac{75}{25 \times 200} = 0.015, \quad n\bar{p} = 200(0.015) = 3.0$$

$$\sigma = \sqrt{n\bar{p}(1 - \bar{p})} = \sqrt{3.0(0.985)} = 1.7190$$

$$UCL = 3.0 + 3(1.7190) = 8.157, \quad CL = 3.0, \quad LCL = 3.0 - 3(1.7190) < 0 \Rightarrow 0$$

(ii) Process average fraction defective

$$\bar{p} = 75/5000 = 0.015, \text{ i.e. } 1.5\% \text{ defective.}$$

(iii) Probability of exactly 7 defectives in the 26th sample (binomial)

$$P(D = 7) = \binom{200}{7}(0.015)^7(0.985)^{193} = 0.0211$$

(iv) Probability of 7 or more defectives

$$P(D \geq 7) = 1 - \sum_{d=0}^6 \binom{200}{d}(0.015)^d(0.985)^{200-d} = 0.0324$$

Answer: np chart: UCL = 8.157, CL = 3, LCL = 0; $\bar{p} = 0.015$; $P(D = 7) = 0.0211$; $P(D \geq 7) = 0.0324$.

47	A factory inspects different samples of a product, recording the number of defects in each sample. The sample size is constant, and the number of defects is provided below: <table><tr><td>Sample No.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>No. of defects</td><td>5</td><td>8</td><td>6</td><td>4</td><td>7</td><td>9</td><td>5</td><td>6</td></tr></table> (i) Draw an appropriate control chart for the data and evaluate the process. (ii) Plot the centreline, upper control limit (UCL) and lower control limit (LCL). Source: Unit II CIE-II, 07/01/2025	Sample No.	1	2	3	4	5	6	7	8	No. of defects	5	8	6	4	7	9	5	6	10	2	CO2
Sample No.	1	2	3	4	5	6	7	8														
No. of defects	5	8	6	4	7	9	5	6														

The number of defects is counted in samples of constant size, so a **c chart** (Poisson) is appropriate.

$$\bar{c} = \frac{\sum c}{m} = \frac{50}{8} = 6.25$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 6.25 + 3\sqrt{6.25} = 13.75$$

$$CL = 6.25, \quad LCL = \bar{c} - 3\sqrt{\bar{c}} = -1.25 < 0 \Rightarrow 0$$

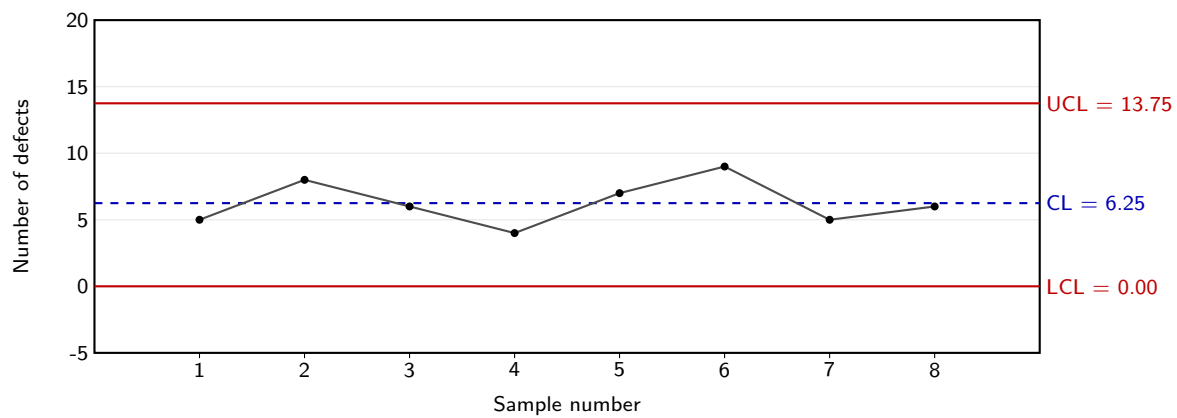


Figure 20: c chart for the number of defects

All eight points (4 to 9 defects) lie between the LCL (0) and the UCL (13.75) and show no run or trend. The process is **in statistical control** with an average of 6.25 defects per sample; any further reduction requires a change to the process itself (common causes).

Answer: c chart: CL = 6.25, UCL = 13.75, LCL = 0; the process is in control.

48	The following table gives the number of missing rivets at the aircraft final inspection: <table><tr><td>Sl No</td><td>Aircraft No.</td><td>Missing rivets</td><td>Sl No</td><td>Aircraft No.</td><td>Missing rivets</td></tr><tr><td>1</td><td>202</td><td>07</td><td>6</td><td>207</td><td>11</td></tr><tr><td>2</td><td>203</td><td>12</td><td>7</td><td>208</td><td>06</td></tr><tr><td>3</td><td>204</td><td>13</td><td>8</td><td>209</td><td>19</td></tr><tr><td>4</td><td>205</td><td>18</td><td>9</td><td>210</td><td>04</td></tr><tr><td>5</td><td>206</td><td>17</td><td>10</td><td>211</td><td>26</td></tr></table> Establish an appropriate control chart for the purpose. Source: Unit II CIE-II, 16/12/2025	Sl No	Aircraft No.	Missing rivets	Sl No	Aircraft No.	Missing rivets	1	202	07	6	207	11	2	203	12	7	208	06	3	204	13	8	209	19	4	205	18	9	210	04	5	206	17	10	211	26	10	4	4
Sl No	Aircraft No.	Missing rivets	Sl No	Aircraft No.	Missing rivets																																			
1	202	07	6	207	11																																			
2	203	12	7	208	06																																			
3	204	13	8	209	19																																			
4	205	18	9	210	04																																			
5	206	17	10	211	26																																			

Each aircraft is one inspection unit and the number of missing rivets (nonconformities) is counted, so a **c chart** is appropriate.

$$\bar{c} = \frac{133}{10} = 13.30, \quad \sqrt{\bar{c}} = 3.647$$

$$UCL = 13.30 + 3(3.647) = 24.24, \quad CL = 13.30, \quad LCL = 13.30 - 3(3.647) = 2.36$$

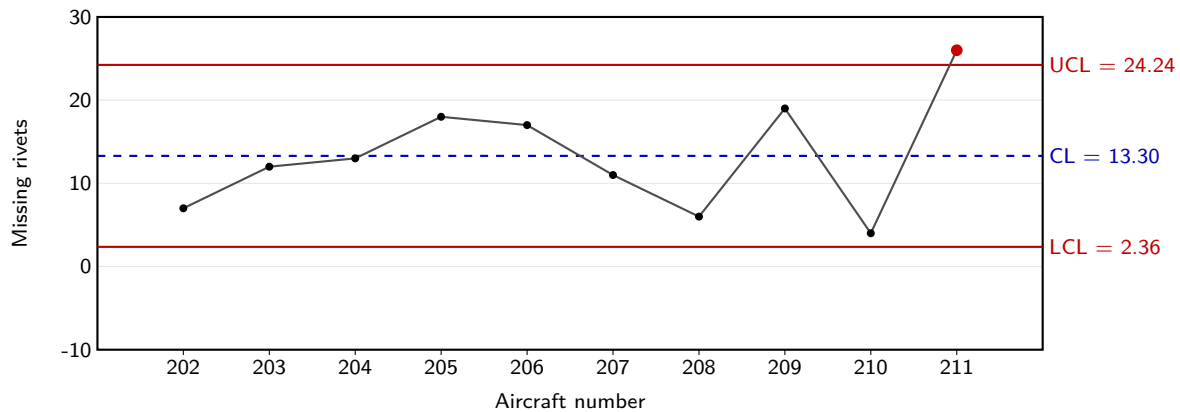


Figure 21: c chart for missing rivets

Aircraft 211, with 26 missing rivets, is above the UCL of 24.24; all other points are within the limits. Aircraft 211 indicates an assignable cause (for example, a new or untrained riveter, a faulty rivet gun or a supply problem) that must be investigated.

Revised limits for future use (aircraft 211 excluded)

$$\bar{c} = \frac{133 - 26}{9} = 11.89$$

$$UCL = 11.89 + 3\sqrt{11.89} = 22.23, \quad LCL = 11.89 - 3\sqrt{11.89} = 1.54$$

Answer: Trial c chart: CL = 13.30, UCL = 24.24, LCL = 2.36 (aircraft 211 out of control). Recommended chart: CL = 11.89, UCL = 22.23, LCL = 1.54.

49	A control chart for nonconformities is maintained on a process producing desk calculators. The inspection unit is defined as two calculators. The average number of nonconformities per machine when the process is in control is estimated to be two. (i) Find the appropriate three-sigma control limits for this size of inspection unit. (ii) What is the probability of type I error for this control chart? Source: Unit II SEE Mar/Apr 2023 (18IM53), Q4b	08	–	–
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(i) Control limits

The average number of nonconformities is 2 per calculator; the inspection unit is two calculators, so $\bar{c} = 2 \times 2 = 4$ per inspection unit.

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 4 + 3(2) = 10, \quad CL = 4, \quad LCL = 4 - 3(2) = -2 \Rightarrow 0$$

(ii) Probability of type I error

$$\begin{aligned} \alpha &= P(c > 10 \mid \lambda = 4) + P(c < 0) = 1 - P(c \leq 10 \mid \lambda = 4) \\ &= 1 - \sum_{x=0}^{10} \frac{e^{-4} 4^x}{x!} = 1 - 0.99716 = 0.00284 \end{aligned}$$

Answer: UCL = 10, CL = 4, LCL = 0 per inspection unit of two calculators; type I error $\alpha = 0.0028$.

50	A cloth manufacturing company is maintaining 100% inspection of all the finished fabric. The past 8 months' data from the inspection indicated that the average number of defects per unit of fabric is 8. (i) Determine the 3σ control limits. (ii) With these control limits, what is the probability that the process is judged out of control when it really is not? Source: Unit II SEE Fast Track Jul 2019 (16IM53), Q3b	04	–	–
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(i) Three-sigma limits (c chart)

$$\bar{c} = 8, \quad UCL = 8 + 3\sqrt{8} = 16.49, \quad CL = 8, \quad LCL = 8 - 3\sqrt{8} = -0.49 < 0 \Rightarrow 0$$

(ii) Probability of a false alarm (type I error)

$$\alpha = P(c > 16.49 \mid \lambda = 8) = 1 - P(c \leq 16) = 1 - \sum_{x=0}^{16} \frac{e^{-8} 8^x}{x!} = 1 - 0.99628 = 0.00372$$

Answer: UCL = 16.49, CL = 8, LCL = 0; probability of concluding out of control when in control = 0.0037.

51	Briefly outline the basic principles of the cusum control chart for monitoring the process mean. Source: Unit III SEE Mar/Apr 2023 (18IM53), Q5a	04	–	–
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A Shewhart chart uses only the last sample and is insensitive to small shifts (about 1.5σ or less). The **cumulative sum (cusum) chart** plots the cumulative sums of the deviations of the sample values from a target value μ_0 :

$$C_i = \sum_{j=1}^i (\bar{x}_j - \mu_0)$$

- If the process stays on target, C_i is a random walk with mean zero, so it hovers near zero. If the mean shifts upward, a positive drift develops in C_i , and if it shifts downward, a negative drift develops; a significant trend is evidence that the mean has shifted.
- Because several samples are combined, the cusum is more effective than the Shewhart chart for small shifts and is especially effective with samples of size $n = 1$.
- **Tabular cusum:** $C_i^+ = \max[0, x_i - (\mu_0 + K) + C_{i-1}^+]$ and $C_i^- = \max[0, (\mu_0 - K) - x_i + C_{i-1}^-]$. K is the reference value

(allowance), usually half the shift to be detected, $K = k\sigma$ with $k = \frac{1}{2}$; $H = h\sigma$ is the decision interval ($h = 4$ or 5). The process is out of control when C^+ or C^- exceeds H .

- The counters N^+ and N^- (number of periods since the cusum was last zero) show when the shift probably began.

52	Give an example of any three situations in which the sample size used for process monitoring is $n = 1$, that is, the sample consists of an individual unit. Source: Unit III SEE Mar/Apr 2023 (18IM53), Q6a	04	–	–
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The chart for individual measurements (individuals chart with the moving range) is used when the sample size is $n = 1$, in situations such as:

- (1) **Automated inspection:** every unit produced is measured automatically (for example, 100% measurement of each part at a machine), so there is no rational subgroup.
- (2) **Slow production or costly testing:** the production rate is so low, or the measurement so expensive or destructive, that it is inconvenient to accumulate a sample of size greater than one (for example, large castings, a batch of chemical product).
- (3) **Process measurements that are repeated reading the same quantity:** such as the thickness or resistivity of a single wafer, the viscosity of a batch, or temperature, where repeat measurements differ only because of laboratory or analysis error.
- (4) **Data available only periodically:** accounting data and similar measurements made once a week or once a month.

53	Explain the meaning of the following terms and their importance: (i) producer's risk, (ii) consumer's risk, (iii) AQL, (iv) LTPD (LQL). Differentiate between producer's risk and consumer's risk, and AQL and LQL. Source: Unit III CIE-II, 07/01/2025, Q5; SEE Feb/Mar 2026 (IM353IA), Q5a	10	3	3
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A good sampling plan must protect both the supplier and the consumer. The OC curve is therefore designed to pass through two specified points: a point $(p_1, 1 - \alpha)$ for good lots and a point (p_2, β) for bad lots.

Term	Meaning	Importance
AQL (acceptable quality level)	The poorest level of quality for the supplier's process that the consumer would consider acceptable as a process average. It is a property of the supplier's process, not of the sampling plan, and it is neither a specification nor a target.	The consumer designs the sampling procedure so that the OC curve gives a high probability of acceptance at the AQL.
LTPD (lot tolerance percent defective), also RQL or LQL	The poorest level of quality that the consumer is willing to accept in an individual lot. It is a level of lot quality specified by the consumer, not a characteristic of the plan.	Gives the protection obtained for individual lots of poor quality: the plan is designed to give a low probability of acceptance at the LTPD.
Producer's risk (α)	The probability of rejecting a lot of AQL quality (p_1). On the OC curve the point (AQL, $1 - \alpha$) is called the producer's risk point.	Measures the risk the supplier runs of having good lots rejected (typically $\alpha = 0.05$).
Consumer's risk (β)	The probability of accepting a lot of LTPD quality (p_2). The point (LTPD, β) is the consumer's risk point.	Measures the risk the consumer runs of accepting bad lots (typically $\beta = 0.10$).

Differences: the producer's risk concerns the rejection of good lots and is read at the AQL (the supplier's concern); the consumer's risk concerns the acceptance of bad lots and is read at the LTPD (the consumer's concern). The AQL is the best process average that is acceptable, whereas the LTPD is the worst quality accepted in a single lot, so $AQL < LTPD$.

Example: for the plan $n = 89$, $c = 2$ the probability of acceptance is 0.940 at $p = 1\%$ (so $\alpha \approx 0.060$) and 0.092 at $p = 6\%$ ($\beta \approx 0.092$).

54	With the help of an OC curve, define the terms producer's risk and consumer's risk in an acceptance sampling plan.	06	2	1
Source: Unit III SEE Feb/Mar 2025 (IM353IA), Q5a				

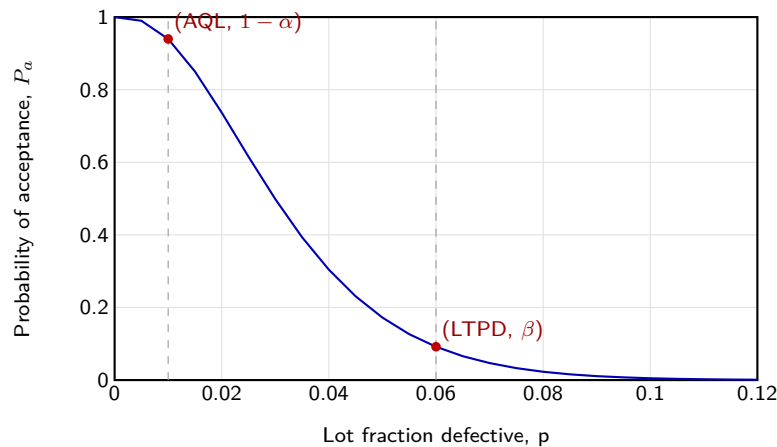


Figure 22: OC curve of the plan $n = 89$, $c = 2$ showing the producer's risk point and the consumer's risk point

- **Producer's risk (α):** the probability that a good lot, with fraction defective equal to the AQL (p_1), is rejected. On the OC curve it corresponds to the point $(p_1, 1 - \alpha)$. For $n = 89$, $c = 2$ at $p_1 = 0.01$, $P_a = 0.940$, so $\alpha = 0.060$.
- **Consumer's risk (β):** the probability that a bad lot, with fraction defective equal to the LTPD (p_2), is accepted. It is the point (p_2, β) of the OC curve. For the same plan at $p_2 = 0.06$, $\beta = P_a = 0.092$.

55	Explain the following indices used in acceptance sampling plans: (i) Acceptable Quality Level, (ii) Rejectable Quality Level.	06	3	3
Source: Unit III SEE Feb/Mar 2026 (IM353IA), Q5b				

(i) Acceptable quality level (AQL): the poorest level of quality for the supplier's process that the consumer would consider acceptable as a process average. It is a property of the supplier's manufacturing process, not of the sampling plan. It is not usually a specification of the product or a target for the supplier; it is a standard against which lots are judged. The consumer usually designs the plan so that the OC curve gives a high probability of acceptance ($1 - \alpha$, typically 0.95) at the AQL; MIL STD 105E is indexed by the AQL.

(ii) Rejectable quality level (RQL): also called the lot tolerance percent defective (LTPD) or the limiting quality level (LQL): the poorest level of quality that the consumer is willing to accept in an individual lot. It is specified by the consumer and is not a characteristic of the plan. The plan is designed to give a low probability of acceptance (the consumer's risk β , typically 0.10) at the RQL, which protects the consumer from individual lots of poor quality.

56	What are the characteristics of a good acceptance sampling plan?	06	3	2
Source: Unit III SEE Feb/Mar 2026 (IM353IA), Q6a				

From the design principles of acceptance sampling, a good plan has these characteristics:

- (1) **It discriminates between good and bad lots:** the OC curve gives a high probability of acceptance at the AQL (low producer's risk α) and a low probability of acceptance at the LTPD (low consumer's risk β); a steeper OC curve (larger sample size) approaches the ideal OC curve.
- (2) **It is specified by two points on the OC curve** $(p_1, 1 - \alpha)$ and (p_2, β) , because one point is not enough to fix n and c .
- (3) **It does not rely on a zero acceptance number:** plans with $c = 0$ reject many lots of acceptable quality; for example, $n = 89$, $c = 0$ accepts only about 41% of 1% defective lots against about 78% for $n = 89$, $c = 1$.
- (4) **The protection does not vary with lot size:** sampling a fixed percentage of the lot gives different protection for different lot sizes, which is illogical.
- (5) **It is based on random samples from homogeneous lots,** formed in a way convenient to handling, so that the sample represents the lot.
- (6) **It is economical:** a small average inspection (ASN or ATI) for the protection it gives; double or multiple plans can

reduce the average inspection, but they are administratively more complex.

57	Compare single, double and multiple sampling plans. Source: Unit III SEE Feb/Mar 2025 (IM353IA), Q6a; SEE Jan/Feb 2021 (18IM53), Q6b (single, double and triple)	06	2	1
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Basis	Single sampling	Double sampling	Multiple sampling
Procedure	One sample of n units; accept if $d \leq c$, reject if $d > c$.	First sample n_1 : accept, reject or take a second sample n_2 ; the combined result decides.	More than two samples; at each stage accept, reject or continue; a decision at the last stage. Sequential sampling takes one unit at a time.
Parameters	n, c	$n_1, c_1, n_2, c_2 (r_1, r_2)$	n and acceptance and rejection numbers for each stage
Sample size	Fixed and the largest	Smaller first sample; average sample number varies with lot quality	Samples at each stage are smaller than in single or double plans
Average inspection	Constant n	Often less than single (very good or very bad lots decided on the first sample); may be more for intermediate quality unless curtailed	Lowest on average
Administration	Simplest	More complex; storing material awaiting a second sample	Most complex
Protection	All can be designed to have equivalent OC curves, i.e. the same risks for a lot of given quality.		

The type of procedure is chosen on administrative efficiency, the information produced, the average amount of inspection and the effect on material flow.

58	What assumptions do you make while designing a single sampling plan? Source: Unit III SEE Jan/Feb 2021 (18IM53), Q6a	05	–	–
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The single-sampling plan (n, c) and its OC curve are developed with the following assumptions:

- (1) **Random sampling:** the n units are selected at random from the lot (for example with random numbers on the serial numbers), so that the sample is representative; otherwise the statistical basis of the plan is lost.
- (2) **Homogeneous lots:** the lots are formed from units produced by the same machines, operators and raw materials at about the same time.
- (3) **Large lot or stream of lots (type-B OC curve):** the lot size N is large (theoretically infinite) compared with n , or the lots come from a continuing process, so the number of defectives in the sample is binomial with parameters n and p ; for an isolated small lot the hypergeometric distribution (type A) is used.
- (4) **Each unit is classified as conforming or defective** (attribute inspection) and the classification is free of inspection error.
- (5) **For rectifying inspection:** rejected lots are 100% inspected and all defectives found, in the sample or in screening, are replaced by good units.
- (6) **The plan is fixed by two points of the OC curve** ($p_1, 1 - \alpha$) and (p_2, β), with the sample size not tied to a fixed percentage of the lot.

59	How do the type-A and type-B OC curves differ? When are they used?	06	—	—
Source: Unit III SEE Jan/Feb 2021 (18IM53), Q6c; SEE Jan/Feb 2021 (16IM53), Q6b				

Basis	Type-A OC curve	Type-B OC curve
Used for	An isolated lot of finite size N.	A continuing stream of lots from a process, or a large lot (sampling from the process).
Distribution	Hypergeometric: $P(d \text{ defectives}) = \frac{C(D, d) C(N-D, n-d)}{C(N, n)}$, $D = Np$.	Binomial with parameters n and p.
Position	Always lies below the type-B curve.	Probabilities of acceptance are always higher than the type-A values.
Difference	Depends on N, n and c.	Depends on n and c only; it is the exact curve for large lots.

When the lot size is at least 10 times the sample size ($n/N \leq 0.10$) the two curves are virtually indistinguishable, and the type-B curve is used as an approximation. Example ($n = 50$, $c = 1$): at $p = 4\%$ the type-A curve for $N = 200$ gives $P_a = 0.362$ against 0.400 from the type-B curve; for $N = 2000$ the type-A curve coincides with the type-B curve.

60	Sketch the ideal operating characteristic curve of a sampling plan. Explain its significance.	03	—	—
Source: Unit III SEE Jan/Feb 2021 (16IM53), Q5b				

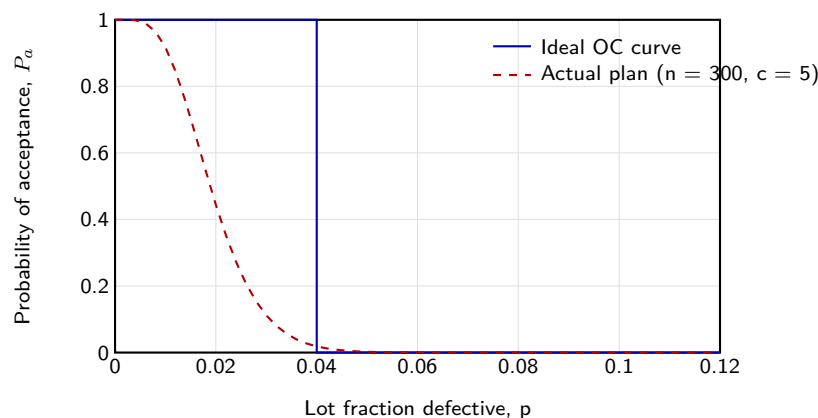


Figure 23: Ideal OC curve compared with the OC curve of an actual plan

The **ideal OC curve** stays horizontal at $P_a = 1.00$ until the quality level that is considered “bad” is reached, at which the curve drops vertically to $P_a = 0.00$ and then runs horizontally for all worse quality. A plan with this curve would accept all lots of good quality and reject all lots of bad quality, with no risk to either party.

Significance: it is the standard against which real plans are judged. It can be realized only by error-free 100% inspection and almost never in practice, but it can be approached by increasing the sample size; the greater the slope of the actual OC curve, the greater the discriminatory power of the plan.

61	Explain the difference between average sample number (ASN) and average total inspection (ATI). State any assumptions made.	03	—	—
Source: Unit III SEE Jan/Feb 2021 (16IM53), Q5c				

Basis	Average sample number (ASN)	Average total inspection (ATI)
Meaning	The average number of units inspected per lot in sampling to reach a decision (sample size depends on the decision stage).	The average number of units inspected per lot, including the 100% inspection (screening) of the rejected lots.
Applies to	Mainly double, multiple and sequential plans (for single sampling $ASN = n$).	Rectifying inspection plans, where the rejected lots are screened.

Formula	Double sampling (complete inspection): $ASN = n_1 + n_2(1 - P_I)$, P_I = probability of a decision on the first sample.	Single: $ATI = n + (1 - P_a)(N - n)$. Double: $ATI = n_1 P_{-a}^1 + (n_1 + n_2) P_{-a}^2 + N(1 - P_a)$.
Varies between	n_1 and $n_1 + n_2$	n (all lots accepted) and N (all lots rejected)

Assumptions: the second sample is inspected completely (no curtailment); rejected lots are 100% inspected and all defective items found, in the sample or in screening, are replaced by good ones; the sampling is random from lots of N units.

62	Write the flowchart for a double sampling plan and explain it with an example.	06	—	—
Source: Unit III SEE Nov/Dec 2019 (16IM53), Q6b				

A double-sampling plan is defined by four parameters: n_1 (first sample size), c_1 (acceptance number of the first sample), n_2 (second sample size) and c_2 (acceptance number for both samples combined).

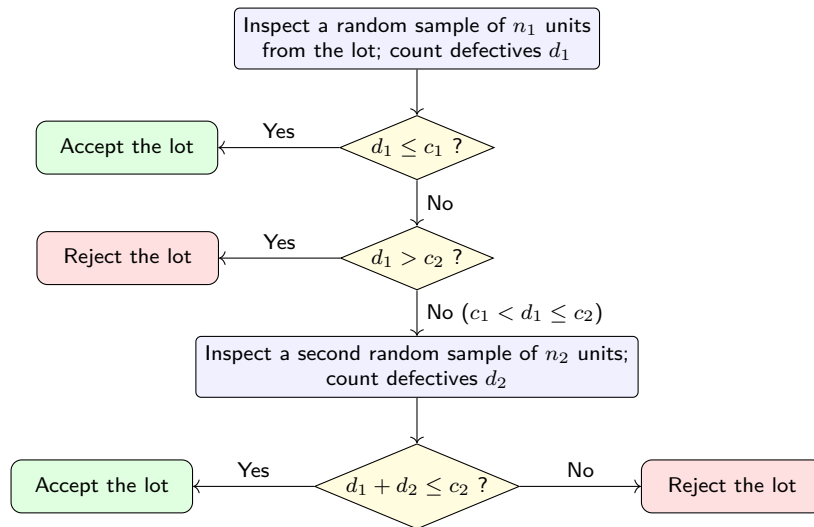


Figure 24: Flowchart of a double sampling plan

Example: $n_1 = 50$, $c_1 = 1$, $n_2 = 100$, $c_2 = 3$. A random sample of 50 items is inspected and d_1 defectives found. If $d_1 \leq 1$ the lot is accepted; if $d_1 > 3$ the lot is rejected; if $d_1 = 2$ or 3 a second sample of 100 items is drawn, and the lot is accepted if $d_1 + d_2 \leq 3$ and rejected otherwise. For incoming lots with $p = 0.05$ the probability of acceptance on the first sample is 0.2794, on the second sample 0.0110, so $P_a = 0.2904$; the probability of rejection on the first sample is 0.2396.

The principal advantage over single sampling is that the total inspection is often reduced, since many lots are decided on the first (smaller) sample, and the supplier gets a “second chance”; the disadvantages are the more complex administration and the possibility of more inspection than single sampling for lots of intermediate quality unless the second sample is curtailed.

63	Draw the OC curve, AOQ curve, ASN curve and ATI curve for the single sampling plan with $N = 5000$, $n = 100$ and $c = 2$.	10	3	2
Source: Unit III CIE-II, 16/12/2025, Q4; SEE Feb/Mar 2025 (IM353IA), Q5b				

The lots are assumed to come from a large process stream, so the number of defectives in the sample is binomial (type-B OC curve). For the single plan with rectifying inspection (rejected lots are 100% inspected and the defectives are replaced):

$$P_a = \sum_{d=0}^c \binom{n}{d} p^d (1-p)^{n-d}, \quad AOQ = \frac{P_a p (N-n)}{N}, \quad ATI = n + (1 - P_a)(N - n), \quad ASN = n$$

$$\text{At } p = 0.02 : P_a = 0.6767, \quad AOQ = \frac{0.6767(0.02)(4900)}{5000} = 0.01326$$

$$ATI = 100 + (1 - 0.6767)(4900) = 1684.2$$

p	P_a	AOQ	ASN	ATI
0.000	1.0000	0.00000	100	100.0
0.005	0.9859	0.00483	100	169.1
0.010	0.9206	0.00902	100	488.9
0.015	0.8098	0.01190	100	1032.0
0.020	0.6767	0.01326	100	1684.2
0.025	0.5422	0.01328	100	2343.3
0.030	0.4198	0.01234	100	2943.1
0.035	0.3159	0.01084	100	3452.1
0.040	0.2321	0.00910	100	3862.5
0.045	0.1672	0.00737	100	4180.9
0.050	0.1183	0.00579	100	4420.5
0.055	0.0824	0.00444	100	4596.3
0.060	0.0566	0.00333	100	4722.6
0.065	0.0384	0.00245	100	4811.7
0.070	0.0258	0.00177	100	4873.6
0.075	0.0171	0.00126	100	4916.1
0.080	0.0113	0.00088	100	4944.8
0.085	0.0074	0.00061	100	4964.0
0.090	0.0048	0.00042	100	4976.7
0.095	0.0031	0.00028	100	4985.0
0.100	0.0019	0.00019	100	4990.5

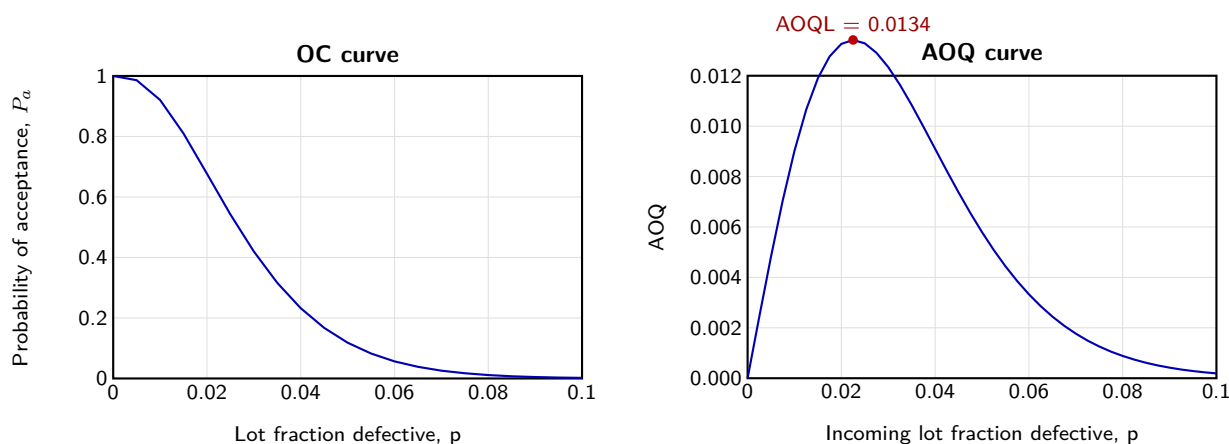


Figure 25: OC curve (left) and AOQ curve (right) of the plan $N = 5000$, $n = 100$, $c = 2$

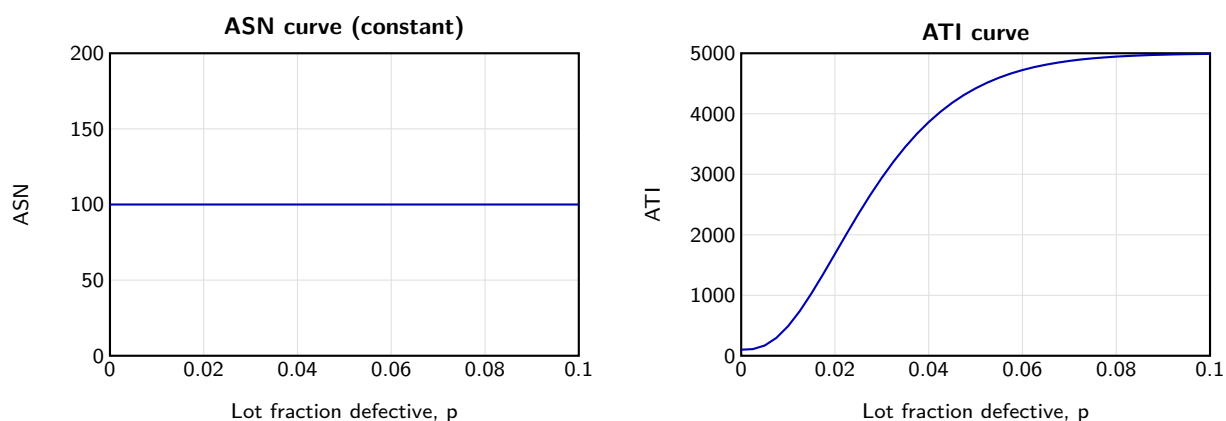


Figure 26: ASN curve (left) and ATI curve (right) of the same plan

The OC curve falls from 1.00 at $p = 0$ to about 0.118 at $p = 5\%$. The AOQ curve rises to a maximum, the **AOQL = 0.0134** at $p \approx 0.0225$, and then falls because more lots are rejected and screened. For a single plan the ASN is constant at $n = 100$; the ATI rises from $n = 100$ for very good lots towards $N = 5000$ for very bad lots.

64	Draw the type-A OC curve for the single sampling plan: $n = 50$, $C = 1$, $N = 200$.	10	2	3
Source: Unit III CIE-II, 07/01/2025, Q1 (Part B)				

The type-A OC curve applies to an isolated lot of finite size N , so the number of defectives in the sample is **hypergeometric**. If the lot contains $D = Np$ defective items:

$$P_a = P(d \leq c) = \sum_{d=0}^c \frac{\binom{D}{d} \binom{N-D}{n-d}}{\binom{N}{n}}$$

$$\text{At } p = 0.05 : D = 200(0.05) = 10, \quad P_a = \frac{\binom{10}{0} \binom{190}{50}}{\binom{200}{50}} + \frac{\binom{10}{1} \binom{190}{49}}{\binom{200}{50}} = 0.2368$$

Lot fraction defective p	Defectives in the lot $D = Np$	Type-A P_a (hypergeometric)	Type-B P_a (binomial)
0.00	0	1.0000	1.0000
0.01	2	0.9384	0.9106
0.02	4	0.7394	0.7358
0.03	6	0.5324	0.5553
0.04	8	0.3623	0.4005
0.05	10	0.2368	0.2794
0.06	12	0.1501	0.1900
0.08	16	0.0561	0.0827
0.10	20	0.0194	0.0338
0.12	24	0.0063	0.0131
0.15	30	0.0011	0.0029
0.20	40	0.0000	0.0002

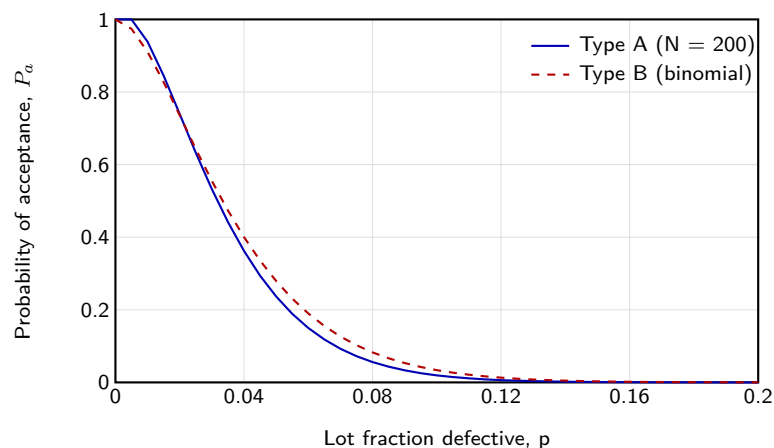


Figure 27: Type-A OC curve for $n = 50$, $c = 1$, $N = 200$ (type-B curve shown for comparison)

Because the sample is a quarter of the lot ($n/N = 0.25 > 0.10$), the type-A curve lies noticeably below the type-B curve; for lots at least ten times the sample size the two curves are practically the same.

65	Suppose a consumer receives lots of 500 candles from a new supplier. To check the quality of the lot, the consumer draws one sample of size 20 and accepts the lot if the inspected sample contains at most one defective candle. Otherwise he/she rejects the lot. Construct the OC curve for this plan. Source: Unit III SEE Feb/Mar 2025 (IM353IA), Q6b	10	3	2
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The plan is $n = 20$, $c = 1$ for an isolated lot of $N = 500$, so the exact (type-A) OC curve uses the hypergeometric distribution with $D = 500p$ defectives in the lot. (Since $n/N = 0.04$, the binomial values are almost identical.)

$$P_a = P(d \leq 1) = \frac{\binom{D}{0} \binom{500-D}{20} + \binom{D}{1} \binom{500-D}{19}}{\binom{500}{20}}$$

Lot fraction defective p	Defective candles $D = 500p$	P_a (type A, hypergeometric)	P_a (type B, binomial)
0.00	0	1.0000	1.0000
0.01	5	0.9858	0.9831
0.02	10	0.9435	0.9401
0.04	20	0.8122	0.8103
0.06	30	0.6594	0.6605
0.08	40	0.5133	0.5169
0.10	50	0.3865	0.3917
0.15	75	0.1698	0.1756
0.20	100	0.0653	0.0692
0.25	125	0.0223	0.0243
0.30	150	0.0068	0.0076

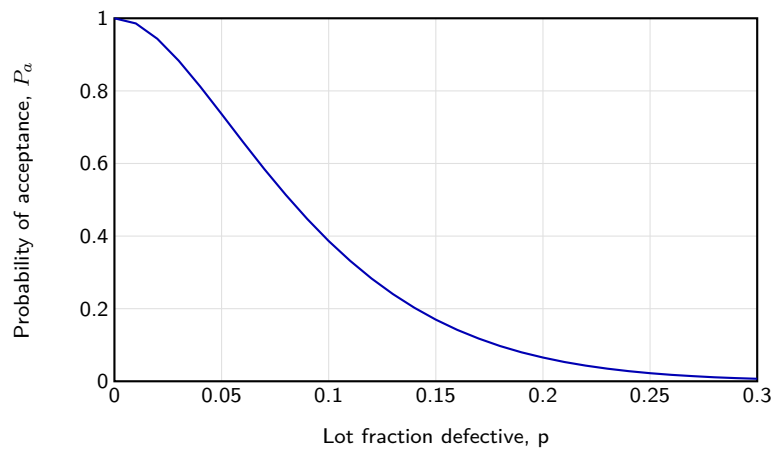


Figure 28: OC curve for the plan $n = 20$, $c = 1$ ($N = 500$)

The consumer accepts 94.4% of the lots that are 2% defective and only 38.7% of the lots that are 10% defective; lots that are 25% defective are accepted only 2.2% of the time.

66	Suppose that a supplier ships components in lots of size 5000. A single sampling plan with $n = 50$ and $C = 2$ is being used for receiving inspection. Rejected lots are screened and all defective items are reworked and returned to the lot. (i) Draw the OC curve for this plan. (ii) Find the level of lot quality that will be rejected 90% of the time. (iii) Management has objected to the use of the above sampling procedure and wants to use a plan with an acceptance number $C = 0$, arguing that this is more consistent with their zero defects program. What do you think of this? Source: Unit III SEE Mar/Apr 2023 (18IM53), Q5b	12	–	–
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(i) OC curve

The lots are rectified (rejected lots are screened), and with $N = 5000 \gg n = 50$ the binomial (type-B) distribution gives $P_a = P(d \leq 2)$.

p	$P_a (n = 50, c = 2)$	$P_a (n = 50, c = 0)$
0.00	1.0000	1.0000
0.01	0.9862	0.6050
0.02	0.9216	0.3642
0.03	0.8108	0.2181
0.04	0.6767	0.1299
0.05	0.5405	0.0769
0.06	0.4162	0.0453
0.08	0.2260	0.0155
0.10	0.1117	0.0052
0.12	0.0513	0.0017
0.15	0.0142	0.0003
0.20	0.0013	0.0000

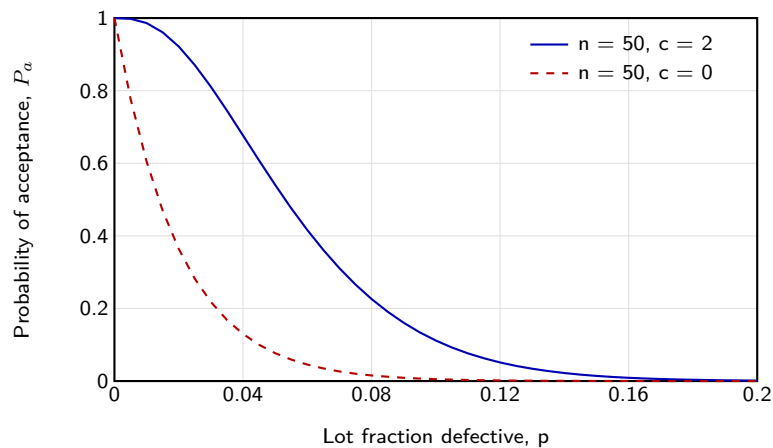


Figure 29: OC curves for $c = 2$ and for the proposed $c = 0$ plan

(ii) Lot quality rejected 90% of the time

A lot is rejected 90% of the time when $P_a = 0.10$. Solving $P(d \leq 2; n = 50, p) = 0.10$ gives

$$p = 0.1030 \text{ (about 10.3\% defective); check: } P_a(0.1030) = 0.100$$

(iii) The zero acceptance number

Plans with $c = 0$ have OC curves that are convex throughout their range; the probability of acceptance begins to drop very rapidly even for small lot fraction defective. This is extremely hard on the supplier and may be uneconomical for the consumer. For $n = 50, c = 0$ the probability of accepting a 1% defective lot is 0.605 against 0.986 with $c = 2$, and at 2%

defective only 0.364 against 0.922; so many lots of acceptable quality would be rejected, and, if rejected lots are screened or returned, this means unnecessary screening and production delays.

A $c = 0$ plan with the same protection against bad lots needs a smaller sample: to reject 90% of lots with $p = 0.1030$, $n \approx 22$; but then a 1% defective lot would be accepted only 80.2% of the time, instead of 98.6% with the present plan. The zero-defects objective is a goal for the supplier's process, not for the acceptance number of a sampling plan; the plan should be chosen from the AQL and LTPD points of the OC curve.

Answer: (ii) Lots about 10.3% defective are rejected 90% of the time. (iii) The change to $c = 0$ is not recommended: it rejects many acceptable lots (only 60.5% of 1% defective lots would be accepted, compared with 98.6%), and the AQL and LTPD points should be used to design the plan.

67	Draw the primary and supplementary OC curves for a double sampling plan with $n_1 = 50$, $C_1 = 2$, $n_2 = 100$, $C_2 = 6$. If the incoming lots have fraction non-conforming $p = 0.05$, what is the probability of acceptance on the first sample? What is the probability of final acceptance? Calculate the probability of rejection on the first sample.	12	–	–
Source: Unit III SEE Mar/Apr 2023 (18IM53), Q6b				

The plan: take $n_1 = 50$. If $d_1 \leq c_1 = 2$ accept; if $d_1 > c_2 = 6$ reject; if $3 \leq d_1 \leq 6$ take $n_2 = 100$ more and accept if $d_1 + d_2 \leq c_2 = 6$. With P_{a^1} and P_{a^2} the probabilities of acceptance on the first and second samples (binomial, type B):

$$P_a = P_a^I + P_a^{II}, \quad P_a^I = P(d_1 \leq 2), \quad P_a^{II} = \sum_{d_1=3}^6 P(d_1) P(d_2 \leq 6 - d_1)$$

At $p = 0.05$

d_1	$P(d_1), n_1 = 50$	$P(d_2 \leq 6 - d_1), n_2 = 100$	Product
3	0.2199	0.2578	0.0567
4	0.1360	0.1183	0.0161
5	0.0658	0.0371	0.0024
6	0.0260	0.0059	0.0002
		Sum = P_{a^2}	0.0754

$$P_a^I = P(d_1 \leq 2) = 0.5405$$

$$P_a^{II} = 0.0754$$

$$P_a = 0.5405 + 0.0754 = 0.6159$$

$$P(\text{reject on first sample}) = P(d_1 > 6) = 0.0118$$

p	P_a on first sample	P_a on second sample	Final P_a (primary OC)	P(reject on first sample)
0.00	1.0000	0.0000	1.0000	0.0000
0.01	0.9862	0.0134	0.9996	0.0000
0.02	0.9216	0.0631	0.9847	0.0001
0.03	0.8108	0.1038	0.9146	0.0007
0.04	0.6767	0.1032	0.7799	0.0036
0.05	0.5405	0.0754	0.6159	0.0118
0.06	0.4162	0.0445	0.4607	0.0289
0.07	0.3108	0.0224	0.3332	0.0583
0.08	0.2260	0.0100	0.2360	0.1019
0.09	0.1605	0.0040	0.1646	0.1596
0.10	0.1117	0.0015	0.1132	0.2298
0.11	0.0763	0.0005	0.0768	0.3091
0.12	0.0513	0.0002	0.0514	0.3935

0.13	0.0339	0.0001	0.0339	0.4789
0.14	0.0221	0.0000	0.0221	0.5616
0.15	0.0142	0.0000	0.0142	0.6387

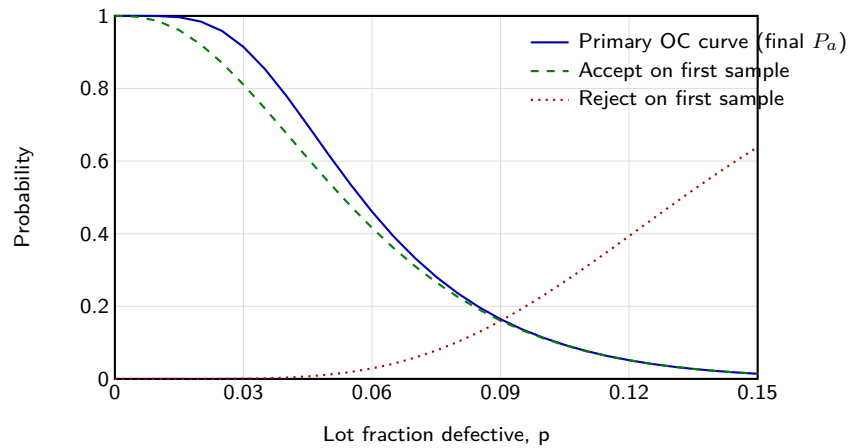


Figure 30: Primary and supplementary OC curves of the double sampling plan $n_1 = 50$, $c_1 = 2$, $n_2 = 100$, $c_2 = 6$

Answer: At $p = 0.05$: probability of acceptance on the first sample = 0.5405; probability of final acceptance = 0.6159; probability of rejection on the first sample = 0.0118.

68	Consider the following data of a double sampling plan: $N = 5000$, $n_1 = 150$, $c_1 = 3$, $r_1 = 6$, $n_2 = 200$, $c_2 = 8$, $r_2 = 9$. Construct the operating characteristic curve and the AOQ curve. Also determine ATI and ASN. Source: Unit III SEE Jan/Feb 2021 (18IM53), Q5	16	–	–
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Operation of the plan: take $n_1 = 150$; accept if $d_1 \leq 3$; reject if $d_1 \geq 6$; if $d_1 = 4$ or 5 take $n_2 = 200$ more and accept if $d_1 + d_2 \leq 8$, reject if $d_1 + d_2 \geq 9$. With rectifying inspection (rejected lots screened, defectives replaced), for incoming quality p :

$$P_a^I = P(d_1 \leq 3), \quad P_a^{II} = \sum_{d_1=4}^5 P(d_1) P(d_2 \leq 8 - d_1), \quad P_a = P_a^I + P_a^{II}$$

$$AOQ = \frac{P_a^I(N - n_1) + P_a^{II}(N - n_1 - n_2)}{N} p$$

$$ASN = n_1 + n_2(1 - P_I), \quad P_I = P_a^I + P(d_1 \geq 6)$$

$$ATI = n_1 P_a^I + (n_1 + n_2) P_a^{II} + N(1 - P_a)$$

The values calculated for a range of incoming quality (binomial probabilities):

p	P _a ^I	P _a ^{II}	P _a	AOQ	ASN	ATI
0.000	1.0000	0.0000	1.0000	0.00000	150.0	150.0
0.005	0.9929	0.0069	0.9998	0.00485	151.4	152.2
0.010	0.9353	0.0561	0.9914	0.00959	162.1	202.9
0.015	0.8106	0.1247	0.9353	0.01353	182.6	488.9
0.020	0.6472	0.1504	0.7976	0.01535	204.2	1161.6
0.025	0.4818	0.1245	0.6063	0.01458	218.7	2084.3
0.030	0.3384	0.0796	0.4180	0.01207	223.2	2988.6
0.040	0.1458	0.0191	0.1649	0.00637	209.3	4203.9
0.050	0.0548	0.0029	0.0576	0.00279	185.9	4721.1
0.060	0.0186	0.0003	0.0189	0.00110	167.9	4908.5

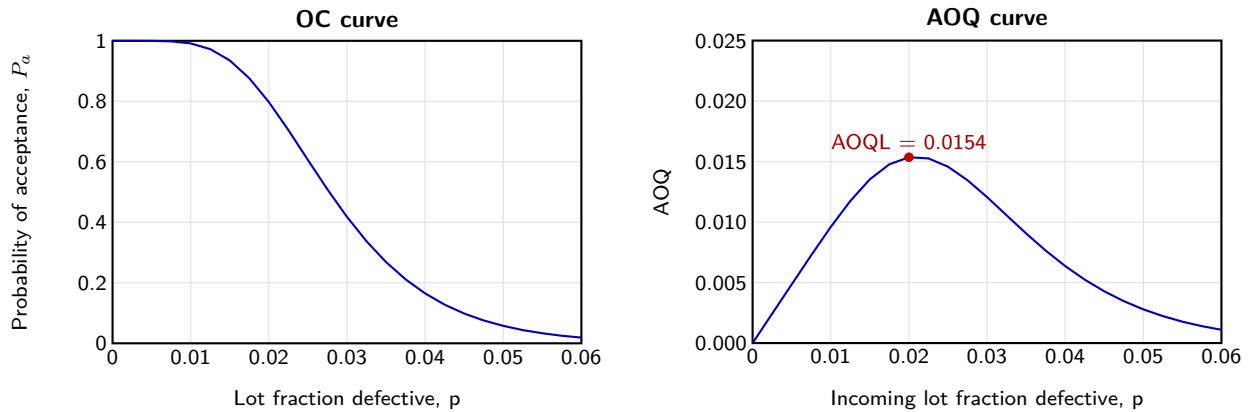


Figure 31: OC curve (left) and AOQ curve (right) of the double sampling plan

The ASN and ATI columns of the table give the average sample number (between $n_1 = 150$ and $n_1 + n_2 = 350$) and the average total inspection (between 150 and $N = 5000$) at each level of incoming quality. The AOQL is about 0.0154, reached when the incoming quality is about 2.00%.

Answer: Operating characteristic and AOQ values are tabulated above (AOQL ≈ 0.0154). At $p = 1.5\%$: $P_a = 0.9353$, ASN = 182.6 units and ATI = 488.9 units.

69	A double sampling plan is as follows: $N = 5000$, $n_1 = 150$, $c_1 = 3$, $r_1 = 6$, $n_2 = 200$, $c_2 = 8$, $r_2 = 9$. The lot fraction defective is 1.5%. Calculate P_a , AOQ, ATI and ASN. Source: Unit III SEE Jan/Feb 2021 (16IM53), Q6a	12	–	–
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Rules of the plan: accept if $d_1 \leq 3$; reject if $d_1 \geq 6$; if $d_1 = 4$ or 5 take the second sample and accept if $d_1 + d_2 \leq 8$. At $p = 0.015$ (binomial probabilities):

$$P_a^I = P(d_1 \leq 3; n_1 = 150, p = 0.015) = 0.8106$$

$$P(d_1 \geq 6) = 0.0263$$

d_1	$P(d_1), n_1 = 150$	$P(d_2 \leq 8 - d_1), n_2 = 200$	Product
4	0.1129	0.8165	0.0922
5	0.0502	0.6472	0.0325
		P_{a^2}	0.1247

$$P_a = P_a^I + P_a^{II} = 0.8106 + 0.1247 = 0.9353$$

$$AOQ = \frac{P_a^I(N - n_1) + P_a^{II}(N - n_1 - n_2)}{N} p = \frac{0.8106(4850) + 0.1247(4650)}{5000} (0.015) = 0.01353$$

$$ASN = n_1 + n_2(1 - P_I) = 150 + 200(1 - 0.8369) = 182.6$$

$$ATI = n_1 P_a^I + (n_1 + n_2) P_a^{II} + N(1 - P_a) = 150(0.8106) + 350(0.1247) + 5000(0.0647) = 488.9$$

Answer: $P_a = 0.9353$; AOQ = 0.01353 (1.353%); ASN = 182.6 units; ATI = 488.9 units.

70	A single sampling plan is as follows: $n = 80, c = 2$. (i) Plot the type-B OC curve (use the distribution which is theoretically correct). (ii) If AQL is 1.5%, find the producer's risk. (iii) If the consumer's risk is 15%, find the LTPD. Source: Unit III SEE Jan/Feb 2021 (16IM53), Q5a	10	–	–
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(i) Type-B OC curve

The type-B OC curve uses the binomial distribution (the theoretically correct distribution when sampling from a stream of lots or a large lot): $P_a = P(d \leq 2)$ with $n = 80$.

p	P_a
0.000	1.0000
0.005	0.9923
0.010	0.9534
0.015	0.8808
0.020	0.7844
0.030	0.5681
0.040	0.3748
0.050	0.2306
0.060	0.1344
0.080	0.0404
0.100	0.0107
0.120	0.0026

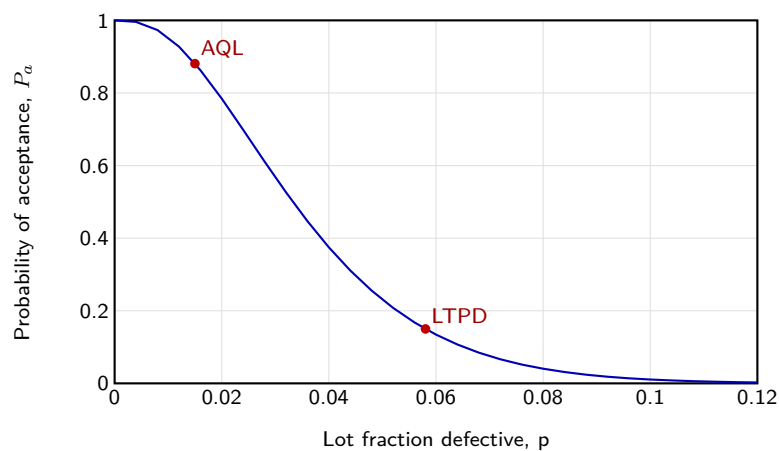


Figure 32: Type-B OC curve of the plan $n = 80, c = 2$

(ii) Producer's risk at AQL = 1.5%

$$P_a(0.015) = 0.8808 \Rightarrow \alpha = 1 - P_a = 1 - 0.8808 = 0.1192$$

(iii) LTPD for a consumer's risk of 15%

The LTPD is the lot quality p_2 at which $P_a = \beta = 0.15$. Solving $P(d \leq 2; n = 80, p) = 0.15$:

$$p_2 = 0.0580 \text{ (LTPD } \approx 5.80\%) \text{ check: } P_a = 0.1500$$

Answer: Producer's risk $\alpha = 0.1192$ (11.9%); LTPD = 5.80% (for $\beta = 15\%$).

71	A single sampling plan of $n = 200$, $c = 4$ and N very large compared with the sample size is used. Draw an AOQ curve. Find the AOQL after computing the probabilities of acceptance of lots of 0.5%, 1%, 2%, 3%, 4% and 5% defectives. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q5a	08	–	–
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When N is very large compared with n , the average outgoing quality is approximately $AOQ \approx P_a \cdot p$ (rejected lots are screened and the defectives replaced, so the outgoing quality of a rejected lot is zero). The probabilities of acceptance are binomial with $n = 200$, $c = 4$.

Lot quality p	$P_a = P(d \leq 4), n = 200$	$AOQ = P_a \times p$
0.5%	0.9965	0.00498
1.0%	0.9483	0.00948
2.0%	0.6288	0.01258
3.0%	0.2810	0.00843
4.0%	0.0950	0.00380
5.0%	0.0264	0.00132

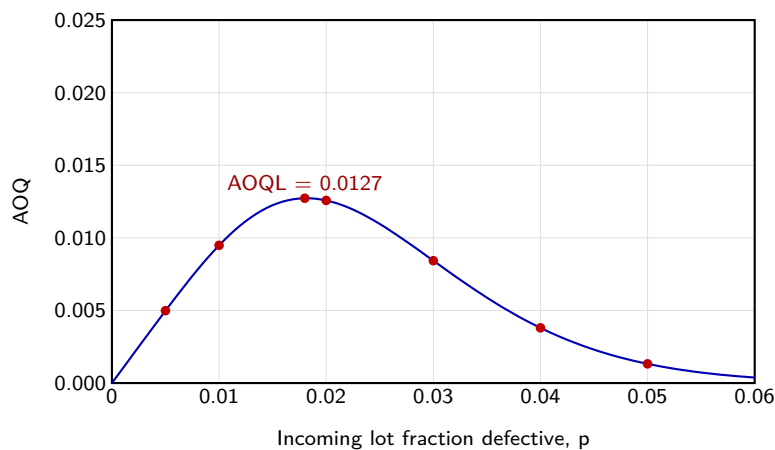


Figure 33: AOQ curve for $n = 200$, $c = 4$

The AOQ rises as the incoming quality worsens, reaches a maximum and then falls as most of the lots are rejected and screened. From the table the curve is highest between $p = 2\%$ and 3% ; refining the calculation gives the maximum **AOQL = 0.0127 at $p \approx 1.80\%$** .

Answer: AOQL ≈ 0.0127 (about 1.27% defective): whatever the quality of the incoming lots, the average outgoing quality will not be worse than this.

72	Construct an OC curve for a double sampling plan of $N = 2600$, $n_1 = 100$, $c_1 = 1$, $n_2 = 200$ and $c_2 = 4$. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q5b	08	–	–
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The plan: take $n_1 = 100$; accept if $d_1 \leq c_1 = 1$; reject if $d_1 > c_2 = 4$; if $d_1 = 2, 3$ or 4 take $n_2 = 200$ more and accept if $d_1 + d_2 \leq c_2 = 4$. The OC curve is the probability of final acceptance as a function of p (binomial, type B):

$$P_a = P(d_1 \leq 1) + \sum_{d_1=2}^4 P(d_1) P(d_2 \leq 4 - d_1)$$

p	P _a ¹ = P(d ₁ ≤ 1)	P _a ² (second sample)	P _a = P _a ¹ + P _a ²
0.000	1.0000	0.0000	1.0000
0.005	0.9102	0.0794	0.9896
0.010	0.7358	0.1518	0.8875
0.020	0.4033	0.0822	0.4854
0.030	0.1946	0.0174	0.2121
0.040	0.0872	0.0024	0.0896
0.050	0.0371	0.0003	0.0373
0.060	0.0152	0.0000	0.0152
0.080	0.0023	0.0000	0.0023
0.100	0.0003	0.0000	0.0003

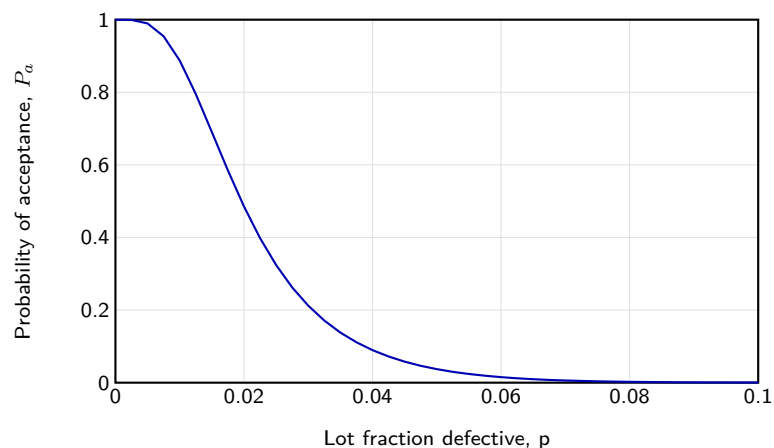


Figure 34: OC curve of the double sampling plan $n_1 = 100$, $c_1 = 1$, $n_2 = 200$, $c_2 = 4$

73	In a single sampling plan $N = 10,000$, $n = 300$, $c = 1$ is used with the stipulation that all rejected lots are to be screened. Compute the AOQ if all lots submitted are 0.50% defective. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q6a	08	–	–
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The lots are 0.50% defective, so $p = 0.005$. The probability of acceptance is binomial with $n = 300$ and $c = 1$:

$$P_a = P(d \leq 1) = (0.995)^{300} + 300(0.005)(0.995)^{299} = 0.5574$$

With rectifying inspection (rejected lots screened, defectives replaced):

$$AOQ = \frac{P_a p (N - n)}{N} = \frac{0.5574(0.005)(10000 - 300)}{10000} = 0.00270$$

Answer: AOQ = 0.00270, i.e. about 0.270% defective in the outgoing lots.

74	Draw the AOQ curve for the data $N = 4000$, $n = 80$, $c = 2$. Source: Unit III SEE Fast Track Oct 2020 (16IM53), Q6b	08	–	–
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p	$P_a = P(d \leq 2), n = 80$	$AOQ = P_a p (N - n)/N$
0.00	1.0000	0.00000
0.01	0.9534	0.00934
0.02	0.7844	0.01537
0.03	0.5681	0.01670
0.04	0.3748	0.01469
0.05	0.2306	0.01130
0.06	0.1344	0.00791
0.08	0.0404	0.00317
0.10	0.0107	0.00105

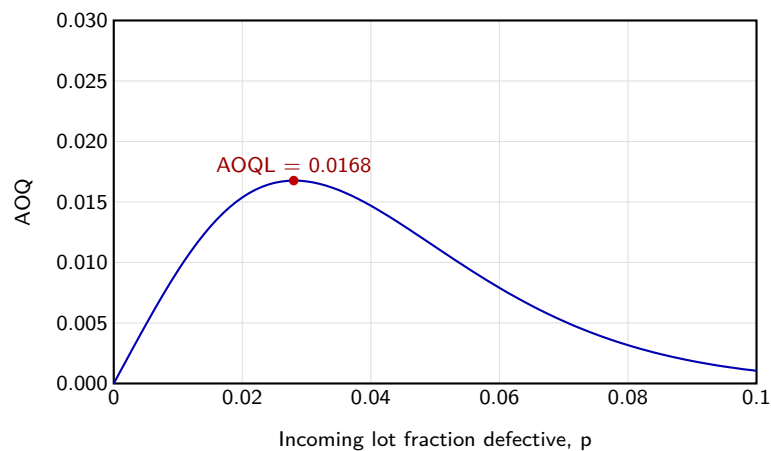


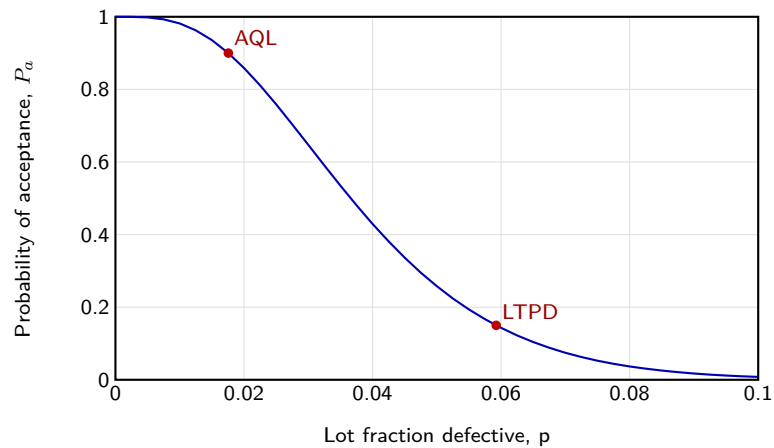
Figure 35: AOQ curve for $N = 4000$, $n = 80$, $c = 2$

$AOQ = P_a p (N - n)/N$ with the binomial P_a . The AOQ is zero for perfect lots and for very bad lots (all rejected and screened) and passes through a maximum, the AOQL = 0.0168, at $p \approx 2.8\%$.

75	A single sampling plan is as follows: $N = 5000$, $n = 100$, $C = 3$. (i) Plot the OC curve. (ii) Determine the AQL and LTPD for a producer's risk $P_r = 10\%$ and a consumer's risk $C_r = 15\%$ respectively. (iii) What is the ATI of the above plan for 1.5% defectives of the incoming lots? Source: Unit III SEE Nov/Dec 2019 (16IM53), Q6a	10	–	–
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(i) OC curve

p	$P_a = P(d \leq 3), n = 100$
0.000	1.0000
0.005	0.9983
0.010	0.9816
0.015	0.9358
0.020	0.8590
0.030	0.6472
0.040	0.4295
0.050	0.2578
0.060	0.1430
0.080	0.0367
0.100	0.0078

**Figure 36:** OC curve of the plan $N = 5000$, $n = 100$, $c = 3$ **(ii) AQL and LTPD**

A producer's risk of 10% means $P_a = 0.90$ at the AQL; a consumer's risk of 15% means $P_a = 0.15$ at the LTPD.

$$P(d \leq 3; 100, p_1) = 0.90 \Rightarrow \text{AQL} = p_1 = 0.0176 \text{ (1.76\%)}$$

$$P(d \leq 3; 100, p_2) = 0.15 \Rightarrow \text{LTPD} = p_2 = 0.0592 \text{ (5.92\%)}$$

(iii) ATI at 1.5% defective

$$P_a(0.015) = 0.9358$$

$$\text{ATI} = n + (1 - P_a)(N - n) = 100 + (1 - 0.9358)(4900) = 414.7$$

Answer: AQL $\approx$ 1.76% (producer's risk 10%); LTPD $\approx$ 5.92% (consumer's risk 15%); ATI at 1.5% defective $\approx$ 415 units per lot.

76	A single sampling plan is as follows: $N = 4000$, $n = 75$, $c = 2$. (a) Construct the OC curve of this plan. (b) If the AQL is 1.5%, find the producer's risk, and if the consumer's risk is 10%, find the LTPD. (c) Construct the AOQ curve and determine the AOQL. (d) Find the ATI of the above plan at 1.5% defective of the incoming lot.	16	–	–
Source: Unit III SEE Fast Track Jul 2019 (16IM53), Q5				

(a) OC curve

The type-B (binomial) OC curve: $P_a = P(d \leq 2)$ with $n = 75$, calculated below for several values of p .

p	P_a	AOQ
0.000	1.0000	0.00000
0.005	0.9935	0.00487
0.010	0.9603	0.00942
0.015	0.8967	0.01320
0.020	0.8101	0.01590
0.030	0.6083	0.01791
0.040	0.4186	0.01643
0.050	0.2697	0.01323
0.060	0.1650	0.00971
0.080	0.0548	0.00430
0.100	0.0161	0.00158

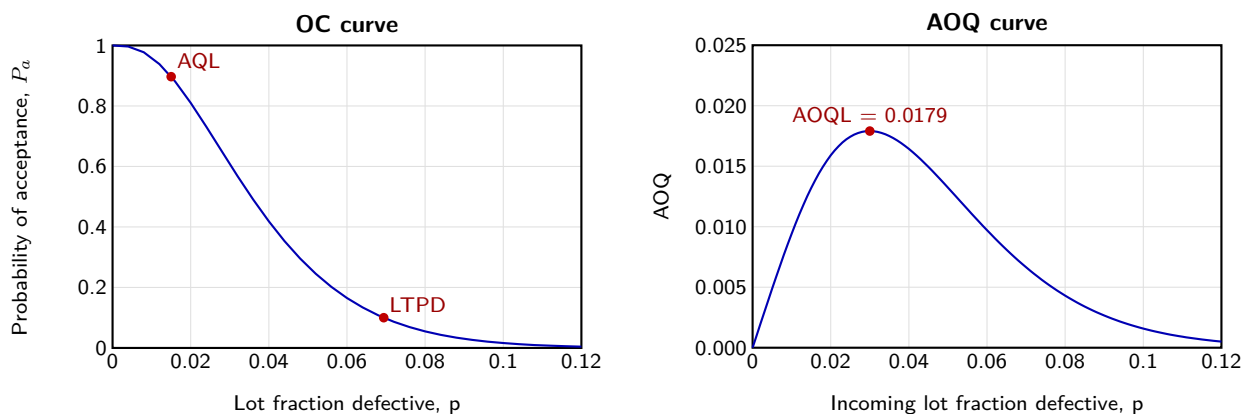


Figure 37: OC curve (left) and AOQ curve (right) of the plan $N = 4000$, $n = 75$, $c = 2$

(b) Producer's risk and LTPD

$$\alpha = 1 - P_a(0.015) = 1 - 0.8967 = 0.1033$$

$$P(d \leq 2; 75, p_2) = 0.10 \Rightarrow \text{LTPD} = p_2 = 0.0694 \text{ (6.94\%)}$$

(c) AOQ curve and AOQL

$\text{AOQ} = P_a p (N - n)/N = P_a p (3925)/4000$ (see the table and the right-hand curve). The maximum of the curve is the AOQL.

$$\text{AOQL} \approx 0.0179 \quad (\text{at } p \approx 0.0300)$$

(d) ATI at 1.5% defective

$$\text{ATI} = n + (1 - P_a)(N - n) = 75 + (1 - 0.8967)(3925) = 480.5$$

Answer: Producer's risk = 0.1033; LTPD = 6.94% ($\beta = 10\%$); AOQL = 0.0179; ATI at 1.5% defective = 480.5 units.

77	The following data refer to a double sampling plan: $N = 50000$, $n_1 = 150$, $c_1 = 0$, $n_2 = 100$, $c_2 = 1$. If 1% defective lots are submitted, compute (a) P_a , the probability of acceptance, (b) AOQ, (c) ASN, (d) ATI, (e) AFI. Source: Unit III SEE Fast Track Jul 2019 (16IM53), Q6	16	–	–
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Operation of the plan: take $n_1 = 150$; accept if $d_1 = 0$; reject if $d_1 > c_2 = 1$, that is $d_1 \geq 2$; if $d_1 = 1$ take $n_2 = 100$ more and accept if $d_1 + d_2 \leq 1$ (so $d_2 = 0$). For $p = 0.01$:

$$P_a^I = P(d_1 = 0) = (0.99)^{150} = 0.2215$$

$$P(d_1 = 1) = 150(0.01)(0.99)^{149} = 0.3355, \quad P(d_2 = 0) = (0.99)^{100} = 0.3660$$

$$P_a^{II} = 0.3355 \times 0.3660 = 0.1228$$

$$P(d_1 \geq 2) = 0.4430$$

(a) Probability of acceptance

$$P_a = P_a^I + P_a^{II} = 0.2215 + 0.1228 = 0.3443$$

(b) Average outgoing quality

$$AOQ = \frac{P_a^I(N - n_1) + P_a^{II}(N - n_1 - n_2)}{N} p = \frac{0.2215(49850) + 0.1228(49750)}{50000} (0.01) = 0.00343$$

(c) Average sample number

$$P_I = P_a^I + P(d_1 \geq 2) = 0.2215 + 0.4430 = 0.6645$$

$$ASN = n_1 + n_2(1 - P_I) = 150 + 100(1 - 0.6645) = 183.6$$

(d) Average total inspection

$$ATI = n_1 P_a^I + (n_1 + n_2) P_a^{II} + N(1 - P_a) = 150(0.2215) + 250(0.1228) + 50000(0.6557) = 32850.5$$

(e) Average fraction inspected

The average fraction inspected is the average total inspection as a fraction of the lot size:

$$AFI = \frac{ATI}{N} = \frac{32850.5}{50000} = 0.6570$$

Answer: $P_a = 0.3443$; $AOQ = 0.00343$; $ASN = 183.6$ units; $ATI = 32850.5$ units; $AFI = 0.6570$.

78	A double sampling plan is as follows: inspect a first sample of 25 items randomly from a lot. The lot is accepted if no defectives are found. If three or more are found, the lot is rejected. Otherwise a second sample of 100 is drawn and the lot is accepted if the combined number of defectives found does not exceed 3. (i) Determine the producer's risk (α) for a lot fraction defective of 0.01. (ii) Determine the consumer's risk (β) for a lot fraction defective of 0.07. (iii) What is the ATI of the above plan for a 2% defective lot? Source: Unit III SEE Feb/Mar 2026 (IM353IA), Q6b (the second acceptance number is not printed in the paper; $c_2 = 3$ and a lot size $N = 1000$ are assumed)	10	2	3
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Plan: $n_1 = 25$, accept if $d_1 = 0$ ($c_1 = 0$); reject if $d_1 \geq 3$; if $d_1 = 1$ or 2 take $n_2 = 100$ more and accept if $d_1 + d_2 \leq c_2$. The paper does not print the second acceptance number, so $c_2 = 3$ is assumed, and for the ATI a lot size $N = 1000$ is assumed (ATI needs N). Binomial probabilities are used.

(i) Producer's risk at $p = 0.01$

$$P_a^I = (0.99)^{25} = 0.7778, \quad P_a^{II} = P(d_1 = 1)P(d_2 \leq 2) + P(d_1 = 2)P(d_2 \leq 1) = 0.1983$$

$$P_a = 0.9762 \Rightarrow \alpha = 1 - P_a = 0.0238$$

(ii) Consumer's risk at $p = 0.07$

$$P_a^I = (0.93)^{25} = 0.1630, \quad P_a^{II} = 0.0096$$

$$\beta = P_a = 0.1725$$

(iii) ATI at 2% defective ($N = 1000$)

$$P_a^I = 0.6035, \quad P_a^{II} = 0.2388, \quad P_a = 0.8422$$

$$ATI = n_1 P_a^I + (n_1 + n_2) P_a^{II} + N(1 - P_a) = 25(0.6035) + 125(0.2388) + 1000(0.1578) = 202.7$$

Answer: $\alpha = 0.0238$ at $p = 0.01$; $\beta = 0.1725$ at $p = 0.07$; ATI ≈ 203 units per lot at 2% defective (for $N = 1000$).

79	List the general practical steps and guidelines for planning and conducting a designed experiment (DOE). Source: Unit IV Improvement CIE, 28/01/2025, Q4; SEE Mar/Apr 2023 (18IM53), Q7b (guidelines for designing experiments)	10	4	4
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Designed experiments are a powerful approach to improving a process, but everyone involved must have a clear idea in advance of the objective of the experiment, the factors to be studied, how the experiment will be conducted and how the data will be analyzed. The recommended procedure has seven steps:

- (1) **Recognition of and statement of the problem:** develop all the ideas about the problem and the objectives of the experiment, soliciting input from engineering, quality, marketing, the customer, management and the operators.
- (2) **Choice of factors and levels:** choose the factors to be varied, the ranges over which they vary and the specific levels at which runs are made, using process knowledge; for screening or characterization keep the number of levels low (most often two).
- (3) **Selection of the response variable:** be sure that the response really gives useful information about the process (usually the average or the standard deviation of a measured characteristic); gauge capability matters.
- (4) **Choice of experimental design:** consider the sample size (number of replicates), the run order of the trials, and whether blocking or other randomization restrictions are involved.
- (5) **Performing the experiment:** monitor the process carefully so that everything is done according to plan; errors in procedure destroy the validity of the experiment.
- (6) **Data analysis:** use statistical methods (ANOVA, graphs, residual analysis and model-validity checking) so that the conclusions are objective rather than judgmental.
- (7) **Conclusions and recommendations:** draw practical conclusions, recommend a course of action, and run follow-up and confirmation tests to validate the conclusions.

Steps 1 to 3 are called **pre-experimental planning** and must be done as well as possible. Experimentation is iterative and sequential: a single large comprehensive experiment at the start is usually a mistake, and as a general rule no more than about 25% of the available resources should be invested in the first experiment. When several factors are of interest, a factorial design should be used.

80	Explain the principles of experimentation (basic principles of experimental design) in detail, with examples. Source: Unit IV Improvement CIE, 13/01/2026, Q4; SEE Feb/Mar 2026 (IM353IA), Q7a; SEE Nov/Dec 2019 (16IM53), Q7a	10	4	4
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A designed experiment is a test or series of tests in which purposeful changes are made to the input variables of a process so that the corresponding changes in the output response can be observed and identified. The basic principles that make the conclusions valid are:

1. Randomization

The runs are made in random order, and the treatments are allocated to the experimental units at random. By randomizing the order of the runs, the effect of any **nuisance variable** that may influence the response is approximately balanced out.

Example: in the paper tensile-strength experiment 24 specimens (4 hardwood concentrations $\times$ 6) are tested in random order; if all the 5% specimens were tested first and the 20% specimens last, a warm-up effect of the tensile tester (strength

rising with the time the machine is on) would be confused with the effect of the hardwood concentration. Randomization also underlies the assumption that the errors are independent.

2. Replication

Each treatment is run more than once (n replicates). The n replicates give the degrees of freedom with which the **experimental error** is estimated ($a(n - 1)$ degrees of freedom in a single-factor experiment, $ab(n - 1)$ in a two-factor factorial), which is needed for the F-tests, and they make the estimates of the effects more precise. **Example:** six specimens at each hardwood level, or $n = 2$ runs of every temperature–pressure combination. Repeat readings of the same unit are not true replicates; each replicate must be a complete, independent run.

3. Blocking (local control)

When it is impossible to run all the observations under constant, homogeneous conditions (for example, not all the tests can be done on one shift or from one batch of material), the runs are grouped into **blocks** that are internally homogeneous, and the block-to-block variation is removed from the experimental error. **Example:** a 2^2 experiment whose runs take 4 hours each is run over two days, two runs on each day, with days as blocks; the average difference between the days is the block effect. Blocking is the principle used to eliminate the effect of a known nuisance factor.

4. Factorial approach: vary the factors together

When several factors are of interest, they are varied together in a factorial design instead of one at a time, so that the interactions between the factors can be estimated and fewer runs are needed. The one-factor-at-a-time method fails when the factors interact (in the textbook optimization example it missed the true optimum, which was 20 yield points higher).

5. Planning and sequential experimentation

The objective of the experiment, factors, levels and response must be decided before the runs (pre-experimental planning), and the experimentation is sequential: not more than about 25% of the resources go into the first experiment, because the factors, ranges and levels are learned as the programme progresses.

81	How do the principles of replication, randomization and local control (blocking) work together to strengthen the validity and reliability of the experimental results in the design of experiments? Source: Unit IV Improvement CIE, 28/01/2025, Q5a (replication and randomization); SEE Feb/Mar 2026 (IM353IA), Q8b	10	3	2
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Principle	What it does	Contribution to the results
Replication	Repeats each treatment so that the experimental error can be estimated from the variation among the replicates.	Reliability: a valid error estimate (mean square error) for the F-tests, and more precise estimates of the effects (the standard deviation of a treatment average is $s/\sqrt{n}$).
Randomization	Assigns the treatments to the experimental units and the run order by chance.	Validity: balances out nuisance variables, avoids systematic bias, and justifies the assumption that the errors are independent.
Local control (blocking)	Groups the units into homogeneous blocks, and every treatment is run in every block.	Precision: the known nuisance variation between blocks (shifts, days, batches) is removed from the error, so smaller effects can be detected.

How they work together: blocking removes the variation that can be foreseen; randomization (within each block) protects against the variation that cannot be foreseen, so that the remaining error is a fair, unbiased representation of chance; replication then measures the size of that error. Without randomization, the error estimate from replication could be biased, and without replication randomization gives no way of judging the size of the error. Together they make the internal validity of the conclusions (that the observed differences are due to the factors) and the reliability (that the effects can be reproduced) as strong as possible, and the results can then be analyzed by ANOVA with its assumptions of independent, normally distributed errors of constant variance.

82	Explain the concept of randomization in the context of the design of experiments. Why is randomization important and how does it help in eliminating bias and improving the validity of the experimental results? Provide an example of how randomization can be applied in a real-world experiment. Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q8a	08	2	4
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Concept: randomization means that both the allocation of the experimental material to the treatments and the order in which the individual runs are made are determined randomly (for example with random numbers), so that every run has the same chance of being made at any position in the sequence.

Why it is important

- It **balances out the effect of nuisance variables** that may influence the response: variables that change with time or order (machine warm-up, tool wear, operator fatigue, ambient temperature) are spread over all the treatments rather than being confounded with one of them.
- It **eliminates systematic bias:** without it any observed difference between treatments could be partly due to the order in which they were run.
- It **supports the statistical analysis:** the ANOVA assumes that the errors are independent random variables, and random order of the runs makes this assumption reasonable. (It can be checked by plotting the residuals against the run order.)
- It improves **validity:** the conclusion that a difference is caused by the factor is credible only when the other influences have been randomized.

Example (paper tensile strength)

The tensile strength of grocery-bag paper is studied at four hardwood concentrations (5, 10, 15 and 20%) with six specimens each, 24 specimens in all. A warm-up effect of the tensile tester makes the observed strength increase the longer the machine is on. If the specimens were tested in order of increasing concentration, the 20% specimens would be tested when the machine is warmest, and the warm-up effect would be mistaken for an effect of the hardwood concentration. The 24 specimens are therefore numbered and tested in a random order (drawn from random numbers), so that each concentration gets early and late test positions and the warm-up effect is approximately balanced out among the treatments.

83	What are the key assumptions behind the application of a full factorial design, and how do they affect the interpretation of the experimental results? Source: Unit IV Improvement CIE, 28/01/2025, Q5b	05	3	2
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The two-factor (full) factorial experiment is analyzed with the fixed-effects model $y(ijk) = \mu + \tau(i) + \beta(j) + (\tau\beta)(ij) + \epsilon(ijk)$. The assumptions and their effect on the interpretation are:

Assumption	Effect on the interpretation
Fixed effects: the levels of the factors are chosen by the experimenter	Conclusions apply only to the levels studied, not to other levels.
The model is correct (each observation = overall mean + factor effects + interaction + error)	If a significant interaction exists the main effects have little meaning alone; the effect of one factor must be read at each level of the other (interaction graph).
The errors $\epsilon(ijk)$ are normally and independently distributed with mean zero and the same variance σ^2 (NID(0, σ^2))	The F-tests and the conclusions on significance are valid only if this holds; it is checked by residual analysis (normal probability plot, residuals against fitted values and against run order); patterns suggest a transformation.
All combinations are run, in random order, and with the same number of replicates n (a completely randomized design)	Randomization protects against nuisance variables, and replication gives the error estimate; if the runs are not randomized the errors are not independent and the results may be biased.

84	Explain six examples of designed experiments in process improvement.	10	4	4
Source: Unit IV Improvement CIE, 13/01/2026, Q5				

Designed experiments are an active method of process improvement: purposeful changes are made to the inputs and the changes in the output are observed. Six applications from the textbook are:

- (1) **Characterizing a process (flow soldering).** An engineer has brought the flow solder process for printed circuit boards into statistical control with u charts and Pareto analysis, giving about 1% defective joints, which is still too many since a board has over 2000 joints. Since the process is in control it is not obvious what to adjust, so a factorial (screening) experiment varies the controllable factors of the machine (such as solder temperature) and the uncontrollable ones, to find which factors affect the defects per board and in which direction they should be adjusted.
- (2) **Optimizing a process (chemical yield).** After a characterization experiment shows that temperature and reaction time are the most important variables, the process is running at 155°F and 1.7 h with a 75% yield. A factorial experiment varying time and temperature together shows that the yield rises with a higher temperature and shorter time; the one-factor-at-a-time method would have missed the interaction and the true optimum, which is at least 20 yield points higher. A response surface experiment can then locate the optimum precisely.
- (3) **Product design (car door hinge).** Engineers designing a door hinge study the check effort (the holding ability of the latch on a hill), which depends on five factors (roller and spring characteristics). A prototype in which the factors can be varied is tested at various combinations of levels to find the most influential factors and improve the design.
- (4) **Determining tolerances (Wheatstone bridge).** To obtain good gauge capability (small standard deviation of measurement error) the engineer experiments with the component tolerances ($\pm 1\%$ for resistors, $\pm 5\%$ for the power supply) as factor levels to find which components have the most critical tolerances, so that only these are tightened, giving a lower-cost design. The experiment is run on a computer model, without building hardware.
- (5) **Aircraft primer paint (two-factor factorial).** Three primers and two application methods (dipping and spraying) are compared with three specimens per combination, 18 runs in random order, with the adhesion force as the response. ANOVA shows no interaction; spraying is superior and primer type 2 is the most effective.
- (6) **Router experiment (2^2 factorial).** A router that cuts registration notches in printed circuit boards produced too much variability in notch dimension although the process was in control. A 2^2 factorial in the bit size and speed, with the vibration as the response, shows that the large effect is the AB interaction, and that low vibration is obtained at the combination of small bit size and low speed, which reduces the variability of the notch dimension.

85	Explicate with illustrations the strategy of experimentation.	08	2	2
Source: Unit IV SEE Feb/Mar 2026 (IM353IA), Q7b				

The strategy of experimentation is the way in which the experimenter plans which combinations of factor levels to run. Three strategies are described:

Strategy	Description and result
One-factor-at-a-time (OFAT)	One factor is varied while all the others are held at fixed levels, then the next factor. It is inefficient (more runs than a factorial) and, since it cannot detect interactions, it can fail dramatically. In the chemical yield example the engineer fixes temperature at 155°F and finds the best time (1.7 h), then fixes the time at 1.7 h and finds the best temperature (155°F): the conclusion “155°F, 1.7 h, yield about 75%” misses the true optimum, which lies at a much shorter time and higher temperature and gives at least 20 points more yield.
Factorial design	All the factors are varied together; in each replicate all possible combinations of the levels are run. It estimates the main effects and the interactions with the same runs, and is the only way to detect interactions. It is used whenever several factors are of interest; for screening, two levels are used (the 2^k design), and with many factors a fractional replication can be used.
Sequential experimentation	Experimentation is iterative: hypotheses are formed, an experiment is run, and new hypotheses follow. A single large experiment at the start is usually a mistake because the important factors, ranges and levels are not known beforehand; no more than about 25% of the resources should be used in the first experiment. Typically a screening (characterization) experiment is followed by an optimization experiment and a confirmation run.

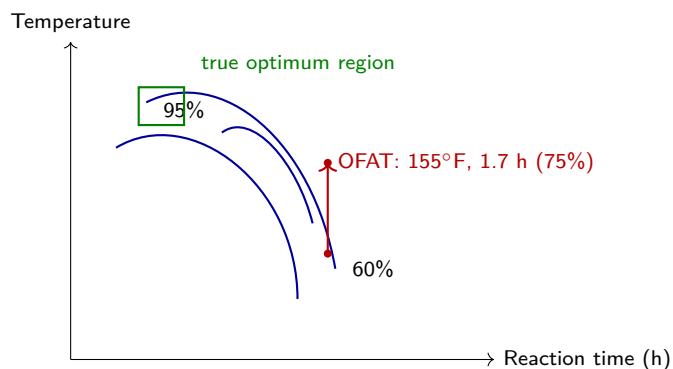


Figure 38: Illustration: the one-factor-at-a-time path stays near 75% yield, whereas a factorial experiment points to the region of higher yield (shorter time, higher temperature)

The experiment itself is planned in the steps of the guidelines (problem statement, factors and levels, response, design, performance, analysis, conclusions), with randomization, replication and blocking built in.

86	Define the Randomized Block Design (RBD). Give an example.	04	2	1
Source: Unit IV SEE Feb/Mar 2026 (IM353IA), Q8a				

Blocking is a design technique for eliminating the unwanted variation caused by non-homogeneous experimental conditions (for example different shifts, days or batches of material). In a **randomized block design (RBD)** the experimental units are first grouped into blocks that are as uniform as possible within themselves; **every treatment is run once (a complete replicate) in every block**, and the run order within each block is randomized. The block-to-block variation (the block effect) is separated out in the analysis, so the experimental error is smaller than in a completely randomized design.

Example: a 2^2 experiment (4 runs, each requiring 4 hours of laboratory analysis) cannot be done in one day. Two days are used as blocks, and each day contains a complete set of the four treatment combinations (1), a, b and ab in random order. The difference between the average results of the two days is the block effect, which is removed from the error. Similarly, the three fertilizers of an agricultural trial are tried on each of several plots (blocks) of uniform soil.

87	Differentiate between CRD and RBD.	04	–	–
Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q7a				

Basis	Completely randomized design (CRD)	Randomized block design (RBD)
Experimental units	Taken as uniform; the environment is as uniform as possible.	Non-uniform; grouped into homogeneous blocks (shifts, days, batches).
Randomization	The treatments are randomized over all the runs.	Randomized within each block (a restriction on randomization).
Layout	Treatments assigned completely at random, equal or unequal replicates.	Each block contains every treatment (a complete replicate).
Model	$y(ij) = \mu + \tau(i) + \varepsilon(ij)$	$y(ij) = \mu + \tau(i) + \text{block effect} + \varepsilon(ij)$
ANOVA	Treatments and error	Treatments, blocks and a smaller error
Use	When the units are homogeneous	When a nuisance factor is known and can be blocked

88	Explain the purpose and procedure of a single-factor experiment in the context of DOE. How is this type of experiment different from a multi-factor experiment and when would each be used? Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q7a	04	2	1
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Basis	Single-factor experiment	Multi-factor (factorial) experiment
Purpose	To compare the mean response at the different levels (treatments) of one factor, for example five heat-treating methods or four hardwood concentrations.	To study the effect of several factors on the response together, and also their interactions.
Procedure	Choose the levels of the factor; make n replicates at each level in random order; fit $y(ij) = \mu + \tau(i) + \varepsilon(ij)$; test equality of treatment means with a one-way ANOVA ($F = MS(\text{treatments})/MS(\text{error})$); check the residuals.	Run all combinations of the levels of the factors (each replicate), in random order; analyze the main effects and interactions with a multi-factor ANOVA or the 2^k contrasts.
Difference	Other factors are held constant; no interaction can be found.	Factors are varied together; interactions are detected; fewer runs than one-factor-at-a-time experiments.
When used	When only one factor is of interest, or in a first comparison of treatments.	When there are several factors (screening, characterization, optimization), and interactions are likely.

89	What is interaction in a factorial design? Source: Unit IV SEE Feb/Mar 2026 (IM353IA), Q8c	02	2	2
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There is an **interaction** between factors A and B when the difference in response between the levels of one factor is not the same at all levels of the other factor; that is, the effect of A depends on the level of B. Example: with the data A effect at $B^- = 20$ and at $B^+ = -10$, the effect of A changes sign with B, so there is interaction. Graphically, the lines of response against the level of A for B^- and B^+ are not parallel. When an interaction is large, the main effects have little meaning, and a significant interaction can mask the significance of the main effects.

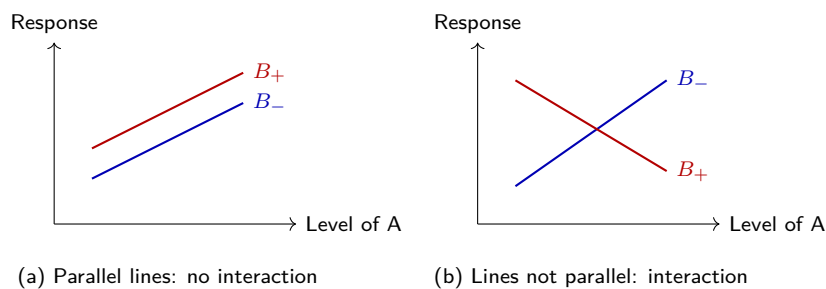


Figure 39: Interaction graph: parallel lines (no interaction) and non-parallel lines (interaction)

90	Define the following terms with an example: (i) factor, (ii) level, (iii) main effect, (iv) interaction effect, (v) treatment combination. Source: Unit IV SEE Jan/Feb 2021 (16IM53), Q8b	06	–	–
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Example used: a chemical process whose yield is studied at two reaction temperatures and two reaction times.

Term	Definition	Example
Factor	A controllable input variable that is varied in the experiment.	Temperature (A) and reaction time (B).
Level	A value or setting of a factor; two levels are called low (–) and high (+).	Temperature: 150°F (–) and 160°F (+); time: 1.0 h (–) and 2.0 h (+).
Main effect	The change in response produced by a change in the level of a factor, averaged over the levels of the other factors: the average response at the high level minus the average at the low level.	Main effect of A = average yield at 160°F – average yield at 150°F.
Interaction effect	The effect when the response difference between the levels of one factor is not the same at all levels of another factor; the effect of A depends on B.	The effect of temperature is large for the short time but small for the long time: AB interaction.
Treatment combination	One combination of levels of all the factors; denoted (1) (all low), a, b and ab for the 2 ² design.	Run “a”: A high (160°F), B low (1.0 h); “ab”: both high.

91	Determine the factors, probable levels of a factor and the response for the situations given below. (i) The Indian cricket team wants to determine which combination of playing XI will enhance the team's performance in the World Cup matches. (ii) A casting company desires to produce good quality castings. Source: Unit IV SEE Jan/Feb 2021 (16IM53), Q8a	10	–	–
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In designing an experiment the response variable (output), the controllable factors and their levels must be identified; the uncontrollable factors (noise factors) are kept constant or randomized.

(i) Cricket team selection

Response	Team performance: runs scored, wickets taken (or the result of the match, win percentage).
Factor A: batting line-up	Levels: strong top order (specialist batsmen) / balanced order with all-rounders.
Factor B: bowling attack	Levels: 3 fast bowlers + 1 spinner / 2 fast bowlers + 2 spinners.
Factor C: number of all-rounders	Levels: 1 / 3.
Factor D: wicket-keeper batsman	Levels: included / not included.
Uncontrollable (noise) factors	Pitch condition, weather, toss, strength of the opposing team.

A 2^k factorial design over these factors (each combination is a possible playing XI) gives the main effects and interactions of the choices on the response.

(ii) Casting company

Response	Quality of the castings: percentage of defective (rejected) castings, or number of defects (blow holes, shrinkage, porosity) per casting.
Factor A: pouring temperature	Levels: 1300°C / 1350°C.
Factor B: moisture content of moulding sand	Levels: 3% / 5%.
Factor C: mould hardness (ramming)	Levels: 60 / 80 (hardness number).
Factor D: binder (clay) content	Levels: 6% / 9%.
Uncontrollable (noise) factors	Ambient humidity, variation of raw material, operator differences.

A 2^4 factorial experiment (16 runs, replicated and run in random order) gives the effects of the factors and their interactions on the defect rate, and the best settings.

92	Write the test matrix for the 2^2 factorial design. Source: Unit IV SEE Fast Track Oct 2020 (16IM53), Q7b	06	—	—
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The 2^2 design has two factors, A and B, each at two levels, low (–) and high (+): $2 \times 2 = 4$ runs (treatment combinations). Each run is denoted by the lower-case letters of the factors at their high levels; (1) means that all factors are at the low level.

Run	Treatment combination	A	B	AB = A × B	Identity (I)	Response
1	(1)	–	–	+	+	y_1
2	a	+	–	–	+	y_2
3	b	–	+	–	+	y_3
4	ab	+	+	+	+	y_4

The effects are the contrasts of the totals (with n replicates per run):

$$A = \frac{1}{2n} [a + ab - b - (1)], \quad B = \frac{1}{2n} [b + ab - a - (1)], \quad AB = \frac{1}{2n} [ab + (1) - a - b]$$

The sum of squares of an effect is $SS = (\text{contrast})^2 / (4n)$. Except for the identity column I, each column has an equal number of plus and minus signs, and the columns are orthogonal (the sum of the products of the signs of any two columns is zero).

93	An experiment was run to determine whether four specific fusing temperatures affect the density of a certain type of brick. The experiment led to the following data: temperature 100: 21.8, 21.9, 21.7, 21.6, 21.7; temperature 125: 21.7, 21.4, 21.5, 21.4; temperature 150: 21.9, 21.8, 21.8, 21.6, 21.5; temperature 175: 21.9, 21.7, 21.8, 21.4. (i) Does the fusing temperature affect the density of the bricks? Use $\alpha = 0.05$. (ii) Analyze the residuals from this experiment. Source: Unit IV SEE Mar/Apr 2023 (18IM53), Q7a	12	—	—
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Single-factor experiment (completely randomized design) with $a = 4$ temperatures (treatments) and unequal numbers of observations (5, 4, 5, 4; $N = 18$). Model: $y(ij) = \mu + \tau(i) + \varepsilon(ij)$. Hypotheses: $H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$ (the temperature has no effect on the density) against H_1 : at least one $\tau(i) \neq 0$.

Temperature	Observations n	Total y(i.)	Average
100	5	108.7	21.740
125	4	86.0	21.500
150	5	108.6	21.720
175	4	86.8	21.700
All	18	390.1	21.672

$$SS_T = \sum y_{ij}^2 - \frac{y^2}{N} = 0.5161$$

$$SS_{Treatments} = \sum \frac{y_i^2}{n_i} - \frac{y^2}{N} = 0.1561$$

$$SS_E = SS_T - SS_{Treatments} = 0.3600$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F ₀	F(0.05, 3, 14)
Temperature (between)	0.1561	3	0.0520	2.02	3.34
Error (within)	0.3600	14	0.0257		
Total	0.5161	17			

Since $F_0 = 2.02 < \text{the critical value } F(0.05; 3, 14) = 3.34$ (P-value = 0.157), the null hypothesis cannot be rejected: at the 5% level there is no evidence that the fusing temperature affects the density of the bricks.

(ii) Residual analysis

The residual is the difference between an observation and its treatment average, $e(ij) = y(ij) - \bar{y}(i.)$. The ANOVA assumes that the errors are normally and independently distributed with the same variance in each factor level; this is checked from the residuals.

Temperature	Observation 1	Observation 2	Observation 3	Observation 4	Observation 5
100	0.060	0.160	-0.040	-0.140	-0.040
125	0.200	-0.100	0.000	-0.100	
150	0.180	0.080	0.080	-0.120	-0.220
175	0.200	0.000	0.100	-0.300	

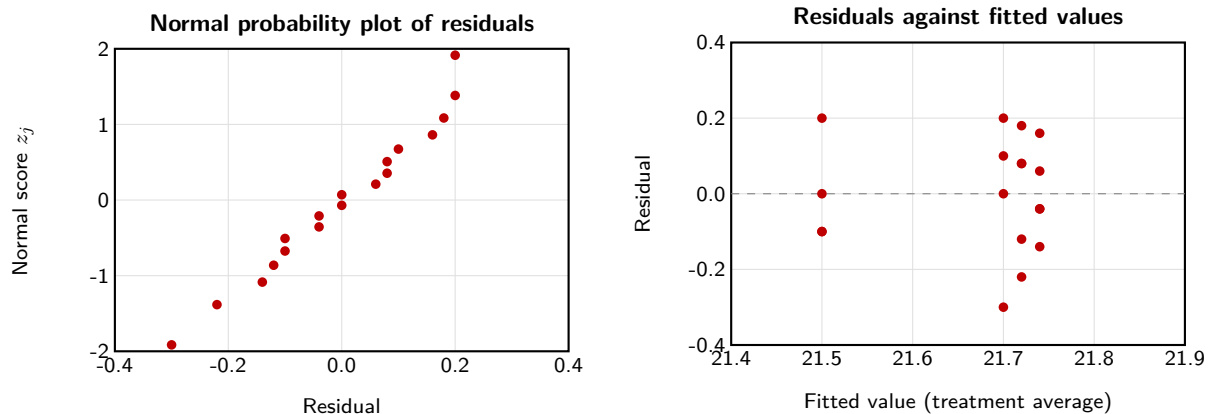


Figure 40: Residual plots for the brick-density experiment

The points of the normal probability plot lie close to a straight line, so the normality assumption is reasonable; the spread of the residuals does not depend on the fitted value or on the treatment (no funnel shape), so the variances appear equal, and no pattern suggests the need for a data transformation. The independence assumption would be checked by plotting the residuals against the run order, which is not available here. The assumptions of the ANOVA are therefore satisfied, and the conclusion of part (i) stands.

94	A researcher is testing the effect of three different fertilizers (AAA, BBB and CCC) on crop yield (in kg). The researcher conducts an experiment with 4 replicates per fertilizer type. The observed yields are: AAA: 22, 24, 20, 23; BBB: 28, 27, 29, 30; CCC: 25, 26, 24, 27. Perform a one-way ANOVA to determine if there is a significant difference in mean yield among the fertilizers at a 5% significance level. Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q8b	08	3	3
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Hypotheses: $H_0: \tau(A) = \tau(B) = \tau(C) = 0$ (mean yield is the same for all fertilizers) against H_1 : at least one mean differs.
 $a = 3$ treatments, $n = 4$ replicates, $N = 12$.

Fertilizer	Observations	Total y(i.)	Average
AAA	22, 24, 20, 23	89	22.25
BBB	28, 27, 29, 30	114	28.50
CCC	25, 26, 24, 27	102	25.50
All		305	25.417

$$\text{Correction factor} = \frac{y_{..}^2}{N} = \frac{305^2}{12} = 7752.083$$

$$SS_T = \sum y^2 - CF = 7849 - 7752.083 = 96.917$$

$$SS_{Treatments} = \frac{89^2 + 114^2 + 102^2}{4} - CF = 78.167$$

$$SS_E = SS_T - SS_{Treatments} = 18.750$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F_0	F(0.05)
Fertilizer (between treatments)	78.17	2	39.08	18.76	4.26
Error (within treatments)	18.75	9	2.0833		
Total	96.92	11			

$F_0 = MS(\text{treatments})/MS(\text{error}) = 39.083/2.083 = 18.76$ is greater than $F(0.05; 2, 9) = 4.26$ (P-value = 0.0006), so H_0 is rejected.

Answer: There is a significant difference in mean yield among the three fertilizers at the 5% level ($F_0 = 18.76 > 4.26$). BBB gives the highest average yield (28.50 kg), then CCC (25.50 kg) and AAA (22.25 kg). (The ANOVA only shows that the means differ, not which; comparison of the treatment averages with the reference distribution of standard deviation $\sqrt{(MSE/n)} = 0.72$ points to BBB as the best.)

95	A chemical engineer has conducted an experiment to study the effect of temperature and pressure on the reaction time of a chemical process. Let the two factors temperature and pressure be denoted by A and B respectively. The following results are obtained from this study; the number of replications is two. Pressure low (100 MPa): temperature low (60°C) 40, 37; temperature high (70°C) 43, 50. Pressure high (150 MPa): temperature low 59, 54; temperature high 37, 43. Estimate the factor effects. Which effects are large? Perform an analysis of variance. What are your conclusions? Source: Unit IV SEE Feb/Mar 2025 (IM353IA), Q7b	12	3	4
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This is a 2^2 factorial design with $n = 2$ replicates. The treatment combination totals are: (1) = A low, B low = 40 + 37 = 77; a = A high, B low = 43 + 50 = 93; b = A low, B high = 59 + 54 = 113; ab = A high, B high = 37 + 43 = 80. Grand total = 363.

Factor effects

$$A = \frac{a+ab-b-(1)}{2n} = \frac{93+80-113-77}{4} = -4.25$$

$$B = \frac{b+ab-a-(1)}{2n} = \frac{113+80-93-77}{4} = 5.75$$

$$AB = \frac{ab+(1)-a-b}{2n} = \frac{80+77-93-113}{4} = -12.25$$

The interaction effect AB (-12.25) is by far the largest in magnitude; the main effect of B (pressure) is 5.75 and that of A (temperature) -4.25. Raising the temperature reduces the reaction time at high pressure (80 against 113, totals) but increases it at low pressure (93 against 77), so the effect of temperature depends on the pressure level.

Analysis of variance

$$SS_A = \frac{[a+ab-b-(1)]^2}{4n} = \frac{(-17)^2}{8} = 36.125$$

$$SS_B = \frac{(23)^2}{8} = 66.125$$

$$SS_{AB} = \frac{(-49)^2}{8} = 300.125$$

$$SS_T = \sum y^2 - \frac{y_{...}^2}{4n} = 461.875$$

$$SS_E = SS_T - SS_A - SS_B - SS_{AB} = 59.500$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F0	F(0.05)	Conclusion
Temperature (A)	36.12	1	36.12	2.43	7.71	Not significant
Pressure (B)	66.12	1	66.12	4.45	7.71	Not significant
Interaction (AB)	300.12	1	300.12	20.18	7.71	Significant
Error	59.50	4	14.88			
Total	461.88	7				

Answer: Only the AB interaction is large and significant ($F_0 = 20.18 > F(0.05; 1, 4) = 7.71$). The main effects of temperature ($F_0 = 2.43$) and pressure ($F_0 = 4.45$) are not significant on their own, but because of the interaction they cannot be interpreted separately: the temperature and pressure must be set together (reaction time is shortest at low temperature with low pressure, or high temperature with high pressure).

96	An experiment investigates the effect of glass type and phosphor type on the brightness of a television tube. The response measured is the current necessary (in microamps) to obtain a specific brightness level. Glass type 1: phosphor type 1: 280, 290, 285; phosphor type 2: 300, 310, 295; phosphor type 3: 290, 285, 290. Glass type 2: phosphor type 1: 230, 235, 240; phosphor type 2: 260, 240, 235; phosphor type 3: 220, 225, 230. Analyze the data and draw conclusions. Source: Unit IV SEE Fast Track Oct 2020 (16IM53), Q7a	10	—	—
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Two-factor factorial (fixed effects): factor A = glass type (a = 2 levels), factor B = phosphor type (b = 3 levels), n = 3 replicates, N = 18. Model: $y(ijk) = \mu + \tau(i) + \beta(j) + (\tau\beta)(ij) + \epsilon(ijk)$. Hypotheses: no effect of glass type, no effect of phosphor type, and no interaction.

	Phosphor 1	Phosphor 2	Phosphor 3	Row total
Glass 1 (cell totals)	855	905	865	2625
Glass 2 (cell totals)	705	735	675	2115
Column total	1560	1640	1540	4740

$$SS_T = \sum y^2 - \frac{y_{..}^2}{abn} = 16150.00$$

$$SS_A = \frac{\sum y_{i..}^2}{bn} - CF = 14450.00$$

$$SS_B = \frac{\sum y_{.j.}^2}{an} - CF = 933.33$$

$$SS_{AB} = \frac{\sum y_{ij.}^2}{n} - CF - SS_A - SS_B = 133.33$$

$$SS_E = SS_T - SS_A - SS_B - SS_{AB} = 633.33$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F0	F(0.05)	Conclusion
Glass type (A)	14450.00	1	14450.00	273.79	4.75	Significant
Phosphor type (B)	933.33	2	466.67	8.84	3.89	Significant
Glass × phosphor (AB)	133.33	2	66.67	1.26	3.89	Not significant
Error	633.33	12	52.78			
Total	16150.00	17				

The effects with F_0 larger than the tabulated F (glass type, P-value 0.0000, and phosphor type, P-value 0.0044) are significant at the 5% level, whereas the glass × phosphor interaction (P-value 0.3178) is not.

Average current (microamps): glass 1: 285.0, 301.7, 288.3; glass 2: 235.0, 245.0, 225.0 for phosphor types 1, 2 and 3. Phosphor type 2 needs the most current on both glasses, and the lines of the interaction graph (cell averages against phosphor type) are roughly parallel, which agrees with the non-significant interaction, so the two factors act approximately additively. Glass type 2 needs much less current than glass type 1 for every phosphor.

Answer: Glass type ($F_0 = 273.8$) and phosphor type ($F_0 = 8.8$) significantly affect the current required for the given brightness; the glass × phosphor interaction ($F_0 = 1.26$) is not significant at the 5% level. The combination that needs the lowest current is glass type 2 with phosphor type 3 (average about 225 microamps).

97	The quality control department of an automobile company is studying the effects of several factors on the surface finish of a spindle that is manufactured using a lathe. Three operators and two machines were selected and two spindles were produced under each combination of the experimental conditions. The results are: operator 1: machine 1: 23, 27; machine 2: 25, 26. Operator 2: machine 1: 36, 34; machine 2: 36, 39. Operator 3: machine 1: 24, 38; machine 2: 37, 34. Assume a significance level of 0.05. (i) Write the model of ANOVA of this experiment. (ii) Check the significance of each of the components of the model. Source: Unit IV SEE Nov/Dec 2019 (16IM53), Q7b	12	—	—
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(i) Model

The experiment is a two-factor factorial ($a = 3$ operators, $b = 2$ machines, $n = 2$ replicates) analyzed with the fixed effects model

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}, \quad i = 1, 2, 3; j = 1, 2; k = 1, 2$$

where μ is the overall mean, $\tau(i)$ the effect of operator i , $\beta(j)$ the effect of machine j , $(\tau\beta)(ij)$ the operator × machine interaction and $\varepsilon(ijk)$ a random error, $NID(0, \sigma^2)$. The hypotheses are H_0 : all $\tau(i) = 0$; H_0 : all $\beta(j) = 0$; H_0 : all $(\tau\beta)(ij) = 0$.

(ii) Analysis of variance

Operator	Machine 1 total	Machine 2 total	Row total
Operator 1	50	51	101
Operator 2	70	75	145
Operator 3	62	71	133
Column total	182	197	379

$$CF = \frac{379^2}{12} = 11970.08, \quad SS_T = \sum y^2 - CF = 402.92$$

$$SS_A = \frac{\sum y_{i..}^2}{bn} - CF = 258.67, \quad SS_B = \frac{\sum y_{.j.}^2}{an} - CF = 18.75$$

$$SS_{AB} = \frac{\sum y_{ij.}^2}{n} - CF - SS_A - SS_B = 8.00, \quad SS_E = 117.50$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F ₀	F(0.05)	Conclusion
Operators (A)	258.67	2	129.33	6.60	5.14	Significant
Machines (B)	18.75	1	18.75	0.96	5.99	Not significant
Interaction (AB)	8.00	2	4.00	0.20	5.14	Not significant
Error	117.50	6	19.58			
Total	402.92	11				

Answer: At the 5% level: operators ($F_0 = 6.60$ against $F(0.05; 2, 6) = 5.14$) are significant; machines ($F_0 = 0.96$ against $F(0.05; 1, 6) = 5.99$) are not significant; operator $\times$ machine interaction ($F_0 = 0.20$ against 5.14) is not significant.

98	The data below are from a cattle experiment in which temperature, protein and gender were evaluated as to the efficiency of weight production (feed per pound of gain). Two replicates of each treatment were run according to a completely randomized design. House temperature 90°: protein high: male 2.96, 3.04; female 2.69, 2.75; protein low: male 2.80, 2.72; female 2.60, 2.70. House temperature 60°: protein high: male 2.89, 2.75; female 2.73, 2.67; protein low: male 2.94, 3.00; female 2.85, 2.75. Conduct ANOVA for main effects and interaction effects. Draw suitable conclusions. Source: Unit IV SEE Jan/Feb 2021 (18IM53), Q7b	12	—	—
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Three factors at two levels each, with $n = 2$ replicates: a 2^3 factorial design in a completely randomized design (16 runs). Coding: temperature A (60° = −, 90° = +), protein B (low = −, high = +), gender C (male = −, female = +). The effects are found from the contrasts of the treatment totals:

$$\text{Effect} = \frac{\text{Contrast}}{4n}, \quad SS = \frac{(\text{Contrast})^2}{8n}, \quad F_0 = \frac{MS}{MS_E}$$

House temperature	Protein level	Gender	Replicate 1	Replicate 2	Total
90°	High	M	2.96	3.04	6.00
90°	High	F	2.69	2.75	5.44
90°	Low	M	2.80	2.72	5.52
90°	Low	F	2.60	2.70	5.30
60°	High	M	2.89	2.75	5.64
60°	High	F	2.73	2.67	5.40
60°	Low	M	2.94	3.00	5.94
60°	Low	F	2.85	2.75	5.60

$$y_{....} = 44.84, \quad CF = \frac{(44.84)^2}{16} = 125.6641, \quad SS_T = \sum y^2 - CF = 0.2511$$

$$SS_E = SS_T - \sum SS_{effects} = 0.0316 \quad (8 \text{ d.f.}), \quad MS_E = 0.00395$$

Source of variation	Contrast	Effect	Sum of squares	d.f.	Mean square	F ₀	Significant at 5%?
Temperature (A)	-0.32	-0.0400	0.0064	1	0.0064	1.62	No
Protein level (B)	0.12	0.0150	0.0009	1	0.0009	0.23	No
Gender (C)	-1.36	-0.1700	0.1156	1	0.1156	29.27	Yes
A $\times$ B	1.12	0.1400	0.0784	1	0.0784	19.85	Yes
A $\times$ C	-0.20	-0.0250	0.0025	1	0.0025	0.63	No

$B \times C$	-0.24	-0.0300	0.0036	1	0.0036	0.91	No
$A \times B \times C$	-0.44	-0.0550	0.0121	1	0.0121	3.06	No
Error			0.0316	8	0.0039		
Total			0.2511	15			

The critical value is $F(0.05; 1, 8) = 5.32$.

Cell averages (feed per pound of gain, lower is better): temperature 90°: high protein 2.860, low protein 2.705; temperature 60°: high protein 2.760, low protein 2.885. Males average 2.888 and females 2.718.

Answer: At the 5% level Gender (C) and $A \times B$ are significant; the remaining effects are not. Females use less feed per pound of gain than males (2.718 against 2.888). The temperature $\times$ protein interaction means that the effect of the protein level depends on the house temperature: at 90° low protein is better (2.705 against 2.860), at 60° high protein is better (2.760 against 2.885); so the best (lowest feed per pound of gain) condition is 90° with low protein.

99	A single factor experiment is to be designed for estimating the tensile strength of a composite material. One of the constituents of the composite material varies in the combination at 5%, 10% and 20%. Six specimens are prepared for each combination. Explain the process of data tabulation for a completely randomized design. Source: Unit IV SEE Fast Track Jul 2019 (16IM53), Q7a	08	—	—
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This is a completely randomized single-factor experiment: one factor (the percentage of the constituent) with $a = 3$ levels (treatments: 5%, 10% and 20%) and $n = 6$ replicates (specimens) per treatment, so $N = a \times n = 18$ specimens.

Step 1: randomize the run order

The 18 specimens (6 per level) are tested in a random order so that the effect of nuisance variables (such as a warm-up effect of the tensile tester) is balanced out. For example, a random order of the tests (test position: constituent level) is:

1: 10%	2: 5%	3: 10%	4: 10%	5: 5%	6: 5%
7: 5%	8: 10%	9: 20%	10: 20%	11: 10%	12: 5%
13: 5%	14: 20%	15: 20%	16: 10%	17: 20%	18: 20%

Step 2: tabulate the data

The tensile strengths are recorded in a table with the treatment (level) in the rows and the replicates in the columns; $y(ij)$ is the j th observation under treatment i :

Constituent level	Replicates 1	2	3	4	5	6	Total $y(i.)$	Average
5%	$y(1,1)$	$y(1,2)$	$y(1,3)$	$y(1,4)$	$y(1,5)$	$y(1,6)$	$y(1.)$	$\bar{y}(1.)$
10%	$y(2,1)$	$y(2,2)$	$y(2,3)$	$y(2,4)$	$y(2,5)$	$y(2,6)$	$y(2.)$	$\bar{y}(2.)$
20%	$y(3,1)$	$y(3,2)$	$y(3,3)$	$y(3,4)$	$y(3,5)$	$y(3,6)$	$y(3.)$	$\bar{y}(3.)$
Grand total							$y(..)$	$\bar{y}(..)$

The notation is: $y(i.) = \sum(j) y(ij)$ is the total of the observations under treatment i , $\bar{y}(i.) = y(i.)/n$ its average, $y(..) = \sum(i)\sum(j) y(ij)$ the grand total and $\bar{y}(..) = y(..)/N$ the grand mean, with $N = 18$. These totals are the starting point of the analysis of variance (the sums of squares are calculated from them).

100	Explain the application of a 'linear statistical model' and ANOVA for the above case (the tensile strength of a composite material at 5%, 10% and 20% of a constituent, six specimens each). Source: Unit IV SEE Fast Track Jul 2019 (16IM53), Q7b	08	—	—
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Linear statistical model

The observations are described by the linear statistical model for the single-factor experiment

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3; \quad j = 1, 2, \dots, 6$$

where y_{ij} is the j th tensile strength observed at treatment i (percentage level), μ is the overall mean common to all treatments, $\tau(i)$ is the effect of the i th treatment (defined as the deviation from the overall mean so that $\sum \tau(i) = 0$) and $\varepsilon(ij)$ is a random error component, assumed normally and independently distributed with mean zero and variance σ^2 . Each treatment therefore defines a normal population with mean $\mu(i) = \mu + \tau(i)$ and the common variance σ^2 . This is the fixed effects model.

Analysis of variance

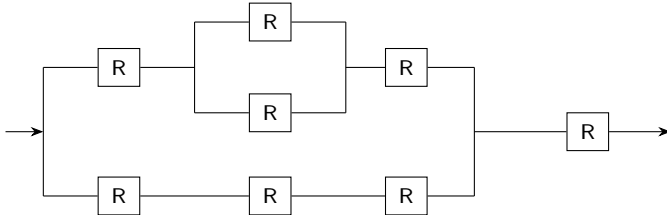
The hypotheses are $H_0: \tau_1 = \tau_2 = \tau_3 = 0$ (the percentage of the constituent has no effect on the tensile strength) against $H_1: \tau(i) \neq 0$ for at least one i . The ANOVA partitions the total variability into a component between treatments and a component within treatments (error):

$$SS_T = SS_{Treatments} + SS_E, \quad SS_T = \sum_i \sum_j y_{ij}^2 - \frac{y_{..}^2}{N}, \quad SS_{Treatments} = \sum_i \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N}$$

Source of variation	Sum of squares	Degrees of freedom	Mean square	F_0
Between treatments (constituent %)	SS(Treatments)	$a - 1 = 2$	MS(Treatments) = SS(Treatments)/2	MS(Treatments)/MS(E)
Error (within treatments)	SS(E)	$N - a = 15$	MS(E) = SS(E)/15	
Total	SS(T)	$N - 1 = 17$		

If H_0 is true, the ratio $F_0 = \text{MS(Treatments)}/\text{MS(E)}$ follows an F distribution with 2 and 15 degrees of freedom, so H_0 is rejected at the 5% level when $F_0 > F(0.05; 2, 15) = 3.68$; that is, the constituent percentage affects the tensile strength. If H_0 is rejected, the treatment averages $\bar{y}(i.)$ are compared (for example with a plot against a reference distribution of standard deviation $\sqrt{\text{MS(E)}/6}$) to find which levels differ.

Model adequacy is checked from the residuals $e(ij) = y(ij) - \bar{y}(i.)$: a normal probability plot (normality), residuals against the fitted values and the factor levels (equal variances) and against the run order (independence).

101	Consider the following series-parallel configuration: the upper branch consists of a component R in series with two parallel components R, R and then a component R; the lower branch consists of three components R in series; the two branches are in parallel and are followed by a component R in series.  (i) Derive the expression for the system reliability in terms of the component reliabilities, stating clearly the assumption made. (ii) Compute the system reliability for $R = 0.9$ and $R = 0.95$ (and $R = 0.96$). Source: Unit V Improvement CIE, 13/01/2026, Q1; Improvement CIE, 28/01/2025, Q3 ($R = 0.96$)	10	2	3
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(i) Expression for the system reliability

Assumption: the components fail independently of one another, and each component has the same reliability R over the period considered. Then the reliabilities of series blocks multiply and the unreliabilities of parallel blocks multiply.

- (1) The two parallel components in the upper branch form a block of reliability $R_p = 1 - (1 - R)^2 = 2R - R^2$.
- (2) The upper branch has the first component, the parallel block and the last component in series: $R_u = R \cdot (2R - R^2) \cdot R = R^3(2 - R)$.
- (3) The lower branch has three components in series: $R_l = R^3$.
- (4) The two branches are in parallel: $R_b = 1 - (1 - R_u)(1 - R_l) = 1 - [1 - R^3(2 - R)](1 - R^3)$.
- (5) The output component R is in series with this block, so

$$R_s = R \left\{ 1 - [1 - R^3(2 - R)](1 - R^3) \right\}$$

(ii) Numerical values

R	Upper branch $R^2(2R - R^2)$	Lower branch R^3	Two branches in parallel	System reliability
0.90	0.8019	0.7290	0.9463	0.8517
0.95	0.9002	0.8574	0.9858	0.9365
0.96	0.9201	0.8847	0.9908	0.9512

$$R = 0.9 : R_u = 0.9^3(2 - 0.9) = 0.8019, R_l = 0.729$$

$$R_b = 1 - (1 - 0.8019)(1 - 0.729) = 0.9463, R_s = 0.9 \times 0.9463 = 0.8517$$

Answer: $R_s = 0.8517$ for $R = 0.9$, 0.9365 for $R = 0.95$ and 0.9512 for $R = 0.96$.

102	Derive and explain the reliability for two components in parallel.	10	4	3
Source: Unit V Improvement CIE, 13/01/2026, Q3				

Two components are in **parallel** when the system works if at least one of them works, so it fails only if **both** fail. Let the reliabilities of the components be R_1 and R_2 , and assume that the components fail independently.

Derivation

- (1) The probability that component 1 fails is $1 - R_1$, and that component 2 fails is $1 - R_2$.
- (2) By independence, the probability that both fail (the system fails) is the product $(1 - R_1)(1 - R_2)$, which is the unreliability of the system.
- (3) The system reliability is the complement:

$$R_s = 1 - (1 - R_1)(1 - R_2) = R_1 + R_2 - R_1R_2$$

For identical components, $R_s = 1 - (1 - R)^2 = 2R - R^2$, and for n components $R_s = 1 - (1 - R)^n$.

Explanation

- R_s is **greater than the larger** of R_1 and R_2 , because $R_s - R_1 = R_2(1 - R_1) > 0$; adding a parallel component always increases the reliability (redundancy), with diminishing returns.
- **Example:** $R_1 = 0.80$ and $R_2 = 0.85$ give $R_s = 1 - (0.20)(0.15) = 0.97$.
- **Exponential components** with failure rates λ_1 and λ_2 : $R_s(t) = e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1 + \lambda_2)t}$. For identical components, $R_s(t) = 2e^{-(\lambda)t} - e^{-2\lambda t}$, and the mean time to failure, $\int R_s(t) dt = 2/\lambda - 1/(2\lambda) = 3/(2\lambda)$, is 1.5 times that of a single component. For $\lambda = 0.001$ per hour and $t = 500$ h: $R = e^{-0.5} = 0.6065$ for one component, and $R_s = 0.8452$ for the pair.
- If the second component is a standby that is switched on when the first fails (perfect switching), the system life is gamma distributed with $r = 2$ and $R(t) = e^{-\lambda t}(1 + \lambda t)$, which is higher than for active parallel redundancy.

103	A participant in a motor rally is required to complete a mission of 2500 km. His vehicle can be imagined to have 3 subsystems, namely the fuel injection system, the ignition system and other systems. The mean distance to repair the 3 subsystems is known to be 6000 km, 8000 km and 10000 km respectively. Find the reliability of the motorist completing the mission without repair. Source: Unit V SEE Feb/Mar 2026 (IM353IA), Q9a	08	3	4
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The distance travelled before a subsystem needs repair is modelled by the exponential distribution (distance plays the role of time), so for a subsystem with a mean distance to repair θ the failure rate is $\lambda = 1/\theta$ per km and the probability of completing a distance x without repair is $R(x) = e^{-x/\theta}$. The three subsystems are in series (a failure of any one of them stops the mission) and are assumed to fail independently, so the reliabilities multiply.

Subsystem	Mean distance θ (km)	$\lambda = 1/\theta$ (per km)	$R(2500) = e^{-2500/\theta}$
Fuel injection	6000	0.000167	0.6592
Ignition	8000	0.000125	0.7316
Other systems	10000	0.000100	0.7788

$$R_s = e^{-2500/6000} e^{-2500/8000} e^{-2500/10000} = e^{-(0.4167+0.3125+0.2500)} = e^{-0.9792} = 0.3756$$

Answer: The reliability of completing the 2500 km mission without repair is 0.3756 (about 37.6%).

104	The MTBF of a certain unit is 100 hours. Calculate the reliability of the unit for 150 hours of operation. If the reliability of the unit is to be increased by 15%, 25%, 35%, 45% and 55%, calculate the corresponding % changes in the MTBF that are necessary. Plot % change in reliability vs % change in MTBF. Source: Unit V SEE Feb/Mar 2026 (IM353IA), Q9b	08	3	4
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For a unit with an exponentially distributed time to failure, the mean (MTBF, mean time to failure) is $\theta = 1/\lambda$, and the reliability for a period t is $R(t) = e^{-t/\theta}$.

$$R(150) = e^{-150/100} = e^{-1.5} = 0.2231$$

To raise the reliability by $x\%$ the new reliability must be $R' = R(1 + x/100)$; the MTBF that gives R' over the same 150 hours follows from $R' = e^{-150/\theta'}$, that is $\theta' = 150/(-\ln R')$.

Increase in reliability	Required reliability $R' = R(1 + x)$	Required MTBF = $150/(-\ln R')$ (h)	Change in MTBF
15%	0.2566	110.3	10.3%
25%	0.2789	117.5	17.5%
35%	0.3012	125.0	25.0%
45%	0.3235	132.9	32.9%
55%	0.3459	141.3	41.3%

$$\text{For 15\% : } R = 0.2231(1.15) = 0.2566, \quad \theta = \frac{150}{-\ln 0.2566} = 110.3 \text{ h}, \quad \text{change} = \frac{110.3-100}{100} = 10.3\%$$

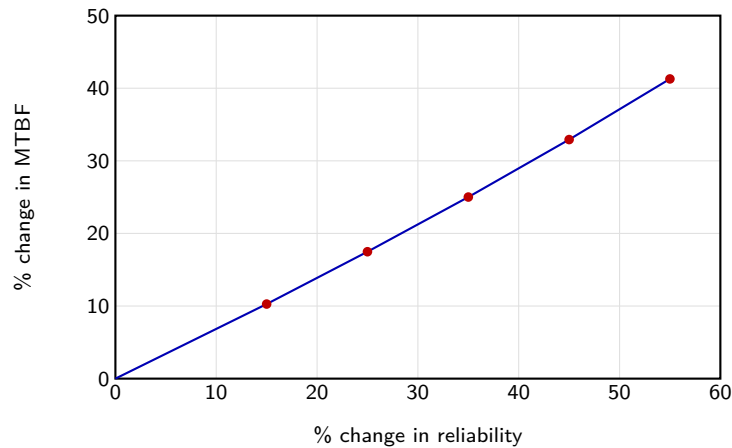
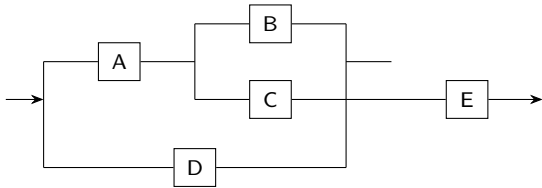


Figure 41: Percentage change in MTBF needed for a given percentage increase in reliability

Answer: $R(150\text{ h}) = 0.2231$. The MTBF must increase by 10.3%, 17.5%, 25.0%, 32.9% and 41.3% for reliability increases of 15%, 25%, 35%, 45% and 55%; the required MTBF grows faster than the reliability gain (the curve is convex).

105	Distinguish between series and parallel systems (configurations of system reliability). Give suitable examples.	04	3	3
Source: Unit V SEE Feb/Mar 2026 (IM353IA), Q10a (ii); SEE Jan/Feb 2021 (18IM53), Q8a				

Basis	Series system	Parallel system
Working condition	The system works only if all the components work.	The system works if at least one component works.
Failure	The failure of any one component fails the system.	The system fails only when all the components fail.
Reliability (independent components)	$R_s = R_1 R_2 \dots R_n$ (product)	$R_s = 1 - (1 - R_1)(1 - R_2) \dots (1 - R_n)$
Effect of adding components	Reliability decreases; it is less than that of the least reliable component.	Reliability increases (redundancy); it is more than that of the most reliable component.
Exponential components	$R_s(t) = e^{-\lambda(s)t}$ with $\lambda(s) = \sum \lambda_i$ (the failure rates add), so $MTTF = 1/\lambda(s)$	$R_s(t) = 1 - \prod (1 - e^{-\lambda_i t})$; a longer mean life than any single component
Examples	Starting system of a car (battery, starter, ignition); production line of several machines; a string of series-wired lights	Two blowers or two pumps of which one is enough; dual hydraulic brake circuits; an aircraft with duplicate computers

106	Consider the following series-parallel configuration: component A is in series with the parallel pair B and C; this block is in parallel with component D; and the combination is in series with component E.  Determine the reliability of the system for 20 hours of operating period. The failure rates per hour are $\lambda_A = 0.01$, $\lambda_B = 0.0125$, $\lambda_C = 0.02$, $\lambda_D = 0.02$, $\lambda_E = 0.025$. Source: Unit V SEE Feb/Mar 2026 (IM353IA), Q10b	08	3	3
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The components have exponentially distributed lives, so each has the reliability $R(t) = e^{-\lambda t}$ over $t = 20$ h; the components fail independently.

Component	λ (per hour)	$R(20) = e^{-20\lambda}$
A	0.01	0.8187
B	0.0125	0.7788
C	0.02	0.6703
D	0.02	0.6703
E	0.025	0.6065

Reduce the diagram step by step:

$$R_{BC} = 1 - (1 - R_B)(1 - R_C) = 1 - (1 - 0.7788)(1 - 0.6703) = 0.9271$$

$$R_{ABC} = R_A R_{BC} = 0.8187 \times 0.9271 = 0.7590$$

$$R_{ABCD} = 1 - (1 - R_{ABC})(1 - R_D) = 1 - (1 - 0.7590)(1 - 0.6703) = 0.9206$$

$$R_s = R_{ABCD} R_E = 0.9206 \times 0.6065 = 0.5583$$

Answer: The reliability of the system for 20 hours of operation is 0.5583.

107	In a rerolling process two blowers are connected to reheat the furnace using pulverized coal. One of the blowers is standby equipment. The blower system includes a stator, motor, coupling and blower with the reliabilities 0.9, 0.92, 0.85 and 0.95 respectively. The pulverized coal system has a reliability of 0.99 and the furnace 0.95 for an operating period of 250 hours. Draw the system configuration and determine the reliability of the total system for 250 hours of operation. Source: Unit V SEE Feb/Mar 2025 (IM353IA), Q9b	08	3	4
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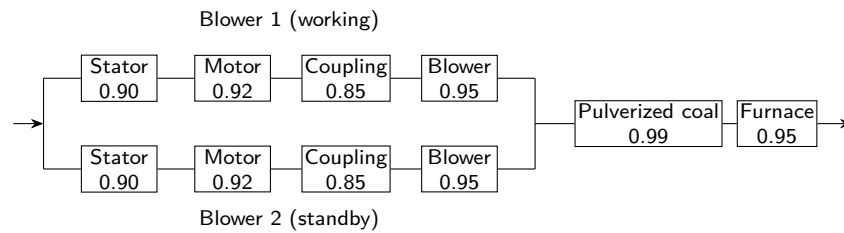
System configuration

Figure 42: Reliability block diagram: two blower systems in parallel (one is standby), in series with the pulverized coal system and the furnace

Reliability

Each blower system is a series chain of four components, assumed to fail independently. The standby blower is treated as a parallel (redundant) unit, with the given reliabilities for the 250-hour period and perfect switching.

$$R_{blower} = 0.9 \times 0.92 \times 0.85 \times 0.95 = 0.6686$$

$$R_{blowers} = 1 - (1 - R_{blower})^2 = 1 - (1 - 0.6686)^2 = 0.8902$$

$$R_s = R_{blowers} \times 0.99 \times 0.95 = 0.8902 \times 0.99 \times 0.95 = 0.8372$$

Answer: The reliability of the total system for 250 hours is 0.8372 (a single blower system alone would give 0.6288).

108	An amplifier has an exponential time-to-failure distribution with a failure rate of 10% per 2000 hours. What is the reliability of the amplifier at 8000 hours? Find the mean time to failure.	06	3	3
Source: Unit V SEE Feb/Mar 2025 (IM353IA), Q10b				

For the exponential distribution $f(t) = \lambda e^{-\lambda t}$, the reliability is $R(t) = e^{-\lambda t}$ and the mean time to failure is $1/\lambda$. A failure rate of 10% per 2000 hours is taken as $\lambda = 0.10/2000$ per hour.

$$\lambda = \frac{0.10}{2000} = 0.00005 \text{ per hour}$$

$$R(8000) = e^{-\lambda t} = e^{-(0.00005)(8000)} = e^{-0.4} = 0.6703$$

$$MTTF = \frac{1}{\lambda} = \frac{1}{0.00005} = 20,000 \text{ hours}$$

Answer: $R(8000 \text{ h}) = 0.6703$; $MTTF = 20,000$ hours. (The probability of failing before the mean life is 0.632, whatever the value of λ .)

109	Define the following terms: (i) reliability, (ii) failure rate, (iii) MTBF.	03	—	—
Source: Unit V SEE Mar/Apr 2023 (18IM53), Q8a				

(i) Reliability: the probability $R(t)$ that a component or system does not fail (survives) beyond time t , $R(t) = P(T > t) = 1 - F(t)$, where T is the time to failure. For the exponential model $R(t) = e^{-\lambda t}$.

(ii) Failure rate: the parameter λ of the exponential time-to-failure distribution, the number of failures per unit time (per hour); it is constant over the life of the item in this model.

(iii) MTBF: the mean time between (successive) failures, i.e. the mean life; in the exponential model it is the mean of the distribution, $1/\lambda$ (the textbook's mean time to failure, MTTF). For an MTBF of 100 hours, $\lambda = 0.01$ per hour.

110	A component has a failure rate which is according to the exponential failure law, $\lambda = 4 \times 10^{-3}$ failures/hour. (i) Determine the reliability of this component for $t = 10$ h, 100 h, 1000 h and 10000 h. (ii) Plot the time versus reliability graph. Comment on the plot that you have obtained. Source: Unit V SEE Mar/Apr 2023 (18IM53), Q8b	05	–	–
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(i) Reliability

$R(t) = e^{-\lambda t}$ with $\lambda = 0.004$ per hour (mean time to failure $1/\lambda = 250$ h).

t (h)	λt	$R(t) = e^{-\lambda t}$
10	0.04	0.9608
100	0.40	0.6703
1000	4.00	0.0183
10000	40.00	4.25e-18

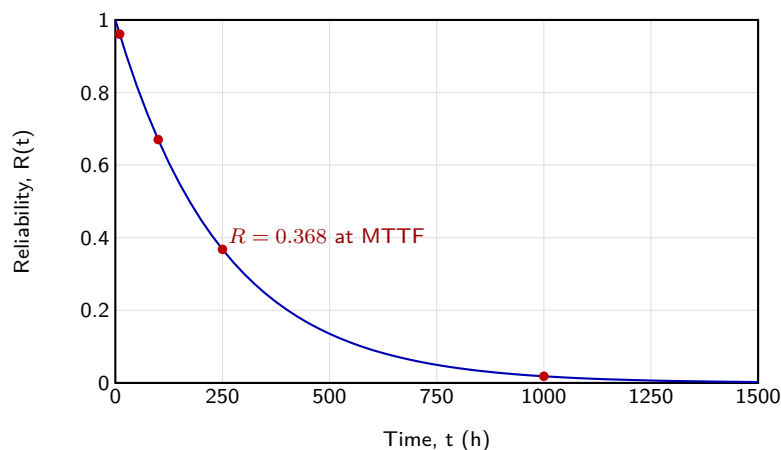
(ii) Time versus reliability graph

Figure 43: Reliability against time for $\lambda = 0.004$ per hour (exponential failure law)

Comment: the reliability falls exponentially with time: it is nearly 1 for a short mission (0.961 at 10 h), 0.670 at 100 h, only 0.018 at 1000 h and practically zero (4×10^{-18}) at 10,000 h. At the mean life $t = 1/\lambda = 250$ h the reliability is only $e^{-1} = 0.368$, so 63.2% of the units have failed by the mean life. The slope of the curve is steepest at the start and becomes flatter later, because the failure rate is constant while the number of survivors decreases.

111	Three components are in series with failure rates: component 1, $\lambda_1 = 4 \times 10^{-6}$ failures/hour; component 2, $\lambda_2 = 4 \times 10^{-4}$ failures/hour; component 3, $\lambda_3 = 6 \times 10^{-6}$ failures/hour. Determine the reliability of this series system configuration. If the three components were arranged in parallel configuration, what would be the system reliability? (Assume $t = 1000$ h.) Source: Unit V SEE Mar/Apr 2023 (18IM53), Q8c	08	–	–
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Component	λ (per hour)	$R(1000) = e^{-1000\lambda}$
1	4e-06	0.996008
2	4e-04	0.670320
3	6e-06	0.994018

Series configuration

$$R_s = R_1 R_2 R_3 = e^{-(\lambda_1 + \lambda_2 + \lambda_3)t} = e^{-(0.000410)(1000)} = e^{-0.41} = 0.663650$$

Parallel configuration

$$R_p = 1 - (1 - R_1)(1 - R_2)(1 - R_3) = 1 - (0.003992)(0.329680)(0.005982) = 0.9999921$$

Answer: Series: $R_s = 0.6637$; parallel: $R_p = 0.999992$. The series system is limited by its weakest component (component 2), whereas the parallel arrangement is almost certain to survive 1000 hours.

112	A system has three units with failure rates of 1.5, 4 and 3.8 failures per 10^6 hours respectively. Determine (i) the MTBF of the system; (ii) the reliability of the system for 10 hours if the components are connected in series; (iii) the reliability if these components are connected in parallel. Source: Unit V SEE Jan/Feb 2021 (18IM53), Q8b	08	–	–
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The units have exponential lives with $\lambda_1 = 1.5 \times 10^{-6}$, $\lambda_2 = 4 \times 10^{-6}$ and $\lambda_3 = 3.8 \times 10^{-6}$ failures per hour and fail independently.

(i) MTBF of the system (series)

For a series system the failure rates add, $\lambda_s = \sum \lambda_i$, and the mean life is $1/\lambda_s$.

$$\lambda_s = (1.5 + 4 + 3.8) \times 10^{-6} = 9.3e-06 \text{ per hour}$$

$$MTBF = \frac{1}{\lambda_s} = 107,527 \text{ hours}$$

(ii) Reliability for 10 hours, series

$$R_s(10) = e^{-\lambda_s t} = e^{-(9.3 \times 10^{-6})(10)} = e^{-0.000093} = 0.999907$$

(iii) Reliability for 10 hours, parallel

$$1 - R_i(10) \approx \lambda_i t = 1.5 \times 10^{-5}, 4 \times 10^{-5}, 3.8 \times 10^{-5}$$

$$R_p(10) = 1 - (1.5 \times 10^{-5})(4 \times 10^{-5})(3.8 \times 10^{-5}) = 1 - 2.3e-14 \approx 0.999999999999998$$

Answer: MTBF of the series system = 107,527 hours; series reliability for 10 h = 0.999907; parallel reliability for 10 h $\approx 1 - 2.3e-14$ (practically 1).

113	A step-down transformer, rectifier (amplifier) and filter constitute a series system. The failure rates of these components are: transformer 1.56 per cent failures per 1000 hours, rectifier 2 per cent failures per 1000 hours, filter 1.7 per cent failures per 1000 hours. The equipment is to operate for 1500 hours. What is the probability of survival of the system? Source: Unit V SEE Jan/Feb 2021 (16IM53), Q7b; SEE Nov/Dec 2019 (16IM53), Q8b	08	–	–
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The failure rates are $\lambda_1 = 0.0156$, $\lambda_2 = 0.02$ and $\lambda_3 = 0.017$ failures per 1000 hours (1.56%, 2% and 1.7% per 1000 hours), that is 1.56×10^{-5} , 2×10^{-5} and 1.7×10^{-5} per hour. The components are in series with exponential lives, so the failure rates add:

$$\lambda_s = 0.0156 + 0.02 + 0.017 = 0.0526 \text{ per 1000 h}$$

$$t = 1500 \text{ h} = 1.5 \text{ (thousand hours)}$$

$$R_s(1500) = e^{-\lambda_s t} = e^{-(0.0526)(1.5)} = e^{-0.0789} = 0.9241$$

Answer: The probability of survival of the system for 1500 hours is 0.9241 (about 92.4%).

114	Find the reliability of a system with three components A , B and C in parallel. The reliabilities of A , B and C are 0.95, 0.92 and 0.90 respectively. Source: Unit V SEE Fast Track Oct 2020 (16IM53), Q8b	06	—	—
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$$R_s = 1 - (1 - R_A)(1 - R_B)(1 - R_C) = 1 - (0.05)(0.08)(0.10) = 1 - 0.0004 = 0.9996$$

Answer: $R_s = 0.9996$

115	The components in the system below are exponentially distributed with the indicated failure rates (per hour): 0.008, 0.002, 0.001 and 0.003, connected in series.  Develop an expression for the reliability of the system. What is the system reliability at the time of 100 hours? Source: Unit V SEE Fast Track Oct 2020 (16IM53), Q8c	06	—	—
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The four components are in series and have exponentially distributed lives, so the system works at time t only if all four survive; for independent components

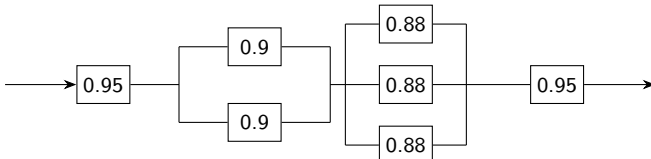
$$R_s(t) = e^{-\lambda_1 t} e^{-\lambda_2 t} e^{-\lambda_3 t} e^{-\lambda_4 t} = e^{-(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)t} = e^{-\lambda_s t}$$

$$\lambda_s = 0.008 + 0.002 + 0.001 + 0.003 = 0.014 \text{ per hour}$$

$$R_s(t) = e^{-0.014t}$$

$$R_s(100) = e^{-0.014(100)} = e^{-1.4} = 0.2466$$

Answer: $R_s(t) = e^{-0.014t}$; $R_s(100 \text{ h}) = 0.2466$. The system behaves like a single exponential unit with failure rate 0.014 per hour (MTTF = $1/0.014 = 71.4$ hours).

116	Calculate the reliability of the system given below from the reliability of the parts: a component of reliability 0.95 in series with two parallel components of reliability 0.9 each, then three parallel components of reliability 0.88 each, then a component of reliability 0.95.  Source: Unit V SEE Fast Track Jul 2019 (16IM53), Q8b	08	—	—
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The blocks are in series, so the system reliability is the product of the block reliabilities; each parallel block is 1 minus the product of the unreliabilities (independent components).

$$R_1 = 0.95$$

$$R_2 = 1 - (1 - 0.9)^2 = 1 - 0.01 = 0.99$$

$$R_3 = 1 - (1 - 0.88)^3 = 1 - (0.12)^3 = 1 - 0.001728 = 0.998272$$

$$R_4 = 0.95$$

$$R_s = R_1 R_2 R_3 R_4 = 0.95 \times 0.99 \times 0.998272 \times 0.95 = 0.8919$$

Answer: The reliability of the system is 0.8919.

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